Associated with document Ref. Ares(2022)8204179 - 28/11/2022

ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

Associated with document Ref. Ares(2022)8204179 - 28/11/2022

DESCRIPTION OF THE ACTION (PART A)

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101092339
Project name:	An Open Innovation Ecosystem for exploitation of materials for building envelopes towards zero energy buildings
Project acronym:	Exploit4InnoMat
Call:	HORIZON-CL4-2022-RESILIENCE-01
Topic:	HORIZON-CL4-2022-RESILIENCE-01-20
Type of action:	HORIZON-IA
Service:	HADEA/B/03
Project starting date:	fixed date: 1 January 2023
Project duration:	48 months

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PROJECT SUMMARY

Project summary

Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.

Use the project summary from your proposal.

Building sector is responsible for 40% of energy consumption and 36% of CO2 emissions in the EU. As nZEB becomes the
new standard, the role of building materials and smart envelope systems is becoming more and more important.
Exploit4InnoMat OITB will make available a high-end Open Innovation Testbed network for building envelopes including roofs
and facades, enabling the replication of prototypes in different buildings taking into consideration the trade-offs between the
three sustainability pillars, the life cycle stages as well as their impacts.
End-User Market Needs for low-cost, flexible, on demand material-based solutions, which will be assessed through an extensive partnership knowledge on materials characterization and modelling, monitoring and process control, environmental and assessment, regulatory and standardization, social acceptance and innovation management, realizing for
nZEB solutions. In order to achieve this target a wide range of expertise has been brought to the OITB covering fields such as Open Innovation Testbeds for nano enabled cement, non cement premixes and ceramics, advanced coatings and glazing
solutions loaded with aerogel, fibers, PCMs and other nanomaterials providing multifunctional properties; pilot lines for nanodispersion, 3D printing and roboting spraying; as well as a network of four real scale living laboratories for nZEB technologies evaluation, will facilitate a realization tool for developing close to market technological solutions. Additionally, a
semi-automated tool combining BIM analysis, fast track modelling and simulation will make enable a digital tool for utilizing
building blocks (structural, solar thermal and BIPV) in order to create a harmonized and aesthetically pleasing urban environment.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	UNIS	UNI SYSTEMS SYSTIMATA PLIROFORIKIS MONOPROSOPI ANONYMI EMPORIKI ETAIRIA	EL	983611914
2	BEN	SINTEF	SINTEF AS	NO	910945140
3	BEN	NTUA	ETHNICON METSOVION POLYTECHNION	EL	999978142
4	BEN	NTNU	NORGES TEKNISK-NATURVITENSKAPELIGE UNIVERSITET NTNU	NO	999977851
5	BEN	RISE	RISE RESEARCH INSTITUTES OF SWEDEN AB	SE	999613422
6	BEN	CSIC	AGENCIA ESTATAL CONSEJO SUPERIOR DE INVESTIGACIONES CIENTIFICAS	ES	999991722

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
7	BEN	USTUTT	UNIVERSITY OF STUTTGART	DE	999974747
8	BEN	UNAV	UNIVERSIDAD DE NAVARRA	ES	999641746
9	BEN	KA	KANSAI ALTAN BOYA SANAYI VE TICARET ANONIM SIRKETI	TR	897213626
10	BEN	ICP	INLECOM COMMERCIAL PATHWAYS COMPANYLIMITED BY GUARANTEE	IE	893373008
11	BEN	STAM SRL	STAM SRL	IT	916965542
12	BEN	AMS	ADVANCED MATERIAL SIMULATION SL	ES	952832553
13	BEN	BIBM	BUREAU INTERNATIONAL DU BETON MANUFACTURE	BE	953676744
14	BEN	IRIS	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	998993786
15	BEN	YSCCL	Y.S. CYPRIOT CATALYSTS LIMITED	CY	895894814
16	BEN	DIADIKASIA	DIADIKASIA BUSINESS CONSULTING SYMVOULOI EPICHEIRISEON AE	EL	983764495
17	BEN	CETMA	CENTRO DI RICERCHE EUROPEO DI TECNOLOGIE DESIGN E MATERIALI	IT	999500223
18	BEN	WHITESTEPS	WHITESTEPS TECHNOLOGIES ANONYMI ETAIREIA YPSILIS TECHNOLOGIAS PARAGOGIS ENERGEIAS KAI SYMMETOXON	EL	888815560
19	BEN	UNIVPM	UNIVERSITA POLITECNICA DELLE MARCHE	IT	999866689
20	BEN	BEIA	BEIA CONSULT INTERNATIONAL SRL	RO	999685687
21	BEN	UNISMART	UNISMART - FONDAZIONE UNIVERSITA DEGLI STUDI DI PADOVA	IT	909789482
22	BEN	FORTH	IDRYMA TECHNOLOGIAS KAI EREVNAS	EL	999995893
23	BEN	SOLITEK	UAB SOLI TEK R&D	LT	937257554
24	BEN	CHX	CROWDHELIX LIMITED	IE	905205165
25	BEN	F6S NIL	F6S NETWORK IRELAND LIMITED	IE	900885658
26	AP	FALU	FACHHOCHSCHULE ZENTRALSCHWEIZ - HOCHSCHULE LUZERN	CH	998686781
27	AP	TEC	TECH INSPIRE LTD	UK	942469364

LIST OF WORK PACKAGES

Work packages *Grant Preparation (Work Packages screen) — Enter the info.*						
Work Package No	**Work Package name**	**Lead Beneficiary**	**Effort (Person-Months)**	**Start Month**	**End Month**	**Deliverables**
WP1	Coordination & Management	1 - UNIS		1	48	D1.1 – Report on Data Management and IPR activities (06) D1.2 – Report on Data Management and IPR activities (M36) D1.3 – Report on Data Management and IPR activities (48)
WP2	Set up of OITB tools & procedures	11 - STAM SRL		1	48	D2.1 – Report on the designs, specifications and requirements for PL modifications D2.2 – Report on the administrative procedures & legal documents of OITB D2.3 – Delivery of 9 upgraded pilot lines D2.4 – Report on the preparatory phase of end users D2.5 – Report on operational procedures of OITB D2.6 – Report on the fine-tuning actions based on the open call selected test cases (M36) D2.7 – Report on the fine-tuning actions based on the open call selected test cases selected test cases (M48) D2.8 – Report on Exploit4InnoMat contribution to standards and regulation
WP3	Materials and modelling processes by design for building Envelope Systems	5 - RISE		1	48	D3.1 – Reports on the optimization of processes (24) D3.2 – Reports on the optimization of processes (36)

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (Person-Months)	Start Month	End Month	Deliverables
						D3.3 – Delivery of BIM objects files for Lighthouses and Demo Building D3.4 – Report on modelling and simulation activities for the selected and Open Call test cases D3.5 – Delivery of fillers produced at VEF and report on the production D3.6 – Delivery of multifunctional paints and report on the production D3.7 – Delivery of lightweight mortars and report on the production D3.8 – Delivery of insulating bricks and report on the production D3.9 – Report on the upscaling of open call TC in Materials PLs
WP4	Characterization and testing of smart building envelope materials component and systems	4 - NTNU		13	36	D4.1 – Delivery of PVT systems and report on the production D4.2 – Delivery of electrochromic windows and report on the production D4.3 – Report on the upscaling of open call TC in Components PLs D4.4 – Delivery of load bearing lightweight prefab elements for new construction and report on the production D4.5 – Delivery of prefab elements for refurbishment/external cladding and report on the production D4.6 – Delivery of 3D printed cladding elements and report on the production D4.7 – Delivery of cement free construction

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (Person-Months)	Start Month	End Month	Deliverables
						materials and components and report on the production
WP5	Characterization & Environmental Services	19 - UNIVPM		13	48	D5.1 – Report on characterization profiles of selected test cases D5.2 – Report on LCA/LCC analysis of the selected test cases (M36) D5.3 – Report on LCA/LCC analysis of the selected test cases (M48) D5.4 – Report on the recyclability assessment of Exploit4InnoMat technologies
WP6	Final Prototypes evaluation and demonstration in Living Laboratories and sustainability	3 - NTUA		1	48	D6.1 – Report on the preparation of the Lighthouses and Demo demonstration activities site (M24) D6.2 – Report on the preparation of the Lighthouses and Demo demonstration activities site (M48) D6.3 – Report on the Lighthouses validation activities D6.4 – Delivery of Exploit4InnoMat database
WP7	Impact maximisation	1 - UNIS		1	48	D7.1 – Dissemination and Communication activities (M12) D7.2 – PEDR D7.3 – Materials Helix development and clustering report (M12) D7.4 – Delivery of open call campaign identity

Work packages *Grant Preparation (Work Packages screen) — Enter the info.*						
Work Package No	**Work Package name**	**Lead Beneficiary**	**Effort (Person-Months)**	**Start Month**	**End Month**	**Deliverables**
						D7.5 – Dissemination and Communication activities (M24) D7.6 – Dissemination and Communication activities (M36) D7.7 – PEDR (M18) D7.8 – PEDR (M36) D7.9 – PEDR (M48) D7.10 – Materials Helix development and clustering report (M24) D7.11 – Materials Helix development and clustering report (M48) D7.12 – Dissemination and Communication activities (M48)

Work package WP1 – Coordination & Management

Work Package Number	WP1	Lead Beneficiary	1. UNIS
Work Package Name	Coordination & Management		
Start Month	1	End Month	48

Objectives

The overall objective is to establish the governance structure for an effective project direction and management, performing the financial, legal and administrative coordination.
WP1 has the following specific objects

1. To perform the financial, legal, administrative and technical coordination of the project
2. To establish the communication flow and methods for reporting, progress monitoring and quality assurance
3. To coordinate activities with the EC and the project officer
4. To control the project advance and time schedules & review of milestones and deliverables
5. To ensure the proper communication within the consortium
6. To develop measures for avoiding risks and establish mitigation measures
7. To promote gender equality

Description

Task 1.1: Administrative and financial project management (Start: 1 End: 48)
Leader: UNIS Main Contributors: all partners
RISE will be the contact point for the EC and the partners, and will be responsible for the administrative, financial, legal and ethical management of the consortium. The objective of this task is to ensure the smooth running of the project including communication between the consortium and the EC, so that the project is progressing in a coordinated and coherent manner and that all management activities, financial and legal aspects and other issues are managed according to the GA terms. To plan, organize & review all consortium financial and administrative activities as well as to report to the EC and the Project Steering Committee (PSC), monitor plans and operational performance of all financial and administrative activities. To review, control & report financial information (budget and costs) to the consortium & the EC. This task covers financial and administrative management/coordination activities, in accordance with article II.16 of the grant agreement. Tasks include: *Communicating between consortium and EC, including the provision of legal and financial information about the partners during the GA preparation phase, and the resolution of administrative or contractual issues during the project. *Preparing and updating the CA, including attention to IPR, in collaboration with the Dissemination-Exploitation-Communication (DEC) committee. *Preparation and post-processing of EC reviews, including support in the implementation of any recommendations from the EC and reviewers, with the support of the WP leaders. *Coordination of payments and the distribution of finances; keeping of records and financial accounts, collation of financial statements. *Day-to-day administrative management of the project, for example chairing the PSC, organizing meetings and circulating minutes. *Ensuring gender equality within the project *Collation of the deliverables and milestone reports, and assistance to the technical coordinator in the preparation of progress and periodic reports.
Task 1.2: Technical project management (Start: 1 End: 48)
Leader: NTUA Main Contributors: all partners
Prof. Maria Taxiarchou will undertake the task of technical management. He will monitor the proper implementation of all WPs on a direct day to day basis securing technical project activities and coordinate activities and collaborations within between all WP Leaders. This will also involve maintaining a risk log and milestone progress reports, which will be collated by the Project Technical Managers and forwarded to the Project Coordinator. The work package leaders will report directly to the project technical manager and will be responsible for: *Dictate workflow scheduling and work plan change control procedure, *Coordinate communication between partners within the work package, *Coordinate the work package specific technical activities, *Review and manage the impact of economical and societal issues. The Technical Manager and the WP leaders will handle technical risk management. Before each stage of the project commences the appropriate manager will complete a full risk assessment. This will follow the process identification, evaluation and response planning. The impact of the risk will then be assessed, and contingency resources planned. The WP Leaders are going to actively participate in this process.
Task 1.3 Quality Assurance and risk management
Leader: UNIS Main Contributors: all partners
Implementation of quality assurance procedures with particular focus on monitoring review activities built into the work programme and submitting quality review reports. A quality plan within the Project Quality Handbook (deliverable D6.3)

will specify the quality procedures, internal and external peer reviews, control, monitoring and reporting activities to be implemented in the project to ensure that satisfactory quality standards are met. Implementation of the plan including the risk library, processes and procedures for risk management and reporting to be performed by all partners in each project WP, covering both the project operational risks and the framework level barriers/risks. Monitor the progress to ensure that proactive measures are taken to mitigate risks which could jeopardize the planned project outcomes. Monitor the risks outlined in Section 3.2.

Work package WP2 – Set up of OITB tools & procedures

Work Package Number	WP2	**Lead Beneficiary**	11. STAM SRL
Work Package Name	Set up of OITB tools & procedures		
Start Month	1	**End Month**	48

Objectives

The major objective of the WP2 is to define the managerial structure and the operational procedures of the Ecosystem. WP2 has following specific objects

1. To define the legal aspects and the management structure of the OITB and its relation to the members
2. To determine internal operational procedures and communication channels including the groups of services, their elements and the interactions between them.
3. To develop methodologies for translation of industrial needs to workflows of the Ecosystem
4. To upgrade and modify the 9 pilot lines of the OITB
5. To proceed in the preparation of the selected test cases by the end users and provision of feedback to PLs
6. To continuously improve the internal and external operational procedures of the Ecosystem

Description

Task 2.1: Set up of the OITB administrative, operational tools and procedures (Start: 1 End: 18)
Leader: F6S Main Contributors: all core ecosystem partners
Subtask 2.2.1 Explot4InnoMat Administrative Procedures(F6S): During this task the administrative structure of the Ecosystem and all legal aspects will be concluded. The consortium has already worked and presented the draft administrative structure for the Ecosystem in Section 2. The target is to work towards the set up of the Single Entry Point of the OITB that will undertake the whole administrative and secretarial tasks. Important aspect will be the founding of a new legal entity (Joint Venture) that will act as the SEP. During this task the description of its mission statement will be made along with its objectives. In parallel to the set up of the SEP the internal operation procedures and protocols that will govern the relationships between the SEP and the members of the OITB as well as with the clients will be arranged. At first place an agreement between the SEP and the members of the OITB will be formed. This Framework Agreement will fix the terms of collaboration and will provide details for the services of each core partner and terms of access, taking also into account possible national regulations and laws.
Subtask 2.2.2: Explot4InnoMat platform (subtask leader: UNIS): Explot4InnoMat Partners Intranet (PI) will be the main collaboration tool among OITB partners. Explot4InnoMat PI will be a roles-based web collaboration platform responsible to manage communication, training and knowledge sharing among ecosystem partners. Its main objective will be to handle, share, process & utilize data managed by Explot4InnoMat DMP providing a friendly user interface to retrieve and visualize raw data. Apart from data processing operation, Explot4InnoMat PI will provide tools for partners communication (notifications, news dashboards, asynchronous chat etc.), project diary, knowledge exchange via e-learning tools, web conference capabilities, survey and voting tools, download center & task assignments mechanism. It will operate under an SSL protected URL, being accessible only from consortium members managed from one or more central administrator(s).
Subtask 2.2.2: Operational procedures (subtask leader: UNIST- SEP): In this subtasks tools and methods that will facilitate the access of potential end users to the test bed will be developed. Potential customers will be able to be informed from the services of the OITB through the dedicated web platform. The website will include analytical presentation of the Pilot Lines and the services. It will include a dedicated page that will be application page via which customers can apply for access to the ecosystem. The initial application will be done via a questionnaire that will contain closed and open type questions concerning the test case. The questionnaire will be developed in a manner that will guide the applicants across the various services based on the needs of the test case. Important output of the subtask will be the development of the methodology for the interpretation of the concepts arrived in the SEP to workflows.

Task 2.2: Upgrade and modification of pilot lines (Start: 1 End: 12)
Leader: Stam Main Contributors: IRIS, Pilot Line owners
During this task, necessary modifications and upgrade of the Pilot lines will take place. Already in Section 1, a brief description reporting the modification of the PLs is presented. Efforts will be dedicated on current evaluation of unit operations, detailed engineering of each pilot upgrade, process control, process and plant design, instrumentation, automation and industrial measuring technology and finally all the relevant safety and health measures which should be taken into account. The first step will be the detailed evaluation of the know how obtained through the existing PLs operation, a procedure which will be the basis for each upgrade plan analysis, which should ensure the safe operation in all phases, to maximize operability and accessibility. In parallel each plan will be designed accordingly in order to minimize the construction, operational and maintenance cost. The road map includes: inventory of existing equipment and sensors; specifications and requirements definitions for the test case preparation and running in terms of production capacity, samples dimensions and QA/QC, data acquisition and monitoring; preliminary design of modifications required, namely: block schemes of lines and modification options; market analysis and quotations of necessary equipment and interventions; final definitions of actions and equipment (fit for pilots' allocated budget and time schedule) and finally follow up of interventions and technical assistance.
In all nine pilot lines there will be interventions targeting in the following aspects
Subtask 2.2.1: Infrastructure upgrades (subtask leader: FORTH):
Task partners will initiate the first month of the project a communication link with the pilot lines, in order to identify their requirements and their capabilities. For each pilot line, a detailed layout in a 3D computer-aided design (CAD) system will be prepared, also including the related piping and instrumentation diagram (P&ID) that has been developed. Through the layouts the partners will be able to optimize each upgrade plan, to process and evaluate alternatives, and thus to effectively produce an instructional drawing for the unit assembly.
Subtask 2.2.2: Monitoring upgrades (subtask leader: STAM, IRIS):
Subtask 2.2.2. will provide a state-of-the-art monitoring and performance monitoring system, including safety issues (ventilation, dust and noise control etc.) and will offer to PLs and the partners access to the event-data needed for innovative research and development. IRIS will consider on-line and at-line approaches with different photonic techniques (NIR spectroscopy, diffractometry, turbidity spectrometry) to monitor the dispersion of PCMs and aerogels, as well as composition in SINT (PL1); moisture, distribution of nanoparticles and composition of mixtures in NTUA (PL3); and dispersion of nano-additives in RISE (PL5).
Subtask 2.2.3: Process control of PLs (subtask leader: STAM):
The centerpiece of the pilot plants upgrade will be a state-of-the-art distributed control system following each case study requirements. The new control system will provide advanced visibility on operational parameters and quality control (like mass and volume feeding rate, electrical parameters, viscosity, moisture etc) which will allow operators to monitor all critical process parameters in one display. Open source software will be used in case of incompatibilities, like PL2 where the commercial 3D control platforms are not able to handle complex processes like simultaneously 3D printing and extrusion.
Task 2.3: Preparatory phase for the test cases (Start: 1 End: 12)
Leader: NTUA Main Contributors: SOLI, UNIS,FORTH, PL owners
This tasks involves all preparatory activities related to the test cases that are already scheduled to access the OITB and work towards its validation. End users will collaborate with core partners in order to specify workflows of activities to be run, as well as provide requirements and define any characterisation techniques and cross sectorial services that will be accessed. Moreover under this task raw materials provision, and consumables will also be dealt with. The test cases will set the desirable specifications for the products, and the pilot line will apply the necessary conditions (quality of materials, formulations and processing conditions) for their production to the definition of the specific requirement related to definition of quantities and specification of materials to be used. The end users will also provide recommendations during the upscaling process of the pilot lines, based on their industrial experience.
Task 2.4: Fine tuning of the OITB tools & procedures (Start: 19 End: 48)
Leader: F6S (SEP) Main Contributors: all partners
During the first period of the project the main administrative and operational procedures will be defined and structured. These procedures will be set to action during the initiation of the implementation of the test cases that is the validation phase. As already discussed, the validation phase consists of selected test cases and new test cases will have access in the OITB through the open call. These test cases will provide the necessary feedback on the ecosystem and thus, possible issues concerning these procedures (e.g. deficient communication between the partners, delays in responses, misleading legal documents, contracts, NDA etc) will be improved and fixed. The target is that by the end of the project, the OITB will have identified all deficient procedures and fix/replace them with more efficient ones.
Task 2.5: Standardization & Regulatory services (Start: 1 End: 48)
Leader: UNIST Main Contributors: all core OITB partners
The aim of this task is to gather data on relevant standards (e.g. ISO 21105 for building envelope assembly, ISO 15712 for acoustics, ISO 16932 for glass in buildings, ISO 6781 for thermal insulation, and more) in order to evaluation

the performance of the developed materials and components for the building envelope. It will also cover product standardization for the Explot4InnoMat end users (e.g. as a basis for CE marking) and standardizing new and adjusted measurement methods as well (e.g. for the scientific partners of the consortium). ASHR will collaborate with all involved partners to make sure that the relevant properties of developed materials are properly represented in standards, being the basis for a full exploitation on the market.
Moreover, all the products manufactured in the PLs will be in alignment with the international regulatory framework to be commercialise in European and non-European countries. While some regulations affect the whole European territory, like REACH, CE marking and the Energy Efficiency of Buildings, other regulations are specific of each country such as the Technical Building Code in Spain, the certifications from CSTB in France or from TCNA in EEUU.

Work package WP3 – Materials and modelling processes by design for building Envelope Systems

Work Package Number	WP3	**Lead Beneficiary**	5. RISE
Work Package Name	Materials and modelling processes by design for building Envelope Systems		
Start Month	1	**End Month**	48

Objectives

Multiphysics and multiscale materials modelling will be conducted in order to estimate mechanical, thermal and hygrothermal properties of materials and components. Parameters, namely elastic modulus, compressive or tensile strength, abrasion resistance or friction coefficient; thermal conductivity, U-value or thermal storage capacity and water absorption or waterproof will be optimized.
WP3 has following specific objects:
1. To develop models of the manufacturing processes to integrate the information from material modelling in the targeted industrial process
2. To develop a monitoring and optimization platform
3. To develop an accurate toolset for materials modelling reducing testing and the trial and error process.
4. To conduct predictive modelling of functional properties of materials.
5. To optimize materials parameters and evaluate uncertainties in modelling.
6. To set up specifications for multifunctional coating pilot line including production of nano-additives, polymer synthesis and coating formulations.
7. To identify and develop surface engineered nano-additives at lab scale for both lightweight insulating building envelops and coatings.
8. To develop a pilot line for scaling up selected nano-additives up to 100L scale and validation.
9. To identify, formulate and testing of nano-enabled multifunctional coatings at lab scale for both lightweight insulating mortars, bricks and metal panels.
10. To develop a pilot line for scaling up multifunctional coatings up to 30L scale and validation.

Description

Task 3.1: Multiphysics and multiscale materials modelling (Start: 13 End: 48)
Leader: AMS Main Contributors: USTUTT, end users, PL owners
Physical and data driven models will support and complement the Pilot Lines services. Dealing with physical models, the combination of multiscale materials modelling with Multiphysics is a powerful tool that permit to link the composition and material structure with the final performance of materials and components. The hierarchical multiscale modelling, from atomistic to structural scale, bridges nanoscale mechanisms with macro mechanical properties as elastic modulus, compressive or tensile strength, abrasion resistance or friction coefficient; thermal as conductivity, U-value or thermal storage capacity and hygrothermal as water absorption or waterproof. The possibility of data driven models based on machine learning or deep learning will be considered as an alternative to physical models. When the access to large amount of data is available, AI methods based on neural networks are applicable and could be an alternative or a complement to physical formulation.
USTUTT will provide modelling and simulation for the design of structural graded lightweight concrete elements and a macro and mesoscale FEM tool (MASA) for the analysis of mechanical test results. A high resolution 3D-µCT-scanner is available within a cooperation at USTUTT to generate volume models of graded parts of the developed elements. The routines for evaluating the interior design of the construction element will be expanded to deal with differing hollow

body types and sizes considering multiple load cases. The option of integrating different reinforcement materials and methods will also be of research interest. All developed models will be fed by data produced in the lighthouses and Demo building, and thus their accuracy and reliability will be enhanced.

Task 3.2: Process modelling and Building Information Modelling Services (Start: 6 End: 36)

Leader: STAM Main Contributors: PL owners

In this Task, the Pilot Lines will be modelled in order to simulate the operations of each process stage. This approach will allow to create detailed models and to make analysis on material workflows based on the available controlled systems. The first step will include a detailed analysis of each PL process and process control evaluation, focused also on the maintenance or control operation problems for tracing where to intervent and to get production more efficient. At this level, historical operational data will be collected for creating create a data warehouse from the process values, operation points, set points collected and wastes produced. Based on the collected information from each PL, processing models will be developed and calibrated through the historical data. Each model will provide a basis for evaluating current production capabilities and limitations, their performances and for identifying a methodology for upscaling the production. After the upgrade completion, the digital models will be revised and calibrated accordingly for supporting the Pilot Line owners, for managing and optimizing production, inventorying policies, as well as reducing possible errors, material flow delays (particular attention will be paid to the identification of any bottlenecks and to the provision of indications based on scheduling simulations) and wastes reduction. The models developed will also envisage digital twin functionalities, in order to visualize and to explore the Pilot Lines environments from different point of views.

The BIM Process is based on standardization of building components. Their main features (geometric, dimensional, material, performance, lifecycle, etc.) will represent the informative attachment of each component. This object-oriented multilayer informative character actually represents one of the main advantages of BIM approach that both the design and renovation phase as well as the control and manage one (Nagel, Stadler & Colbe, 2009). After all the pertinent information about the building have been stored in the BIM model, it is fundamental to retrieve and interact with it through a 3D graphic interface able to respond to different requests about the material involved: on one hand, communicate of course the project itself but, at the same time, extract any data necessary for its detailed description, components checking and problems finding (Krygiel, Nies,& Green, 2008). From this standpoint, the nature of the material will be a coordinated set of basic technological features. It is also clear that the deep knowledge of any new artifact implies the correct control of the variable typology and nature of such components.

The framework will be organized in 4 steps – plan/execution/validation/result – involving: BIM AECO Market, BIM regulation context, Impact of BIM, BIM adoption/procurement/uses, BIM LOD (level of Details/accuracy), BIM interoperability, BIM Information Exchanges (open format), BIM Application.

Task 3.3: Multifunctional coatings for building envelope (Start: 13 End: 24)

Leader: RISE Main Contributors: KANS,

The development of multifunctional coatings will include the following steps. Initially the specifications of multifunctional coating pilot line including nano-additives, polymers and coatings will be set. Based on existing standards, policies and guidelines requirements for nano-additives, the new coating systems will be established. At first stage the development of surface engineered nano-additives will be evaluated. This will include commercially available nanomaterials (nano-SiO2, nano-Al2O3, nano-ZnO, CNF, carbon black, nano-clays, nano-TiO2, nano-FeOOH, nano-Cu etc.) suitable for coatings and building envelop applications. The target will be to be screened for the development of surface-engineered nano-additives to enhance their compatibility with matrix as well as for the enhanced performance of products. These formulated nano-additives will be scaled up in the pilot line at 100L or 10kg batches. After this the development of nano-enabled multifunctional coatings will take place. Functionalized nano-additives will be mixed with the main coatings formulation in order to produce paints for galvanized steel sheet and composite mortar panel surfaces. The painted surfaces expected to have hydrophobic, anti-soiling, anti-corrosion, UV resistant, and IR- reflectance properties. Developed multi-functional coating formulations will enhance the performance of materials, increase the material efficiency and thus decrease the maintenance costs. The developed formulation will be provided to the end users KANS for validation. The developed multi-functional coatings will be provided relevant partners for the validation as well as to lighthouses and the demo site.

KPIs: Delivery of 800L of multifunctional coatings for walls

Task 3.4: Lightweight insulating mortars (Start: 13 End: 24)

Leader: NTUA Main Contributors: CETMA, BIBM, CSIC

Main objective of this task is the development of advanced and high-quality lightweight mortars and cement-based rendering premix products with improved multiple functionalities under PL3 following user requirements and masonry applications. The research will be distinguished into two (2) subtasks:

Subtask 3.4.1 development of additives: Hybrid perlite-based lightweight aggregates (i.e. closed structure perlite microsphere) will be produced on a PL3 subunit, equipped with a multi-parametric, innovative Vertical Expansion Furnace (VEF) at the premises of NTUA. The mechanical preparation of raw perlite towards obtaining of a grade with mean size distribution between 100 and 300 μm. Expansion of perlite constitutes will be implemented in the VEF at temperature between 900 and 1100°C. Following preparation of expanded perlite microspheres, the material will be

characterized in terms of the characteristic properties (e.g. density, thermal conductivity, water and oil absorption, LOI). Afterwards, additives of multifunctional agents (e.g. PCMs and nano metakaoline) will be also developed within PL3 in order to be applied in the new mortars for enhancing added-value properties such as hydrophobic behaviour, reflective, low density, low thermal conductivity, high compressive strength, low water uptake, fire resistance, and lower embodied energy to the final material.
KPIs: Delivery of 2m3 of hybrid perlite based lightweight aggregates produced at VEF
Subtask 3.4.2 development of final products: In the second subtask, the development and optimization of the mortars will be performed. It will be focused on granulometry, density, thermal conductivity, compressive/flexural strength and water up-take, whereas it will be based on the investigation of a number of parameters: formulation and raw materials type (lightweight aggregate type and multifunctional agents), variable mixture proportion/aqueous solution solids content, curing conditions, granulometry (i.e. particle size distribution), as well as dispersion solutions and water percentages to obtain the desired consistency and rheological properties. Additional tests will be performed in order to evaluate the workability, application and performance of the new mortars under severe temperature conditions (low or high temperatures) in respect with their thermal and structural performance. The new mortars should have λ value between 0.048-0.160 W/mK, density 230-1100 kg/m3 and compressive strength higher than 1.5 MPa. As the optimum PL3 operational parameters will be identified and monitored, reliable data for LCA and techno economic analysis of each solution delivered will be provided regarding energy consumption and productivity of the line and system efficiency. The optimum formulations identified at PL3 will be used for the production of the required samples to be used for the complete characterisation in the frame of WP7 and the appropriate quantities for the demo scale activities foreseen in WP9.
KPIs: Delivery of 4m3 of lightweight insulation mortars
Task 3.5: High performance insulating bricks (Start: 13 End: 24)
Leader: CSIC Main Contributors: CETMA, AMS
This task involves all activities related to the development of new bricks with improved functionalities, especially in respect with insulation, acoustic and mechanical properties by applying a cost effective and low energy consuming process. The innovative ceramics will be developed by embedding CDW, PCMs and inorganic loose fill materials on their main body and their internal geometries. The overall approach will be focused on advanced ceramic body composition, geometrical and structural characteristics towards the achievement of good ceramic stability and improvement of their thermal characteristics, infrared blocking and microporous structure. The new ceramics will consist of the inorganic paste and additives; their proper combination will be studied aiming to improve these properties. The shape and geometry of the brick is mainly characterised by its dimensions (EN 772-16, 2000). The external dimensions (height and length) should respect the existing regulations (BS EN 771-1). In the frame of this study, CSIC with the technical support from SCHL will work on the characterization and advice of the different brick thicknesses, aiming to achieve with one brick the same technical and insulation efficiency as the currently applied combination of brick and insulation material. On this framework the pilot line after the upgrade and commissioning will be tested in relation to the operational parameters, the data acquisition process and it's overall performance.
Characterization tests for these advanced materials will be carried out by means of X-ray diffraction techniques, scanning electron microscopy, infrared spectroscopy, mercury intrusion porosimetry, etc. Moreover, CSIC will also study nanomaterials effect on the microstructure of ceramic and cementitious materials using adsorption-desorption isotherms, laser technology, nanoindentation and measurement of reflectance and / or transmittance, among others. Evaluation of resistance to freeze and thaw cycles, resistance to carbonation and to chloride attack will be analyzed too. Novel approaches to embed high performance insulation materials within traditional construction components will be also addressed in this task with the technical support from SCHL.
KPIs: Delivery of 1000 units of medium sized high performance insulating bricks, adapting two geometrical solutions
Task 3.6: Open call test cases for Materials PLs (Start: 25 End: 36)
Leader: CSIC Main Contributors: RISE, NTUA, AMS
Under this task, test cases resulting from the open call will be introduced into the OITB to produce, test and validate innovative solution in the field of multifunctional advance materials for the building envelope. It is expected that at least one new test case will be introduced in each one of the pilot lines of this group (a total of 3 new test cases). Based on the requests of the end users, the OITB will develop the corresponding workflows and will implement the test cases including validation.

Work package WP4 – Characterization and testing of smart building envelope materials component and systems

Work Package Number	WP4	**Lead Beneficiary**	4. NTNU

Work Package Name	Characterization and testing of smart building envelope materials component and systems		
Start Month	13	**End Month**	36

Objectives

This WP will focus in the design and development of the building's envelope components (PVTs lightweight structural elements and smart windows), the redesign of PV cells repositioning, the identification of the optimum materials for the element assembly and the identification of plug and play solutions for both electrical and liquid flow circuits. Moreover, WP 4 is focused on the design and pilot line production of innovative insulating lightweight walls, 3D printed cladding elements with advanced functionalities and environmental friendly cement free inorganic polymers for construction applications.

The two pilot lines will evaluated in relation to their performance and will be optimizing for adapting and evaluating new technologies requested through external services.

WP4 has the following specific objectives:

1. To identify the specifications and to design the PVTs elements and pilot manufacturing procedure.
2. To select the optimum methodology for enabling electrochromic effect and to design the manufacturing process.
3. To produce the corresponding components and commissioning to the lighthouses and demonstration site.
4. To identify the specifications and to design the manufacturing procedure for each technological solution.
5. To optimise pilot lines production process
6. To develop the corresponding system for delivering to the lighthouses and demonstration site

Description

Task 4.1: PVTs embedded in lightweight structural elements (Start: 13 End: 24)

Leader: SINT Main Contributors: SOLI, UNST

For the cost-efficient production of PV modules necessary in the project, SOLI based on modularity of the product line will develop and deliver glass/glass monocrystalline PVs optimised for both and electric and solar energy harvesting. SINTEF in close collaboration with SOLI will identify the optimum cell and junction boxes positioning for commissioning processing equipment based on the final design. Selection of appropriate materials and methods to facilitate the process of dismantling, recycling and reusing will follow. In relation to fluid circulation system SINTEF will design the and select the appropriate materials for enabling optimum heat exchange between PV cells and circulated fluid. Easy adaptable commercial solutions like water-tight fit locks will be applied for coupling PVTs strings and extending the low-pressure water circulation system. The PVT modules will have less than 1cm thickness and reduced weight. The PV cells and heat exchanging system will be mechanically or chemical bonded. The integrated PVT module will be evaluated in relation to • IV characteristics (Voc, Isc, FF, p), and degradation rate according to IEC standard 61215 (Damp heat test and thermos-cycle tests). Small scale prototypes will be assessed ed in relation to their performance, reliability and stability. Based on the optimum production process the pilot unit will be upscaled for delivering PVTs able to be adapted on commercial façade solutions and for delivering large size prefabricated elements. At this stage the optimum geometries, materials and assembly procedure will be identified, delivering a detailed plan for optimising PL production capacity for delivering active prefabricated elements embedding PVTs.

KPIs: Delivery of 50 m2 of prefabricated PVT systems of total power capacity 100 kWp (both electrical and thermal energy).

Task 4.2: Smart windows (Start: 13 End: 24)

Leader: NTNU Main Contributors: SOLI

The main objective of this task is the development of nanostructured electrochromic (EC) materials enabled smart window components, following an innovative method both for production and assembly For this reason, EC nanomaterials will be synthesized and produced following a combination of variable chemical composition, crystal structure, size, dimensionality at nanometer-scale and consequently different properties in order to result in high performance EC smart windows. The applied EC nanomaterials will be prepared as well as assembled via solution-based methods that are usually less costly and easily scalable. In this way, cost-efficient and fast method will be used for preparing homogeneous thin films, which can be adjusted to achieve structurally uniform thin films containing tiny EC nanoparticles, based on the fact that the interaction between EC nanoparticles and the substrate need to be strong enough to keep the produced films from peeling off in order to obtain efficient high end products. Additionally, the development of nanoelectrochromics will be performed and applied in order to provide improved switching times, higher coloration efficiency, and most importantly, improved stability/durability against electrochemical cycling, compared to already in-market products. This task will be performed, according to the following steps:

a. Development and ECO design optimization of the structural board, including the proper anchoring system

b. Synthesis and development of nanoelectrochromic materials
c. Nanoassembly for high performance EC in respect with coloration efficiency, switching time, and cycling durability
d. Testing, evaluation and optimization of the overall proposed solution (i.e. nanostructured electrochromic materials enabled smart window)
e. Preparation of upscaled production for application on demonstration sites
KPIs: Delivery of 50m2 of electrochromic glazing windows
Task 4.3Prefabricated insulating lightweight walls (Start: 13 End: 24)
Leader: UNST Main Contributors: FORTH, YCCM, NTNU
Subtask 6.1.1: Load bearing lightweight prefabricated elements for new construction
Load-bearing lightweight cement based panels will be develoled using CDWs and a steel fiber reinforcement process. All the materials (both aggregates and binding systems) will be identified so as to ensure the optimum compatibility between the ingredients and strong cohesion and coherence properties leading to improved mechanical properties. The investigation of lightweight panel development will not be limited to material level but will also include the manufacturing priceless ss investigation. Thus, the optimum mixing procedure will be also identified and applied to the robotic spraying pilot line. Furthermore, special focus will be given to the shaping of the building blocks, delivering advanced geometries following end user requirements. PL7 will deliver prefabricated element with a ratio of CDW higher than 15%, U-value less than 0.26 W/m2K, surface weight less than 360 kg/m2 for a thickness less than 32 cm and EI60 rating fire resistance
KPIs: Delivery of 100m2 of load bearing prefab elements (maximum unit size: 2x2 m)
Subtask 6.1.2. Prefabricated elements for refurbishment/external cladding
In this subtask, the synthesis and process optimisation for the new lightweight prefabricated element for refurbishment will take place, by combining cement as a binder, a minimum steel fibers reinforcement ratio, for delivering concrete with insulation properties in order to substitute insulation panels and reduce thickness. This subtask will deliver lightweight panels for external cladding with a ratio of CDW higher than 15%, λ less than 2,30 W/mK, surface weight less than 160 kg/m2 for a thickness less than 13 cm and EI60 rating fire resistance.
KPIs: Delivery of 180m2 of of external cladding systems
Task 4.4: Multifunctional cladding systems (Start: 13 End: 24)
Leader: NTUA Main Contributors:, UNST, FORTH, ULIG, NTNU
Subtask 4.4.1: 3D printed cladding system
Aim of this task is to develop multifunctional cladding elements embedding complex close cellular structures filled with additives, able to provide enhanced thermal properties. The production line includes a robotic system handling two heads, from which the first one follows a 3D printing process using a semi inorganic filament (PLA mixed with industrial minerals) and the second head working in parallel is an extrusion system able to fill the 3D printer geometries with loose fill materials or pastes. The first step will be the identification, testing and evaluation of the optimum filament synthesis with the minimum organic ratio, for delivering semi-inorganic elements able to fulfil construction products requirements. The second step and following the modelling results delivered from AMS, will be the design of 3D printing models (like STL, VRML, Gcode) for routing both printing and extrusion head. The production time between the dierent congurations will be evaluated by adapting both simple and complex shapes in relation with the proper thickness selection. After the selection of the most promising geometrical models, a group of additives (loose fill or pastes) will be used to fill the close cellular structures and depending the end user requirements. Following this production method, Task 6.2 will develop and provide medium sized (80x80cm) lightweight cladding elements (300-500 kg/m3) with enhanced insulating properties ($\lambda < 0.12$ W/m K) by filling with close cellular structure-loose fill perlite and panels with similar geometries where the cells are filled PCMs. The elements will be evaluated in relation to their structural and thermal properties, whereas their durability will be tested on accelerated weathering chambers
KPIs: Delivery of 200 units of 3D printed cladding elements with size 50x50 cm
Subtask 4.4.2: Cement-free construction materials and components
The aim of this WP is the development of various formulations of inorganic precursor polymers, which upon suitable processing will to the direct synthesis of formed building components/elements. Therefore, the selection of the optimum raw materials, the identification of the proper process conditions as well as the identification of the most promising polymerization synthesis are most important parameters for the formation of inorganic polymers with the optimum properties. On this framework selected formulations will be processed on the pilot line for manufacturing cement free construction products. The addition of selected waste materials and CDW aggregates from PL9 will be evaluated, for further improving the embodied energy of the new products and their environmental footprint. The effect of the curing temperature and time on the properties of the solid final polymeric material will be evaluated in detail and will be optimized accordingly. From the operation of the pilot scale system reliable results will be obtained concerning its technical performance and energy and materials consumption, which will be used for the detailed calculation of the embodied energy and the environmental footprint of the inorganic polymers.
KPIs: Delivery of 180m2 of cement free components (of 1m maximum dimension)
Task 4.5: 'Open call test cases for test cases for Components and smart manufacturing PLs (Start: 25 End: 36)

Leader: NTNU Main Contributors: SINT, UNST
Task 4.3 focuses on producing, testing and validating innovative solution in the field of multifunctional component for the building envelop during the open call of the Explot4InnoMat OITB. The ecosystem's services will also be validated during the open call to ensure its fast and efficient operation after the completion of the project. At least one new test case will be include in the tow pilot lines of the group.
Moreover,this task focuses on providing to interested clients the opportunity to access the Explot4InnoMat OITB and its services in order to develop, test and validate their concepts in the field of smart manufacturing of materials and components for the building envelope. At least one new test case will be included in each one of the three pilot lines of this group

Work package WP5 – Characterization & Environmental Services

Work Package Number	WP5	**Lead Beneficiary**	19. UNIVPM
Work Package Name	Characterization & Environmental Services		
Start Month	13	**End Month**	48

Objectives

WP5 will be dedicated to the services for the characterization of the new materials and components solutions as well as to the assessment of their environmental profile in terms of LCA/LCC and recyclability. A wide range of testing infrastructures will be utilized for in-detail characterisation of materials and final products in terms of their physical and thermal, mechanical and other properties, structure and morphology. Of pivotal importance will be the assessmentof LCA/LCC, the environmental impact and economic risks related to the production, installation, maintenance and disassembly procedures as well as to the potential re-use/recycle.
WP5 has the following objectives:
1. To characterize the materials' and components' characteristics of all Explot4InnoMat technologies.
2. To further develop and adjust the characterisation techniques for the new materials as a pre-requisite to standardization of measurement methods and products
3. To Evaluate the environmental impact of the process and products developed, and the potential uses, thanks to LCA/ LCC tools and methodologies following ISO 14040 and 14044
4. To investigate the recyclability of the new materials and components and their after-life value

Description

Task 5.1: Structural & morphological characterization (Start: 13 End: 36)
Leader: UNIPVM Main Contributors: PL owners
This task involves all activities related to the structural and morphological characterization of material and structures developed. A network of partners will collaborate in order to result in fast characterization workflows, for the determination of mechanical and morphological parameters (material tensile, bending, compression, strength, durability and appearance) but also load capacity, resistance to wind or snow load, deformation, impact loads and earthquake resistance. These characterizing parameters may be determined on a material level, on a component level (fixings, anchorage) and on a full scale level.
In a first step all relevant parameters have to be identified and a state of the art of existing characterization and testing regulations and standards (EN, ISO, ASTM) will be established. This will be coordinated with the works in WP8. In a next step a state of the art of all existing testing infrastructure within the network will be elaborated. Suitable and accurate characterization and test methods for the specific product, material or component will be defined and if necessary new test methods have to be developed.
Task 5.2: Functional properties characterization (Start: 13 End: 48)
Leader: FALU Main Contributors: PL owners, end users
This task involves all activities related to the functional properties characterization of materials, components and full size building envelopes developed. A network of partners will collaborate in order to result in fast characterization workflows, for the determination of all functional properties, which are mandatory for nZEB building envelopes. All materials and elements that will be developed in PLs will follow Construction Products Regulation (Regulation (EU) No 305/2011). An indicative list for materials evaluation includes Aggregates water absorption (EN 13055), thermal conductivity (EN 12667), assessment of surface properties of the coating: Reflectivity (EN 903),·assessment of plaster/mortar for thermal conductivity (EN 12667). In relation to component and envelop technological solutions, properties like watertightness

(EN 12154, EN 12155), airtightness (EN 12152, EN 12152) level, acoustic insulation (EN ISO 16283, EN ISO 10140-2, etc.) and fire performance (EN 13501-1, EN 13501-2, etc.) will be also evaluated. BIPVs will evaluated accordingly to EN 50583-1:2016 Photovoltaics in buildings - Part 1", and IEC 63092 Photovoltaics in buildings categories. Testing and characterization of materials, components and full-scale façade elements will then be coordinated and carried out by FALU and USTUT already equipped with the appropriate infrastructure. In a few cases certified laboratories will perform the test needed. Based on the test results, materials and components may be further improved and optimized. Results of this task will also provide valuable input for standardization committees in the area of building envelopes and regulatory agencies.

Task 5.3: LCA/LCC analysis (Start: 13 End: 48)

Leader: CETMA Main Contributors: CETRI, all partners

Subtask 5.3.1 Life Cycle Analysis: Sustainability assessment covers life cycle assessment (LCA) studies, where the methodology identifies the environmental impacts of a product, production system or a service throughout its life cycle. This task consists on the definition of goal and scope of the LCA to be performed. In defining the scope of the LCA, focus will be given to the identification and characterizations of selected components and materials, taking into account their functions, properties and compositions. After this characterization, all life cycle stages will be assessed considering different scenarios, in order to evaluate qualitatively the influence of each component and material on the environment, to analyse the impact pathway on the environment and to define the product system boundaries. For the EC construction industry a value chain life cycle covers manufacturing processes for materials including the acquisition of raw materials required (cradle phase), construction of materials and components, use of manufacturing process, transportation, installation and end of life scenarios. Varying the materials has different potentials in nZEB concepts and also different carbon footprints for production and recycling. To conduct a comprehensive comparison between differentmaterials, the full life cycle will be taken into account. LCA will be carried out according to ISO 14040: Environmental management -- Life cycle assessment -- Principles and framework and ISO 14044: Environmental management -- Life cycle assessment -- Requirements and guidelines. The study will use the multicriteria and holistic approach offered by LCA tools and methodology and following the previous mentioned standards. LCA studies will be performed using the SimaPro software packages.

Subtask 5.3.2 Life Cycle Cost Analysis: LCC follows a similar approach to LCA in order to provide an economic evaluation, which considers not only apparent costs but also hidden costs such as those related to environment. The lifecycle cost is often referred to as the sum of all costs incurred during an asset's useful life and allows for a more appropriate cost-benefit analysis. This method will assist in the reduction of the total cost, identification of high-cost stages in a product's lifecycle, and comparison of competing products. The LCC analysis will be based 6 steps: analysis plan, model development, model analysis, analysis documentation, review of results, and implementation & update. Both LCA and LCC methodologies involve an iterative structure and the initial goal and scope definition will be revised throughout the Project. Based on the project activities in other WPs, a project specific Life Cycle Inventory will be developed to be utilized in both LCA and LCC. Finally, life cycle impact assessment will be carried out consider the most relevant impact categories and metrics.

Task 5.4: Recyclability assessment and circular economy aspects (Start: 13 End: 45)

Leader: NTUA Main Contributors: Pilot Line owners

Construction and demolition waste is receiving more attention recently following new circular economy policies. In order to maximize the sustainability of the new solutions that will be proposed by Explot4InnoMat clients, recyclability assessment will be carried out. The assessment will be carried out following the below steps:

Propose appropriate disassembly scenarios from the building structures

Proceed in compositional analysis of waste streams, by products and scrap materials

Carry out tests to see if new products and materials have the potential to produce a useable secondary raw material

Examine the financial values of the materials at end of life as secondary raw materials or for potential reuse of them.

Report on a recyclability profile for the new materials and components

NTUA will utilize the Pilot Line 9 for materials separation and reprocessing in order to acquire up-to-date data about the recyclability of the materials and their potential reuse as secondary raw materials inside the Explot4InnoMat open innovation ecosystem. This task will be implemented following the EU CDW protocols and guidelines as published on 18/9/2018.

Work package WP6 – Final Prototypes evaluation and demonstration in Living Laboratories and sustainability

Work Package Number	WP6	**Lead Beneficiary**	3. NTUA
Work Package Name	Final Prototypes evaluation and demonstration in Living Laboratories and sustainability		

Start Month	1	End Month	48

Objectives

The purpose of this phase is the long term (up to 24 months) validation of the products and materials solutions developed in the previous phases on the three lighthouses and the public demo building.
WP6 has the following objectives:

1. To install and calibrate sensors and monitoring systems in the lighthouses and demo building
2. To define all the safety procedures and acquire appropriate licences for the installation of the components
3. To install the fabricated prototypes in the lighthouses and the demo building
4. To perform tests on the integrated prototypes
5. To monitor the efficiency of the new building solutions in terms of energy harvesting, energy savings of building, durability and indoor air quality
6. To develop a building envelope's materials database, based on the materials tested in the lightghouses

Description

Task 6.1: Set up of lighthouses (Start: 1 End: 24)
Leader: SINT Main Contributors:, NTUA, PL owners, FORTH, SOLI, TECH, BEIA, YCCM
This task is focused on all preparatory activities related to the Living Labs – Lighthouses and Demo Site which will host the developed products, in order to validate their performance and to enable a continuous data flow for model's evaluation and calibration. The relevant partners will immediately start the analysis of the requirements, regulatory aspects for the products and their use for the effective preparation of the Lighthouses. The guidelines and protocols identified for each material/element will be studied in detail in order to organise informative seminars for the technicians who will perform the installation, ensuring accurate and optimum positioning and safe working conditions. End users will collaborate with core partners to specify workflows of activities to be run, as well as provide requirements and define all evaluation techniques and cross sectorial services that will be accessed. On this direction the existing infrastructure will be commissioned for adapting new sensors and monitoring devices, revising also accordingly the data logging software. Living-Labs will develop gateways for providing under a uniform format the data needed for continuous evaluation of each technological solution and models calibration. PLs and Living Labs will have a continuous collaboration for ensuring accurate follow-up of the time plan scheduled. This task will also aim at monitoring the relevant parameters according to the specifications and requirements. The target is to analyse, schedule the monitoring scenarios spanning across the entire season (heating/cooling season, intermediate season) or selected representative periods implementing monitoring objectives, control strategies, methodologies for the analysis of monitored data and compile them to exploitable results.
Task 6.2: Demonstration activities of Explot4InnoMat products (Start: 25 End: 48)
Leader: UNIS Main Contributors: all partners
According to the requirements identified, the products manufactured to the PLs and the end users will be delivered and installed. Following task 9.1 technical seminars, the installation procedure will be performed following the exact procedure identified in the previous tasks for promoting the effectiveness in relation to the fastness, the workability and the ratio of the wastes produced. The task leader, with the support of the technology providers, will conduct monitoring of the installed components for a period of two years. Monitoring and assessment will target aspects related to products effectiveness through online measuring, temperature variations in relation to environmental conditions, heat flow, humidity, water tightness and condensation. Possible material degradation which can affect their main properties like thermal conductivity and durability, which may result thermal bridges will occasionally be monitored through infrared sensors. Explot4InnoMat solutions will be evaluated following IPMVP protocol and will include scenarios for evaluating independently components, models, prediction algorithms, software solutions and the overall solution. For each scenario monitored parameters, calibration procedure and KPIs will be fully identified at the earlier stage. The know how obtained from the Light houses will create a concrete base for fast, efficient application, under real market conditions in (Agioi Anargiroi) demonstration site.
Task 6.3: Development of the Explot4InnoMat products database (Start: 25 End: 48)
Leader: NTUA Main Contributors: UNAV, all partners
During the implementation of Task 9.2, the developed prototypes will be tested in the 3 living labs as well as in the public demo building. The outcome of this long-term validation will include data regarding the products and building performance, technical, environmental and human impact on the building and the components as well as feedback and insights of potential consumers of the novel materials. Part of these data will feed the developed models and simulations in order to further improve their accuracy. All data derived from measurements, models, simulations, characterization, LCA/LCC analysis, recyclability assessment and feedback from inhabitants, will create the profile of developed solutions. Thus, the target of this task is to organize a data base in which comprehensive profiles of the

materials or components developed in the Explot4InnoMat OITB will be saved and will be open (under specific terms) for further exploitation in the frames of the open innovation environment of the ecosystem.

Work package WP7 – Impact maximisation

Work Package Number	WP7	**Lead Beneficiary**	1. UNIS
Work Package Name	Impact maximisation		
Start Month	1	**End Month**	48

Objectives

The aim of this WP is to disseminate and communicate appropriate information of the project and more specifically of the OITB raising awareness about Exploit4ZBand its services, and thus to fully exploit its results .
WP7 has the following specific objectives:
1. To disseminate the project results to targeted workshops & conferences and sectors
2. To design and implement a powerful communication campaign for engagement of stakeholders and clients as well as to the wider public.
3. To develop a proper plan for the management and protection of the foreground knowledge generated.
4. To further develop and evolve the initial business plan of the OITB and its penetration strategy among SMEs, Start-ups and industry of the relevant sectors.
5. To monitor the market situation and assess the business opportunities.
6. To define a commercialisation model and an exploitation route of the OITB
7. To launch an open call and select 6-8 new test cases

Description

Task 7.1: Dissemination & Communication activities (Start: 1 End: 48)
Leader: UNIS Main Contributors: all partners
The sustainability of the Ecosystem is based on the raise of awareness of the European industry for added value services in the sector, offered by accessing the OITB. The preliminary dissemination and communication plan that is presented in Section 2.2 will be updated frequently during the project based on the progress of the project and specific situations. Activities will vary, from basic communication (project web platform, brochures, leaflets, social media) to stakeholder engagement (idea pitching in events, publications, technical conferences, trade fairs). An international conference on advance materials building envelope will be organized in which, a showcase of Explot4InnoMat cases, workshops, training courses be presented. Attendance in conferences and further enrichment of previous activities will also take place. The conference, showcase, and webinars organization will continue on a regular base (annually or biannually) after the project completion, in addition to maintaining the autonomous operation of the SEP portal, the regular attendance to conferences and events, and the enrichment of social media profiles. The dissemination and communication activities will build and solidify a business-driven and customer-oriented brand identity for Explot4InnoMat OITB. The exploitation of results will be achieved through inbound (e.g. blogs, social media, etc.) and outbound (press releases, direct email, etc.) marketing activities targeting relevant industrial sectors and professional associations. Furthermore, the web platform will serve as a main exploitation channel for the OITB, with dedicated pages for the presentation of the Exploi4ZEB (e.g. the 9 Pilot Line and the services), the success stories (test cases that have been successful validated at TRL 7) the lighthouses and the terms of access for new clients. Finally, a separate page will be dedicated for attracting investments.
Task 7.2: Exploitation activities (Start: 1 End: 48)
Leader: DIAD Main Contributors: all partners
This task will cover the exploitation activities of the OITB. The ultimate exploitable result of the Explot4InnoMat project is the OITB itself that will offer in fair conditions and open access services to its clients. Thus, this task has as a priority to work towards the exploitation of this result, setting the base to obtain a clientele, initially across Europe in the field of advance materials and components for the building envelope. The exploitation strategy will comprise different phases such determination of the value proposition, market analysis, risks, operation costs, pricing policy, expected revenues, marketing plan and strategic alliances. A study will be carried out of the market landscape and the opportunities for each service included in the OITB. This process will also include the competition analysis and the characterization of the configuration of the value creation (partners, value change, customer relationship, etc.) Based on the results of this exercise, economic analysis will be performed to evaluate the profitability of the revenue model, understand future cash flow and highlight the need for additional funding or external investments. The whole analysis will be summarized in

the business plan report of the OITB. Already the core partners of the Ecosystem have worked on an initial business plan, presented in section 2. This business plan will be further updated and will include the SEP as well as dedicated customized exploitation plans for all the core members of the OITB.

Task 7.3 Collaborative Ecosystem: Building the "Materials" Helix (L:CHX; C:All. M1-M48)

CHX, working closely with partners, will build and manage the existing Materials Helix on the CHX platform that will act as the backbone of the dissemination and exploitation strategies developed both during and after the project. Thanks to the Exploit4InnoMat project more than 150 new international, interdisciplinary and strategic members (key stakeholders and end-users, details in section 2.2) will join the helix by M36, including Exploit4InnoMat partners, members from relevant existing helixes (i.e., Nature Building, Raw Minerals, Manufacturing, Digital, Smart Cities, Circular Plastics, Mission Climate, Energy helixes, etc.), researchers, industry actors, ethic and policy makers, civil society (end users), investors, participants of other relevant projects (to promote and encourage clustering activities) and relevant actors across Europe/International. The following activities will take place: a) By M6 all project partners will be invited to join the helix, the helix description will be updated (with the project logo and information of the project) and the helix news will be communicated to the membership base); b) Organise 2 Helix events and launch/implementation of its strategy (M24, M48); c) Profile >150 members (at researcher and organisational level) in the CHX platform (M36); d) Deliver impact in line with the Helix strategy (including supporting dissemination events, and building collaboration through the platform). The Helix will be managed by a Helix Manager, will contain information about the project and the members, and a resources tool with links to the project website and social media sites (Facebook, Twitter and LinkedIn), and any dissemination and communication material (presentations, scientific publications, etc.) produced during the project. To ensure sustainability, CHX will commit to maintaining the helix for at least 2 years after the project ends.

Task 7.4: Training activities (Start: 1 End: 48)

Leader: WHITE Main Contributors: all partners

In order to promote the new knowledge, processes and solutions generated during the project implementation as well as to create new links with the interested audiences towards the building envelope market, a two orientations training approach will be carried out. This training will consist of internal and external activities.

1) Internal OITB Training will cover the needs of the OITB members. The core partners will build a communication process which vision will be surrounded by their future needs and requirements in developing training courses. Offering training opportunities to take traineeships at all levels will boost their competences. The training topics will be related to the operation of the OITB, its services and offers, open innovation management philosophy, technological developments, targeting in an effective market uptake, enhancement of the innovation capacity and business transformation and overall boosting the OITB competitiveness at a pan-European scale.

2) External Training. The target of these training activities here will be to reach cross sectorial audiences related to materials and solutions for the building envelope. Start-ups have different needs in terms of training, as they usually have sufficient skills, and they may require training on different topics, such as management, marketing and branding. At the same time, innovative services offered by Explot4InnoMat for sure will attract costumers but from the other side intellectual property rights concerns could be an obstacle to enter the ecosystem, thus effective training courses explaining and showcasing the benefits will be implemented into the external training courses. The training schedule will address coaching sessions which aim at knowledge transfer, while typical training courses on e.g. new technologies and materials for the building envelope will be delivered as part of a training program. At least six training events will be held (3 internal and 3 external) during the project.

Task 7.5: Data Management and IPR Protection Activities (Start: 1 End: 48)

Leader: ICP Main Contributors: all partners

Data Management Plan (DMP) [ICP]: Development of the DMP, which will contain information related to the types of data the project will generate and collect, the standards that will be used to represent the data during the project and how partners might exploit the data resulting from the project. The DMP will set robust management procedures that will protect personal data collected as a result of project activities, as well as client owned data, from unauthorised use or sharing, also supporting GDPR compliance. The DMP will be produced in month M06 and finalised in M34. INSP will act as the project's Data Protection Officer (DPO) who will assist in decision-making as the project is developed to identify potential risks, mitigation solutions and support compliance. All partners internal DPOs will closely collaborate with ORIGINS DPO. FAIR principles will be taken into account.

Innovation management and IPR activities [ICP]: Identify and manage innovation from the project, advising the Project Coordinator (PC) and partners on the scope and potential of the innovations arising and paying attention to discerning IPR at regular checkpoints. The work also includes identifying, assisting and overseeing the filing of at least 3 strategic patent filings. All IP produced will be monitored and the potential IPR attached to them (e.g., copyright, patent) will be also verified and cleared for the project's smooth execution. Further activities include: a) Design the IPR protection strategy to promote the sustainability of project results, building on existing assets, considering aspects defined in Section 2.2.2.

b) Establish IPR management procedures and associated activities. c) Conduct a search on the prior-art for potential inventions, trademarks and other IPRs when necessary and evaluate the costs for potential IP protection. d) Monitor the project from the IPR point of view so that confidentiality and all the partners' defined IPRs in the CA are fully respected.

Task 7.6 Clustering with other EU projects and initiatives (L: CHX; C:All. M1-M48)
A clustering plan will be included in the DEC plan (details in section 2.2). The following activities will be developed in order to maximise cross-project communication, to exchange knowledge and experiences and to facilitate synergies and strategic partnering with other European organisations: a) Identification of the projects linked and relevant to Exploit4InnoMat (i.e. sister projects and other EC-chaired or funded initiatives in the relevant domains, and Knowledge Networks set-up by European Commission services: EIT Manufacturing, EIT Digital, EIT Raw Materials, S3P-Industry, S3 Partnership on Sustainable Buildings, Cluster Eco Construction, EBC, etc.). b) First contact with the project directors. Coordinators will be invited to the Exploit4InnoMat Kick-off meeting. c) Development of a Joint Clustering Board and Plan (with the sister projects), including the following activities: Creation of a common identity (logo), mailing list, promotion of other projects in Exploit4InnoMat website, yearly meetings, invitation to Exploit4InnoMat events and to join the Materials helix, etc. (details in section 2.2). Partners will be encouraged to create consortium and develop new ideas for future funding opportunities.
Task 7.7: Launch of the open call (Start: 14 End: 36)
Leader: F6S (SEP) Main Contributors: all partners
The open call procedure will be concluded in three phases. During the first phase (14-22) the core members of the OITB will provide analytical reports on the resources spent until that time on the selected test cases and a projection until the end of the implementation of them. Thus, any surplus resources will be transferred on the resources already projected for this WP. Based on this, an open call for 6-8 new end users will be launched through the web platform of the OITB. The call will last for 4 months. During this period focused dissemination activities will be undertaken by the whole consortium in order to disseminate the call. The target is to receive at least 30 applications for access in the Ecosystem. The applications will consist of a 10 pages proposal where the applicant will have to provide specific data according to the guidelines of the open call. The open call will be launched at M18 and will have a deadline at M22.
During the second phase (M22-24), the applications will be evaluated by the OITB members based on the following two, equal weight, criteria:
(a) TRL of the proposed test cases. The OITB will set as entry level the TRL4-5. Concepts of lower TRL will be rejected as the upscaling process cannot be based on valid preliminary data,
(b) market and innovation potential of the new test cases based on a preliminary draft feasibility study as presented by the applicants
Finally, during the third phase (M25-36) the implementation of the new test cases will take place as described in the relevant tasks of WP 3 – 10 starting from month 25.

Associated with document Ref. Ares(2022)8204179 - 28/11/2022

Associated with document Ref. Ares(2022)8204179 - 28/11/2022

LIST OF DELIVERABLES

Deliverables

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	Report on Data Management and IPR activities (06)	WP1	10 - ICP	R — Document, report	C-UE/EU-C - EU Classified	6
D1.2	Report on Data Management and IPR activities (M36)	WP1	10 - ICP	R — Document, report	C-UE/EU-C - EU Classified	36
D1.3	Report on Data Management and IPR activities (48)	WP1	10 - ICP	R — Document, report	C-UE/EU-C - EU Classified	48
D2.1	Report on the designs, specifications and requirements for PL modifications	WP2	11 - STAM SRL	R — Document, report	C-UE/EU-C - EU Classified	6
D2.2	Report on the administrative procedures & legal documents of OITB	WP2	25 - F6S NIL	R — Document, report	C-UE/EU-C - EU Classified	12
D2.3	Delivery of 9 upgraded pilot lines	WP2	11 - STAM SRL	DEM — Demonstrator, pilot, prototype	C-UE/EU-C - EU Classified	12
D2.4	Report on the preparatory phase of end users	WP2	11 - STAM SRL	DEM — Demonstrator, pilot, prototype	C-UE/EU-C - EU Classified	12
D2.5	Report on operational procedures of OITB	WP2	25 - F6S NIL	R — Document, report	C-UE/EU-C - EU Classified	18
D2.6	Report on the fine-tuning actions based on the open call selected test cases (M36)	WP2	25 - F6S NIL	R — Document, report	C-UE/EU-C - EU Classified	36
D2.7	Report on the fine-tuning actions based on	WP2	5 - RISE	R — Document, report	C-UE/EU-C - EU Classified	48

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
	the open call selected test cases selected test cases (M48)					
D2.8	Report on Exploit4InnoMat contribution to standards and regulation	WP2	7 - USTUTT	R — Document, report	PU - Public	48
D3.1	Reports on the optimization of processes (24)	WP3	11 - STAM SRL	R — Document, report	C-UE/EU-C - EU Classified	24
D3.2	Reports on the optimization of processes (36)	WP3	11 - STAM SRL	R — Document, report	C-UE/EU-C - EU Classified	36
D3.3	Delivery of BIM objects files for Lighthouses and Demo Building	WP3	19 - UNIVPM	R — Document, report	C-UE/EU-C - EU Classified	24
D3.4	Report on modelling and simulation activities for the selected and Open Call test cases	WP3	12 - AMS	R — Document, report	C-UE/EU-C - EU Classified	36
D3.5	Delivery of fillers produced at VEF and report on the production	WP3	3 - NTUA	DEM — Demonstrator, pilot, prototype	C-UE/EU-C - EU Classified	18
D3.6	Delivery of multifunctional paints and report on the production	WP3	5 - RISE	DEM — Demonstrator, pilot, prototype	C-UE/EU-C - EU Classified	24
D3.7	Delivery of lightweight mortars and report on the production	WP3	3 - NTUA	DEM — Demonstrator, pilot, prototype	S-UE/EU-S - EU Classified	24
D3.8	Delivery of insulating bricks and report on the production	WP3	6 - CSIC	DEM — Demonstrator, pilot, prototype	C-UE/EU-C - EU Classified	24

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D3.9	Report on the upscaling of open call TC in Materials PLs	WP3	6 - CSIC	R — Document, report	C-UE/EU-C - EU Classified	36
D4.1	Delivery of PVT systems and report on the production	WP4	2 - SINTEF	DEM — Demonstrator, pilot, prototype	C-UE/EU-C - EU Classified	24
D4.2	Delivery of electrochromic windows and report on the production	WP4	4 - NTNU	DEM — Demonstrator, pilot, prototype	C-UE/EU-C - EU Classified	24
D4.3	Report on the upscaling of open call TC in Components PLs	WP4	4 - NTNU	R — Document, report	C-UE/EU-C - EU Classified	36
D4.4	Delivery of load bearing lightweight prefab elements for new construction and report on the production	WP4	7 - USTUTT	DEM — Demonstrator, pilot, prototype	C-UE/EU-C - EU Classified	24
D4.5	Delivery of prefab elements for refurbishment/external cladding and report on the production	WP4	7 - USTUTT	DEM — Demonstrator, pilot, prototype	C-UE/EU-C - EU Classified	24
D4.6	Delivery of 3D printed cladding elements and report on the production	WP4	3 - NTUA	DEM — Demonstrator, pilot, prototype	C-UE/EU-C - EU Classified	24
D4.7	Delivery of cement free construction materials and components and report on the production	WP4	17 - CETMA	R — Document, report	C-UE/EU-C - EU Classified	36
D5.1	Report on characterization profiles of selected test cases	WP5	19 - UNIVPM	R — Document, report	C-UE/EU-C - EU Classified	36

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D5.2	Report on LCA/LCC analysis of the selected test cases (M36)	WP5	17 - CETMA	R — Document, report	C-UE/EU-C - EU Classified	36
D5.3	Report on LCA/LCC analysis of the selected test cases (M48)	WP5	17 - CETMA	R — Document, report	C-UE/EU-C - EU Classified	48
D5.4	Report on the recyclability assessment of Exploit4InnoMat technologies	WP5	26 - FALU	R — Document, report	C-UE/EU-C - EU Classified	45
D6.1	Report on the preparation of the Lighthouses and Demo demonstration activities site (M24)	WP6	2 - SINTEF	R — Document, report	C-UE/EU-C - EU Classified	24
D6.2	Report on the preparation of the Lighthouses and Demo demonstration activities site (M48)	WP6	2 - SINTEF	R — Document, report	C-UE/EU-C - EU Classified	48
D6.3	Report on the Lighthouses validation activities	WP6	2 - SINTEF	R — Document, report	C-UE/EU-C - EU Classified	48
D6.4	Delivery of Exploit4InnoMat database	WP6	3 - NTUA	R — Document, report	C-UE/EU-C - EU Classified	48
D7.1	Dissemination and Communication activities (M12)	WP7	1 - UNIS	R — Document, report	PU - Public	12
D7.2	PEDR	WP7	16 - DIADIKASIA	R — Document, report	SEN - Sensitive	6
D7.3	Materials Helix development and clustering report (M12)	WP7	24 - CHX	R — Document, report	PU - Public	12

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D7.4	Delivery of open call campaign identity	WP7	25 - F6S NIL	R — Document, report	PU - Public	18
D7.5	Dissemination and Communication activities (M24)	WP7	1 - UNIS	R — Document, report	PU - Public	24
D7.6	Dissemination and Communication activities (M36)	WP7	1 - UNIS	R — Document, report	PU - Public	36
D7.7	PEDR (M18)	WP7	16 - DIADIKASIA	R — Document, report	SEN - Sensitive	18
D7.8	PEDR (M36)	WP7	16 - DIADIKASIA	R — Document, report	SEN - Sensitive	36
D7.9	PEDR (M48)	WP7	16 - DIADIKASIA	R — Document, report	SEN - Sensitive	48
D7.10	Materials Helix development and clustering report (M24)	WP7	24 - CHX	R — Document, report	PU - Public	24
D7.11	Materials Helix development and clustering report (M48)	WP7	24 - CHX	R — Document, report	PU - Public	48
D7.12	Dissemination and Communication activities (M48)	WP7	1 - UNIS	R — Document, report	PU - Public	48

Deliverable D1.1 – Report on Data Management and IPR activities (06)

Deliverable Number	D1.1	**Lead Beneficiary**	10. ICP
Deliverable Name	Report on Data Management and IPR activities (06)		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	6	**Work Package No**	WP1

Description
The DMP will set robust management procedures that will protect personal data collected as a result of project activities, as well as client owned data, from unauthorised use or sharing, also supporting GDPR compliance. The DMP will be produced in month M06 and finalised in M34.

Deliverable D1.2 – Report on Data Management and IPR activities (M36)

Deliverable Number	D1.2	**Lead Beneficiary**	10. ICP
Deliverable Name	Report on Data Management and IPR activities (M36)		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	36	**Work Package No**	WP1

Description
The DMP will set robust management procedures that will protect personal data collected as a result of project activities, as well as client owned data, from unauthorised use or sharing, also supporting GDPR compliance. The DMP will be produced in month M06 and finalised in M34. Identify and manage innovation from the project, advising the Project Coordinator (PC) and partners on the scope and potential of the innovations arising and paying attention to discerning IPR at regular checkpoints. The work also includes identifying, assisting and overseeing the filing of at least 3 strategic patent filings

Deliverable D1.3 – Report on Data Management and IPR activities (48)

Deliverable Number	D1.3	**Lead Beneficiary**	10. ICP
Deliverable Name	Report on Data Management and IPR activities (48)		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	48	**Work Package No**	WP1

Description
The DMP will set robust management procedures that will protect personal data collected as a result of project activities, as well as client owned data, from unauthorised use or sharing, also supporting GDPR compliance. The DMP will be produced in month M06 and finalised in M34. Identify and manage innovation from the project, advising the Project Coordinator (PC) and partners on the scope and potential of the innovations arising and paying attention to discerning IPR at regular checkpoints. The work also includes identifying, assisting and overseeing the filing of at least 3 strategic patent filings

Deliverable D2.1 – Report on the designs, specifications and requirements for PL modifications

Deliverable Number	D2.1	**Lead Beneficiary**	11. STAM SRL
Deliverable Name	Report on the designs, specifications and requirements for PL modifications		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	6	**Work Package No**	WP2

Description
Perform necessary modifications and upgrade of the Pilot lines will take place. Efforts will be dedicated on current evaluation of unit operations, detailed engineering of each pilot upgrade, process control, process and plant design, instrumentation, automation and industrial measuring technology and finally all the relevant safety and health measures which should be taken into account.

Deliverable D2.2 – Report on the administrative procedures & legal documents of OITB

Deliverable Number	D2.2	**Lead Beneficiary**	25. F6S NIL
Deliverable Name	Report on the administrative procedures & legal documents of OITB		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	12	**Work Package No**	WP2

Description
During this task the administrative structure of the Ecosystem and all legal aspects will be concluded. The target is to work towards the set up of the Single Entry Point of the OITB that will undertake the whole administrative and secretarial tasks. Important aspect will be the founding of a new legal entity (Joint Venture) that will act as the SEP. During this task the description of its mission statement will be made along with its objectives

Deliverable D2.3 – Delivery of 9 upgraded pilot lines

Deliverable Number	D2.3	**Lead Beneficiary**	11. STAM SRL
Deliverable Name	Delivery of 9 upgraded pilot lines		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	12	**Work Package No**	WP2

Description
During this task, necessary modifications and upgrade of the Pilot lines will take place. Efforts will be dedicated on current evaluation of unit operations, detailed engineering of each pilot upgrade, process control, process and plant design, instrumentation, automation and industrial measuring technology and finally all the relevant safety and health measures which should be taken into account

Deliverable D2.4 – Report on the preparatory phase of end users

Deliverable Number	D2.4	**Lead Beneficiary**	11. STAM SRL
Deliverable Name	Report on the preparatory phase of end users		

Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	12	**Work Package No**	WP2

Description
This task involves all preparatory activities related to the test cases that are already scheduled to access the OITB and work towards its validation.

Deliverable D2.5 – Report on operational procedures of OITB

Deliverable Number	D2.5	**Lead Beneficiary**	25. F6S NIL
Deliverable Name	Report on operational procedures of OITB		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	18	**Work Package No**	WP2

Description
During this task and within the first period of the project the main administrative and operational procedures will be defined and structured. These procedures will be set to action during the initiation of the implementation of the test cases that is the validation phase.

Deliverable D2.6 – Report on the fine-tuning actions based on the open call selected test cases (M36)

Deliverable Number	D2.6	**Lead Beneficiary**	25. F6S NIL
Deliverable Name	Report on the fine-tuning actions based on the open call selected test cases (M36)		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	36	**Work Package No**	WP2

Description
This deliverable involves the validation phase consists of selected test cases and new test cases will have access in the OITB through the open call. These test cases will provide the necessary feedback on the ecosystem and thus, possible issues concerning these procedures (e.g. deficient communication between the partners, delays in responses, misleading legal documents, contracts, NDA etc) will be improved and fixed. The target is that by the end of the project, the OITB will have identified all deficient procedures and fix/replace them with more efficient ones.

Deliverable D2.7 – Report on the fine-tuning actions based on the open call selected test cases selected test cases (M48)

Deliverable Number	D2.7	**Lead Beneficiary**	5. RISE
Deliverable Name	Report on the fine-tuning actions based on the open call selected test cases selected test cases (M48)		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	48	**Work Package No**	WP2

Description
This deliverable involves the validation phase consists of selected test cases and new test cases will have access in the OITB through the open call. These test cases will provide the necessary feedback on the ecosystem and thus, possible issues concerning these procedures (e.g. deficient communication between the partners, delays in responses, misleading legal documents, contracts, NDA etc) will be improved and fixed. The target is that by the end of the project, the OITB will have identified all deficient procedures and fix/replace them with more efficient ones.

Deliverable D2.8 – Report on Exploit4InnoMat contribution to standards and regulation

Deliverable Number	D2.8	**Lead Beneficiary**	7. USTUTT
Deliverable Name	Report on Exploit4InnoMat contribution to standards and regulation		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP2

Description
The deliverable will gather data on relevant standards (e.g. ISO 21105 for building envelope assembly, ISO 15712 for acoustics, ISO 16932 for glass in buildings, ISO 6781 for thermal insulation, and more) in order to evaluation the performance of the developed materials and components for the building envelope.

Deliverable D3.1 – Reports on the optimization of processes (24)

Deliverable Number	D3.1	**Lead Beneficiary**	11. STAM SRL
Deliverable Name	Reports on the optimization of processes (24)		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	24	**Work Package No**	WP3

Description
In this deliverable based on the collected information from each PL, processing models will be developed and calibrated through the historical data. Each model will provide a basis for evaluating current production capabilities and limitations, their performances and for identifying a methodology for upscaling the production.

Deliverable D3.2 – Reports on the optimization of processes (36)

Deliverable Number	D3.2	**Lead Beneficiary**	11. STAM SRL
Deliverable Name	Reports on the optimization of processes (36)		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	36	**Work Package No**	WP3

Description
In this deliverable based on the collected information from each PL, processing models will be developed and calibrated through the historical data. Each model will provide a basis for evaluating current production capabilities and limitations, their performances and for identifying a methodology for upscaling the production.

Deliverable D3.3 – Delivery of BIM objects files for Lighthouses and Demo Building

Deliverable Number	D3.3	**Lead Beneficiary**	19. UNIVPM
Deliverable Name	Delivery of BIM objects files for Lighthouses and Demo Building		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	24	**Work Package No**	WP3

Description
In this deliverable the main features of building components(geometric, dimensional, material, performance, lifecycle, etc. will represent the informative attachment of each component. This object-oriented multilayer informative character actually represents one of the main advantages of BIM approach that both the design and renovation phase as well as the control and manage one

Deliverable D3.4 – Report on modelling and simulation activities for the selected and Open Call test cases

Deliverable Number	D3.4	**Lead Beneficiary**	12. AMS
Deliverable Name	Report on modelling and simulation activities for the selected and Open Call test cases		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	36	**Work Package No**	WP3

Description
In this Task, thePilot Lines will be modelled in order to simulate the operations of each process stage. This approach will allow to create detailed models and to make analysis on material workflows based on the available controlled systems. The first step will include a detailed analysis of each PL process and process control evaluation, focused also on the maintenance or control operation problems for tracing where to intervent and to get production more efficient

Deliverable D3.5 – Delivery of fillers produced at VEF and report on the production

Deliverable Number	D3.5	**Lead Beneficiary**	3. NTUA
Deliverable Name	Delivery of fillers produced at VEF and report on the production		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	18	**Work Package No**	WP3

Description
In the second subtask, the development and optimization of the mortars will be performed. It will be focused on granulometry, density, thermal conductivity, compressive/flexural strength and water up-take, whereas it will be based on the investigation of a number of parameters: formulation and raw materials type (lightweight aggregate type and multifunctional agents), variable mixture proportion/aqueous solution solids content, curing conditions, granulometry (i.e. particle size distribution), as well as dispersion solutions and water percentages to obtain the desired consistency and rheological properties

Deliverable D3.6 – Delivery of multifunctional paints and report on the production

Deliverable Number	D3.6	**Lead Beneficiary**	5. RISE
Deliverable Name	Delivery of multifunctional paints and report on the production		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	24	**Work Package No**	WP3

Description
Initially the specifications of multifunctional coating pilot line including nano-additives, polymers and coatings will be set. Based on existing standards, policies and guidelines requirements for nano-additives, the new coating systems will be established. At first stage the development of surface engineered nano-additives will be evaluated

Deliverable D3.7 – Delivery of lightweight mortars and report on the production

Deliverable Number	D3.7	**Lead Beneficiary**	3. NTUA
Deliverable Name	Delivery of lightweight mortars and report on the production		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	S-UE/EU-S - EU Classified
Due Date (month)	24	**Work Package No**	WP3

Description
Main objective of this deliverable is the development of advanced and high-quality lightweight mortars and cementbased rendering premix products with improved multiple functionalities under PL3 following user requirements and masonry applications. The research will be distinguished into two (2) subtasks

Deliverable D3.8 – Delivery of insulating bricks and report on the production

Deliverable Number	D3.8	**Lead Beneficiary**	6. CSIC
Deliverable Name	Delivery of insulating bricks and report on the production		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	24	**Work Package No**	WP3

Description
This deliverable involves all activities related to the development of new bricks with improved functionalities, especially in respect with insulation, acoustic and mechanical properties by applying a cost effective and low energy consuming process. The innovative ceramics will be developed by embedding CDW, PCMs and inorganic loose fill materials on their main body and their internal geometries.

Deliverable D3.9 – Report on the upscaling of open call TC in Materials PLs

Deliverable Number	D3.9	**Lead Beneficiary**	6. CSIC
Deliverable Name	Report on the upscaling of open call TC in Materials PLs		

Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	36	**Work Package No**	WP3

Description
Under this task, test cases resulting from the open call will be introduced into the OITB to produce, test and validate innovative solution in the field of multifunctional advance materials for the building envelope. It is expected that at least one new test case will be introduced in each one of the pilot lines of this group (a total of 3 new test cases)

Deliverable D4.1 – Delivery of PVT systems and report on the production

Deliverable Number	D4.1	**Lead Beneficiary**	2. SINTEF
Deliverable Name	Delivery of PVT systems and report on the production		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	24	**Work Package No**	WP4

Description
For the cost-efficient production of PV modules necessary in the project, SOLI based on modularity of the product line will develop and deliver glass/glass monocrystalline PVs optimised for both and electric and solar energy harvesting. SINTEF in close collaboration with SOLI will identify the optimum cell and junction boxes positioning for commissioning processing equipment based on the final design. Selection of appropriate materials and methods to facilitate the process of dismantling, recycling and reusing will follow.

Deliverable D4.2 – Delivery of electrochromic windows and report on the production

Deliverable Number	D4.2	**Lead Beneficiary**	4. NTNU
Deliverable Name	Delivery of electrochromic windows and report on the production		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	24	**Work Package No**	WP4

Description
The main objective of this deliverable is the development of nanostructured electrochromic (EC) materials enabled smart window components, following an innovative method both for production and assembly For this reason, EC nanomaterials will be synthesized and produced following a combination of variable chemical composition, crystal structure, size, dimensionality at nanometer-scale and consequently different properties in order to result in high performance EC smart windows.

Deliverable D4.3 – Report on the upscaling of open call TC in Components PLs

Deliverable Number	D4.3	**Lead Beneficiary**	4. NTNU
Deliverable Name	Report on the upscaling of open call TC in Components PLs		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	36	**Work Package No**	WP4

Description
Load-bearing lightweight cement based panels will be develoled using CDWs and a steel fiber reinforcement process. All the materials (both aggregates and binding systems) will be identified so as to ensure the optimum compatibility between the ingredients and strong cohesion and coherence properties leading to improved mechanical properties.

Deliverable D4.4 – Delivery of load bearing lightweight prefab elements for new construction and report on the production

Deliverable Number	D4.4	**Lead Beneficiary**	7. USTUTT
Deliverable Name	Delivery of load bearing lightweight prefab elements for new construction and report on the production		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	24	**Work Package No**	WP4

Description
Load-bearing lightweight cement based panels will be develoled using CDWs and a steel fiber reinforcement process. All the materials (both aggregates and binding systems) will be identified so as to ensure the optimum compatibility between the ingredients and strong cohesion and coherence properties leading to improved mechanical properties.

Deliverable D4.5 – Delivery of prefab elements for refurbishment/external cladding and report on the production

Deliverable Number	D4.5	**Lead Beneficiary**	7. USTUTT
Deliverable Name	Delivery of prefab elements for refurbishment/external cladding and report on the production		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	24	**Work Package No**	WP4

Description
Aim of this deliverable is to develop multifunctional cladding elements embedding complex close cellular structures filled with additives, able to provide enhanced thermal properties.

Deliverable D4.6 – Delivery of 3D printed cladding elements and report on the production

Deliverable Number	D4.6	**Lead Beneficiary**	3. NTUA
Deliverable Name	Delivery of 3D printed cladding elements and report on the production		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	24	**Work Package No**	WP4

Description
Aim of this deliverable is to develop multifunctional cladding elements embedding complex close cellular structures filled with additives, able to provide enhanced thermal properties.

Deliverable D4.7 – Delivery of cement free construction materials and components and report on the production

Deliverable Number	D4.7	**Lead Beneficiary**	17. CETMA
Deliverable Name	Delivery of cement free construction materials and components and report on the production		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	36	**Work Package No**	WP4

Description
The aim of this deliverable is the development of various formulations of inorganic precursor polymers, which upon suitable processing will to the direct synthesis of formed building components/elements. Therefore, the selection of the optimum raw materials, the identification of the proper process conditions as well as the identification of the most promising polymerization synthesis are most important parameters for the formation of inorganic polymers with the optimum properties

Deliverable D5.1 – Report on characterization profiles of selected test cases

Deliverable Number	D5.1	**Lead Beneficiary**	19. UNIVPM
Deliverable Name	Report on characterization profiles of selected test cases		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	36	**Work Package No**	WP5

Description
This deliverable involves all activities related to the structural and morphological characterization of material and structures developed. A network of partners will collaborate in order to result in fast characterization workflows, for the determination of mechanical and morphological parameters (material tensile, bending, compression, strength, durability and appearance) but also load capacity, resistance to wind or snow load, deformation, impact loads and earthquake resistance.

Deliverable D5.2 – Report on LCA/LCC analysis of the selected test cases (M36)

Deliverable Number	D5.2	**Lead Beneficiary**	17. CETMA
Deliverable Name	Report on LCA/LCC analysis of the selected test cases (M36)		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	36	**Work Package No**	WP5

Description
The aim of this deliverable is the sustainability assessment covers life cycle assessment (LCA) studies, where the methodology identifies the environmental impacts of a product, production system or a service throughout its life cycle. This task consists on the definition of goal and scope of the LCA to be performed. In defining the scope of the LCA, focus will be given to the identification and characterizations of selected components and materials, taking into account their functions, properties and compositions.

Deliverable D5.3 – Report on LCA/LCC analysis of the selected test cases (M48)

Deliverable Number	D5.3	**Lead Beneficiary**	17. CETMA
Deliverable Name	Report on LCA/LCC analysis of the selected test cases (M48)		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	48	**Work Package No**	WP5

Description
The aim of this deliverable is the sustainability assessment covers life cycle assessment (LCA) studies, where the methodology identifies the environmental impacts of a product, production system or a service throughout its life cycle. This task consists on the definition of goal and scope of the LCA to be performed. In defining the scope of the LCA, focus will be given to the identification and characterizations of selected components and materials, taking into account their functions, properties and compositions.

Deliverable D5.4 – Report on the recyclability assessment of Exploit4InnoMat technologies

Deliverable Number	D5.4	**Lead Beneficiary**	26. FALU
Deliverable Name	Report on the recyclability assessment of Exploit4InnoMat technologies		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	45	**Work Package No**	WP5

Description
Sustainability assessment covers life cycle assessment (LCA) studies, where the methodology identifies the environmental impacts of a product, production system or a service throughout its life cycle. This task consists on the definition of goal and scope of the LCA to be performed. In defining the scope of the LCA, focus will be given to the identification and characterizations of selected components and materials, taking into account their functions, properties and compositions

Deliverable D6.1 – Report on the preparation of the Lighthouses and Demo demonstration activities site (M24)

Deliverable Number	D6.1	**Lead Beneficiary**	2. SINTEF
Deliverable Name	Report on the preparation of the Lighthouses and Demo demonstration activities site (M24)		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	24	**Work Package No**	WP6

Description
In this deliverable the developed prototypes will be tested in the 3 living labs as well as in the public demo building. The outcome of this long-term validation will include data regarding the products and building performance, technical, environmental and human impact on the building and the components as well as feedback and insights of potential consumers of the novel materials.

Deliverable D6.2 – Report on the preparation of the Lighthouses and Demo demonstration activities site (M48)

Deliverable Number	D6.2	**Lead Beneficiary**	2. SINTEF
Deliverable Name	Report on the preparation of the Lighthouses and Demo demonstration activities site (M48)		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	48	**Work Package No**	WP6

Description
In this deliverable the developed prototypes will be tested in the 3 living labs as well as in the public demo building. The outcome of this long-term validation will include data regarding the products and building performance, technical, environmental and human impact on the building and the components as well as feedback and insights of potential consumers of the novel materials.

Deliverable D6.3 – Report on the Lighthouses validation activities

Deliverable Number	D6.3	**Lead Beneficiary**	2. SINTEF
Deliverable Name	Report on the Lighthouses validation activities		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	48	**Work Package No**	WP6

Description
In this deliverable the installation procedure will be performed following the exact procedure identified in the previous tasks for promoting the effectiveness in relation to the fastness, the workability and the ratio of the wastes produced

Deliverable D6.4 – Delivery of Exploit4InnoMat database

Deliverable Number	D6.4	**Lead Beneficiary**	3. NTUA
Deliverable Name	Delivery of Exploit4InnoMat database		
Type	R — Document, report	**Dissemination Level**	C-UE/EU-C - EU Classified
Due Date (month)	48	**Work Package No**	WP6

Description
The developed prototypes will be tested in the 3 living labs as well as in the public demo building. The outcome of this long-term validation will include data regarding the products and building performance, technical, environmental and human impact on the building and the components as well as feedback and insights of potential consumers of the novel materials

Deliverable D7.1 – Dissemination and Communication activities (M12)

Deliverable Number	D7.1	**Lead Beneficiary**	1. UNIS
Deliverable Name	Dissemination and Communication activities (M12)		
Type	R — Document, report	**Dissemination Level**	PU - Public

Due Date (month)	12	**Work Package No**	WP7

Description
The sustainability of the Ecosystem is based on the raise of awareness of the European industry for added value services in the sector, offered by accessing the OITB.

Deliverable D7.2 – PEDR

Deliverable Number	D7.2	**Lead Beneficiary**	16. DIADIKASIA
Deliverable Name	PEDR		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP7

Description
This deliverable will cover the exploitation activities of the OITB. The ultimate exploitable result of the Explot4InnoMat project is the OITB itself that will offer in fair conditions and open access services to its clients. Thus, this task has as a priority to work towards the exploitation of this result, setting the base to obtain a clientele, initially across Europe in the field of advance materials and components for the building envelope.

Deliverable D7.3 – Materials Helix development and clustering report (M12)

Deliverable Number	D7.3	**Lead Beneficiary**	24. CHX
Deliverable Name	Materials Helix development and clustering report (M12)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	12	**Work Package No**	WP7

Description
In this deliverable it will be build and managed the existing Materials Helix on the CHX platform that will act as the backbone of the dissemination and exploitation strategies developed both during and after the project

Deliverable D7.4 – Delivery of open call campaign identity

Deliverable Number	D7.4	**Lead Beneficiary**	25. F6S NIL
Deliverable Name	Delivery of open call campaign identity		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP7

Description
In this deliverable the open call procedure will be concluded in three phases. During the first phase (14-22) the core members of the OITB will provide analytical reports on the resources spent until that time on the selected test cases and a projection until the end of the implementation of them.

Deliverable D7.5 – Dissemination and Communication activities (M24)

Deliverable Number	D7.5	**Lead Beneficiary**	1. UNIS
Deliverable Name	Dissemination and Communication activities (M24)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP7

Description
The sustainability of the Ecosystem is based on the raise of awareness of the European industry for added value services in the sector, offered by accessing the OITB.

Deliverable D7.6 – Dissemination and Communication activities (M36)

Deliverable Number	D7.6	**Lead Beneficiary**	1. UNIS
Deliverable Name	Dissemination and Communication activities (M36)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP7

Description
The sustainability of the Ecosystem is based on the raise of awareness of the European industry for added value services in the sector, offered by accessing the OITB.

Deliverable D7.7 – PEDR (M18)

Deliverable Number	D7.7	**Lead Beneficiary**	16. DIADIKASIA
Deliverable Name	PEDR (M18)		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP7

Description
This deliverable will cover the exploitation activities of the OITB. The ultimate exploitable result of the Explot4InnoMat project is the OITB itself that will offer in fair conditions and open access services to its clients. Thus, this task has as a priority to work towards the exploitation of this result, setting the base to obtain a clientele, initially across Europe in the field of advance materials and components for the building envelope.

Deliverable D7.8 – PEDR (M36)

Deliverable Number	D7.8	**Lead Beneficiary**	16. DIADIKASIA
Deliverable Name	PEDR (M36)		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP7

Description

This deliverable will cover the exploitation activities of the OITB. The ultimate exploitable result of the Explot4InnoMat project is the OITB itself that will offer in fair conditions and open access services to its clients. Thus, this task has as a priority to work towards the exploitation of this result, setting the base to obtain a clientele, initially across Europe in the field of advance materials and components for the building envelope.

Deliverable D7.9 – PEDR (M48)

Deliverable Number	D7.9	**Lead Beneficiary**	16. DIADIKASIA
Deliverable Name	PEDR (M48)		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP7

Description
This deliverable will cover the exploitation activities of the OITB. The ultimate exploitable result of the Explot4InnoMat project is the OITB itself that will offer in fair conditions and open access services to its clients. Thus, this task has as a priority to work towards the exploitation of this result, setting the base to obtain a clientele, initially across Europe in the field of advance materials and components for the building envelope.

Deliverable D7.10 – Materials Helix development and clustering report (M24)

Deliverable Number	D7.10	**Lead Beneficiary**	24. CHX
Deliverable Name	Materials Helix development and clustering report (M24)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP7

Description
In this deliverable it will be build and managed the existing Materials Helix on the CHX platform that will act as the backbone of the dissemination and exploitation strategies developed both during and after the project

Deliverable D7.11 – Materials Helix development and clustering report (M48)

Deliverable Number	D7.11	**Lead Beneficiary**	24. CHX
Deliverable Name	Materials Helix development and clustering report (M48)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP7

Description
In this deliverable it will be build and managed the existing Materials Helix on the CHX platform that will act as the backbone of the dissemination and exploitation strategies developed both during and after the project

Deliverable D7.12 – Dissemination and Communication activities (M48)

Deliverable Number	D7.12	**Lead Beneficiary**	1. UNIS

Deliverable Name	Dissemination and Communication activities (M48)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP7

Description
The sustainability of the Ecosystem is based on the raise of awareness of the European industry for added value services in the sector, offered by accessing the OITB.

LIST OF MILESTONES

Milestones *Grant Preparation (Milestones screen) — Enter the info.*					
Milestone No	**Milestone Name**	**Work Package No**	**Lead Beneficiary**	**Means of Verification**	**Due Date (month)**
1	Kick of meeting	WP1	1-UNIS	KoM minutes approved by partners	1
2	Stam, Iris & Polm conclude the visits to the PLs	WP2	11-STAM SRL	Final Designs and modifications approved by the PL owners	4
3	Formation of Advisory Board	WP1	1-UNIS	First meeting of AB members	6
4	Draft Statute of Joint Venture delivered	WP2	1-UNIS	Draft statute delivered to notary	8
5	Final tests of upgraded PLs	WP2	11-STAM SRL	Report on the running tests and yields	11
6	Delivery of first batch of perlite based aggregates	WP4	3-NTUA	Report on the characterization of batch	14
7	3D scanning of DAAK building	WP3	19-UNIVPM	Delivery of 3D scanned images including IR scanning	16
8	Launch of the open call	WP7	25-F6S NIL	Publication of the call in website and social media	18
9	First successful application of electrochromic layers in glazing	WP5	4-NTNU	Report on the characterization of the glass	19
10	First successful integration of PVTs to elements	WP5	2-SINTEF	Report on the assembly process	20
11	Fabrication of a 2x2 m load bearing prefab element by	WP6	7-USTUTT	Video of the spraying process and characterization profile	22
12	Delivery of prepared materials and components in the 3 lighthouses and Demo Building	WP6	2-SINTEF	Delivery notes for materials transfer	24
13	Installation of Materials in Demo Building	WP6	2-SINTEF	Delivery notes for materials transfer	25

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
14	Delivery of the workflows of the new test cases through the open call	WP5, WP3, WP6, WP4, WP7	7-USTUTT	Report on the workflows	26
15	First successful batches of the open call test cases products	WP5, WP6, WP4	3-NTUA	Characterization report on the materials	30
16	First report on the monitoring of installed materials	WP4	7-USTUTT	Delivery of Report and feeding of models	32
17	Delivery of new products from open call test cases to the 3 lighthouses	WP6, WP4	3-NTUA	Delivery notes for materials transfer	37
18	Event for investor groups, showcasing the concepts implemented in the Exploit4InnoMat OITB	WP2, WP7	25-F6S NIL	Minutes of event, participation list	42
19	Exploit4InnoMat OITB ready for clients	WP7	25-F6S NIL	First applications by clients	48
20	Second report on the monitoring of installed materials	WP4	4-NTNU	Delivery of Report and feeding of models	45

LIST OF CRITICAL RISKS

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
1	Management issues & Financial risks.	WP1	Management activities will not be limited to reporting but it will also include a close financial

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
			monitoring process so as to constantly assess financial progress and be able to identify early signs of concern.
2	Problems in respecting the planned activities and resource allocation	WP1	PSC will decide on possible re-allocation of resources for surpassing the possible delays, after the agreement of the PO
3	Delays or problems in PL site installation or commissioning	WP2	Strong commitment of experienced partners involved gives confidence that all potential issues during the upgrading process will be solved utilizing their knowhow on previous similar installations.
4	In line monitoring system present low accuracy	WP3	The accuracy of the in-line spectroscopic techniques might be low due to the increased concentration mixtures. By installing an in-line, by-pass system with automated dilution can solve the issue.
5	Predicting behavior of materials properties, is computationally and time intensive. P m; I m	WP3	In order to mitigate this risk a simulation methodology, will be developed investigating fidelity and accuracy of predictions versus the complexity of the model.
6	Failing to meet the desired insulating properties, based on designed synthesis and aggregates used (like VEF Hybrid perlite-based lightweight aggregates) P: l, I: h	WP4	Proper evaluation of the material performance in all the development stages. In case of deviations, PLs will consider alternative formulations in terms of loose bulk density, λ and compressive strength in order to select and produce the most suitable aggregate
7	Difficulties in measuring the required improvements in pilot lines and smart manufacturing. P: l, I: h	WP5, WP6, WP4	Experience in monitoring and knowledge of processes will lower the risk and will enable effective measuring systems through sound evaluations parameters
8	PVT panels do not supply enough power. P: l. I: h.	WP5	Means to mitigate the risk include precise design and energy prediction using BIM supported by real time operation monitoring and optimizations
9	Failing to achieve the targeted reduction of the U value with the solutions applied P: l, I h	WP5, WP6, WP4	In order to eliminate the risk in extreme cases the thickness of the materials and components can be increased adding the proper reinforcement solution or implementing an additional insulation panel.
10	Inconsistency of measurement results at various characterization facilities. P: m, I: h	WP7	Train characterization personnel at PLs and at specific labs to perform measurements in a uniform way. Document procedure, maintain equipment and perform round robin tests regularly. Discuss topic in characterization group.

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
11	LCA results are not positive vs. industry benchmark of the project demonstration. P m; I: m	WP7	All test cases have successfully reached TRL4-5 and initial LCA analysis has shown positive results. The LCA during Exploit4InnoMat will include a cost benefit analysis of the proposed technologies, which in turn will determine the "boundary conditions" for their successful introduction to the market.
12	Delays on installation licensing due to application of on commercial certified products	WP7	A well-structured installation strategy will be identified at earl stage. Lighthouses are excluded from official licensing procedures
13	Disagreements in the consensus for the statute of the Joint Venture P: l. I: h.	WP2	Core OITB partners have been committed for the foundation of the JV acting as a SEP and thus, the probability of this risk is minimum. Partners not involved in the JV will be covered by the Framework Agreement
14	Not enough interest from stakeholders on the Open Call P: l. I: m.	WP2, WP5, WP1, WP3, WP6, WP4, WP7	Promotion of the project as well as its outcomes and benefits of entering the OITB are essential to achieve the engagement of SMEs from different sectors. The benefits of accessing the OITB will be properly disseminated
15	Non-agreement in intellectual property rights related to exploitation P: m; I: h	WP7	IPR Management is core for this project taking into account the different practices of the PLs owners as well as the peculiarities of SMEs in this respect. A Consortium Agreement will be signed prior to the Project initiation arranging these issues. Also an Mid-Term Exploitation Plans will be presented, where the rights over the different KERs will be updated to reach exploitation agreements
16	Difficulties in adoption of new envelope solutions by the market P: m; I: h	WP7	Marketing strategies, enhancing the contact with industrial companies will be maximized. Open door meetings and a dedicated events with stakeholders will be organized for the promotion of new products.
17	Development of an efficient network with a relevant number of members and Associations	WP7	CHX will provide the already existing network that constitutes the Materials Helix and based on the methodology and expertise they are bringing to the project, they will mitigate this risk and will guarantee the creation of a Robust Network through their Ecosystem Management Platform.

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PROJECT REVIEWS

Project Reviews *Grant Preparation (Reviews screen) — Enter the info.*			
Review No	**Timing** (month)	**Location**	**Comments**
RV1	18	TBC	
RV2	36	TBC	
RV3	48	TBC	

HISTORY OF CHANGES

TABLE OF HISTORY OF CHANGES			
Version (date)	Section	Page	Explanation
1.1	**Table 3.1h**	32	Addition of Table 3.1h
	Ethics and Security	33	Addition of the comments from the PO
	ANNEX A - Participation of UK & Switzerland partners	34	Inclusion of the Associated Partners Cost
1.2	**GANTT Chart**	29	Inclusion of the updated GANTT Chart with the new deliverables and Milestone N.20
	ANNEX A - Participation of UK & Switzerland partners	37	Correction of the Swish National Funding Agency
	ANNEX A - Participation of UK & Switzerland partners	37	Correction of the max funding request for FALU
	***Table 3.1a* List of work packages**	29	Correction of the total number of PMs and effort
	Table 3.1a: List of work packages	29	Correction of the total number Deliverables
	Table 3.1g: 'Subcontracting costs' items	33	Addition of Table 3.1g for Partner SINTEF
	Table 3.1i Other costs categories' items (e.g. internally invoiced goods and services)	36	Addition of Table 3.1 I for Partner RISE
	Table 3.1h: 'Purchase costs' items	33-36	Explanation of the acronym LL for Partner 4/SINTEF (regional travels to LL) and the rest of the tables
	Table 3.1h: 'Purchase costs' items	33-36	Insertion of a line with the total of purchase costs in each beneficiary's table 3.1h
1.3	**Table 3.1h: 'Purchase costs' items**	33-36	Change of the term external experts with Partner Personnel
	Table 3.1h: 'Purchase costs' items	35	Change of the term external experts with Partner Personnel for partner B23 SOLITEK,
	Table 3.1h: 'Purchase costs' items	32	Removal fom table 3.1g the names of the subcontractors from SINTEF
	Table 3.1h: 'Purchase costs' items	36	Addition of Partner B21 UNISMART along with the respective explanation of the internally invoiced goods and services in the budget (€ 13 000)
	Table 3.1h: 'Purchase costs' items	32	Correction of the Travel and subsistence budget for partner 05. RISE

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1 Excellence

1.1 Objectives and ambition

1.1.1 Context and strategic objectives

Building sector is responsible for 40% of energy consumption and 36% of CO_2 emissions in the EU. The 2018 Energy Performance of Buildings Directive (EPBD) and the 2012 Energy Efficiency Directive (EED), the main EU's legislation regarding reduction of energy consumption of buildings, have set the target that all new buildings must be nearly Zero Energy Buidlings (nZEB) by 31st December 2020 [1]. Moreover, Art. 5 in the EED has set a 3% annual renovation target for public buildings owned and occupied by the central government from the beginning of 2014 onwards. Central government buildings must be renovated to meet at least the national minimum energy performance requirements set in application of Art. 4 of the EPBD. With the ambition to become the world's first climate-neutral continent by 2050, the European Commission presented on 11 December 2019 the European Green Deal. Among the proposed measures, the European Grean Deal includes the ambition "engaging in a 'renovation wave' of private and public buildings". Thus, as nZEB becomes the new standard, the role of building materials and embodied energy or related CO_2-emission is becoming more and more important. It is recognized that a focus on energy efficiency only, entails a clear risk of having buildings not necessarily performing very well with respect to other environmental criteria [2,3]. Also the EC has identified the importance of building materials when it states that: "*The construction and use of buildings in the EU account for about half of all our extracted materials and energy consumption and about a third of our water consumption. The sector also generates about one third of all waste and is associated with environmental pressures that arise at different stages of a building's life-cycle including the manufacturing of construction products, building construction, use, renovation and the management of building waste*" [4].

The **Exploit4InnoMat OITB** (or Ecosystem) aspires to provide a sustainable solution across Europe for the **design, upscaling and validation of new material concepts for the building envelope** that will facilitate in the achievement of the **EC target for nZEB**. The Exploit4InnoMat will provide services across the supply chain of materials, starting from the design and upscaling of them, and going to installation, validation and assessment of scenarios for the end of life. The core of the Exploit4InnoMat is the **9 Pilot Lines**, situated in 7 EU countries, able to process different type of advance and/ornano-enabled multifunctional materials, including sorting and processing of construction and demolition wastes (CDW) tobe used as secondary raw materials. The target is to develop sustainable materials and products that not only will **reduce the operational energy demand** (e.g. by providing better insulation properties, reducing thermal bridges, and harvesting or storage of energy) but will be also of **lower embodied energy** (e.g. due to the use of CDW or other secondary raw materials and optimize their production process), providing a **healthier and more comfortable indoor environment** (thermal comfort, cleaner air). In order to achieve this target a wide range of expertise has been brought to the OITB covering fields such as **materials characterization and modelling, monitoring and process control, environmental andassessment, regulatory and standardization, social acceptance and innovation management**.

The Exploit4InnoMat OITB will provide its services through a Single Entry Point (SEP), a **new legal entity, (joint venture) that will be founded at month 12 of the project.** The SEP will undertake the whole administrative effort for signing contracts on behalf of the OITB members with its clients. In this way, a one stop shop will be formed, facilitating the accessof the clients to the Pilot Lines and the other services of the OITB. The Exploit4InnoMat OITB targets in strengthening the cooperation between various stakeholders (e.g. technology and service providers, SME, Large enterprises) and develop new business models for the construction sector, based on Open Innovation Philosophy.

The Exploit4InnoMat project focuses on the development of a sustainable open innovation test bed (OITB) consisting of 6 Groups of services offered to SMEs or Industry. The first Group contains the 9 pilot plants distributed across Europe (Norway, Sweden, Germany, Spain, Greece, Italy and Belgium) dedicated to the development of materials and components for building's envelope. The other five groups will offer technical and non-technical services (e.g. materials characterization and modelling, validation of products in living hubs, regulatory and standardization consultancy and innovation management) supporting the Pilot Lines. **Figure 1.1** provides a schematic of the Exploit4InnoMat OITB

[1] https://ec.europa.eu/energy/en/topics/energy-efficiency/energy-performance-of-buildings/energy-performance-buildings-directive
[2] Herczeg et al; Resource efficiency in the building sector - Final report; Ecorys; Rotterdam, 23 May 2014
[3] Belussi L. et al. (2019), A review of performance of zero energy buildings and energy efficiency solutions, Journal of Building engineering, vol. 25, 100772, DOI: 10.1016/j.jobe.2019.100772
[4] EC, On Resource Efficiency Opportunities in the Building Sector, Communication from the Commission to the European Parliament, The Council, The European Economic and Social Committee and The Committee Of The Regions, COM(2014) 445 final, Brussels 2014.

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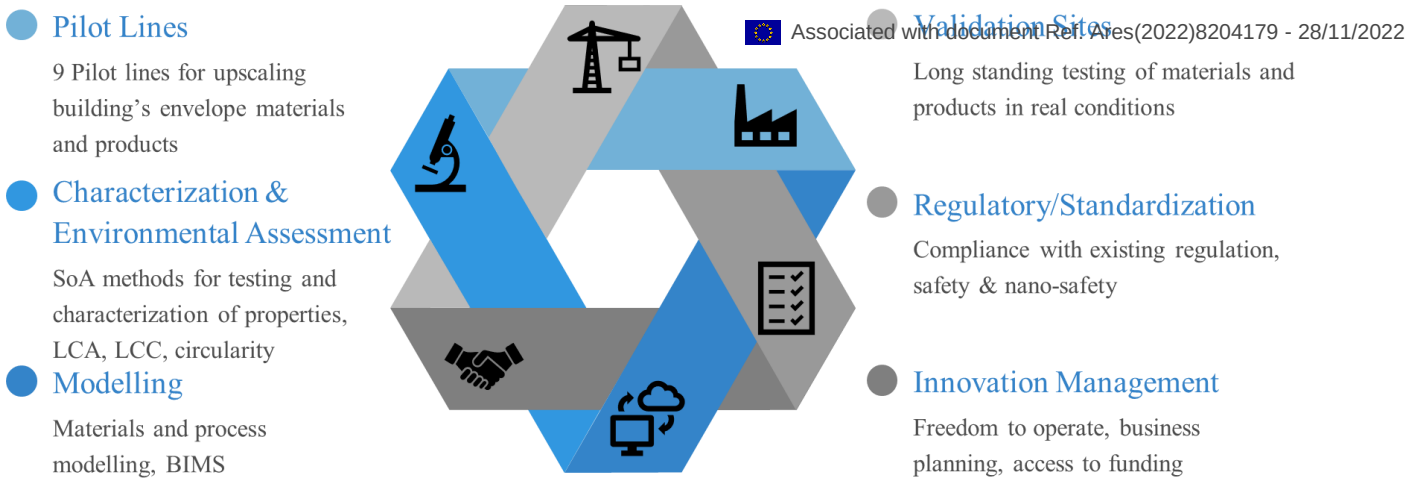

Figure 1.1: *Exploit4InnoMat OITB Groups of Services*

1.1.2 Implementations Objectives, Expected Results and Measurable Outputs

The objectives of Exploit4InnoMat are in accordance with the challenges raised by the topic NMBP-05-2020 and will be implemented through building an OITB comprising of a group of Pilot lines and fast prototyping units, together with 5 groups of services (Characterization & Environmental Assessment, Modelling, Validation Sites, Regulatory & Standardization, Innovation management). The target of the OITB is to assist innovative concepts to reach the market and thus, contribute to the targets set by the EC for achieving significant reduction of energy consumption in buildings, (nZEB). The consortium of Exploit4InnoMat project has split the objectives in the following categories.

Table 1.1: *Exploit4InnoMat project Objectives*

#	Objective	WP	KPIs
A. Objectives related to the set up and operation of the Exploit4InnoMat test bed			
1	Set up a sustainable OITB for efficient and fast upscaling of sustainable advance, nano-enabled materials and components for building's envelopes	2	- Organize 6 Group of services consisting of 22 core partners
2	To upgrade the existing PLs through integrating peripheral systems, process control, routines for the interior design of components, monitoring and quality control systems, targeting in the enhancement and optimization of the production processes.	2	- Upgrade of 9 Pilot Lines - Increase productivity and efficiency of PL by 20%
3	To establish a Single Entry Point (SEP) from which the clients will access the OITB.	2	- Foundation of 1 new joint venture legal entity
4	To determine the administrative structure and define operational procedures and roles in the OITB – avoiding any overlapping that can potentially lead to internal conflicts – securing a smooth operation of the OITB	2	- Establish 2 main agreements governing the relationships of SEP with core members and clients
5	To understand the needs of test cases brought by end users and translating them in upscaling, characterization/testing and validation service workflows.	2-8	- Develop 7 workflows for the end users - Develop 6-8 additional workflows during the open call
B. Objectives related to the development, validation and demonstration of materials and components, upscaled through the Exploit4InnoMat OITB			
1	To develop new advance and nano-enabled materials and components concepts exhibiting enhanced thermal and mechanical properties, energy capture and harvesting and high aesthetics, as well as recyclability	3-8	- Implement 7 selected test cases - 30% reduction in energy consumption across life cycle: production, installation operation end of life
2	Validate the proper operation of the ecosystem in regards to upscaling, testing and validating in operation environment of new materials concepts.	3-10	- 30% faster validation of new concepts compared to SoA validation methodologies
3	Launch an open call of interest for new test cases that will be able to access the ecosystem	3-5, 10, 12	- Include 6-8 additional test cases in the OITB
4	Mobilize a network of 3 lighthouses facilities (Living Labs) and one demo site (public building) in 3 different climate zones for long term evaluation of nZEB materials technologies, creating a link between validation, dissemination and showcasing actions of the materials developed.	9	- Installation of 800 m^2 of building envelope components and materials - Monitor the performance of installed solutions for a period of 2 years providing data for feeding predictive modelling and simulations
C. Objectives related to the Exploit4InnoMat sustainability			
	To establish clustering and networking channels with EC Initiatives (e.g. AMANAC, EPPN, EMMC-ABSL and other	11-12	- Organize 4 clustering events with stakeholders

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	OITB or projects), for cooperation and developing synergies towards a pan European integrated and sustainable open innovation ecosystem.		- Develop 1 open innovation platform for collaboration of stakeholders with OITB
	To collaborate with European standardization and regulatory agencies (e.g. EOTA).	8, 11	- Participate in 3 related events and submit **Exploit4InnoMat** propositions
	To undertake communication/dissemination activities for promoting the ecosystem in targeted audiences, advertising the "success case" stories that will arise from the test cases.	12	- Reach at least 500 potential clients - Show case 13-15 success stories during the project - Organize a world conference related to **Exploit4InnoMat** technologies
	To continuously update the initial business plan of the Exploit4ZEB ecosystem and of its clients	10, 12	- Develop ~15 business plans (1 for OITB 7 end users and 6-8 end users through open call)
	To liaise with potential investors (venture capitals and business angels) for seeking additional funding.	12	- Organize 3 events for investors - Attract a total of 50 investing groups

1.1.3 Ambition - Targeted innovations beyond the current state-of-the-art

The **Exploit4InnoMat** OITB is characterized by a series of important innovations related not only to the developed materials, but also to the wider impact in the European construction market.

Innovation 1: Added value of the Exploit4InnoMat OITB to European market - The Exploit4InnoMat will form European networks of competences along the entire value chain and match the needs of industry by providing users with easy access to pilot infrastructures and services. These networks will reach out to users across different regions of Europe, sharing best practices and experiences, identifying new innovative technologies that can benefit from this scheme, brokering between users and suppliers; leveraging further investment by creating linkages with regional/national initiatives and by stimulating business growth. Increased profitability will result from the collaboration between experts in different technology and services fields, while shortened time to market, especially for SMEs that have a lack of resources for R&D activities and thus they can reduce their financial asset commitment and therefore enhance their flexibility. Moreover, through transferring their technological know-how and resources, the ecosystem participants will be able to expand their knowledge bases and competences, enhancing innovation capability and learning. Finally, the OITB will assist in recognizing the next innovation actors: The following steps will be followed: i. identify the potential technology change and contact the relevant stakeholders, ii. take into consideration research insights, iii. Follow changes in customer demand, iv. Identify competitors. v. Consider policies and legal environment.

Innovation 2: Ecosystem advantages on materials development - Enhanced performance: Exploit4InnoMat will target to sustainable materials for achieving the nZEB target, promising to deliver the optimization of material behaviour and performance needed to improve significantly the insulation capacity, energy performance, durability, maintenance cost and embodied energy. **Fast screening**: Through the database that will be developed under Exploit4InnoMat, based on the holistic testing/characterization/validation of the materials developed in the OITB, fast screening and benchmarking comparison of new materials will be achieved. Predictive modelling of materials, elements and components within the Exploit4InnoMat project will aim to optimize products based on market requirements and to ensure their proper functionality. A network of experienced RTDs in the field of characterisation, the simulating tools and the lighthouse facilities will help to provide fast and accurate profile of the upscaled materials and components. Finally, designing of elegant solutions through a modular approach utilizing building blocks (structural, solar thermal and BIPV) will create a harmonized and aesthetically pleasing urban environment.

Innovation 3: Intelligent synergy of materials, components and Smart Envelope Systems - The core innovation of Exploit4InnoMat OITB is based on the synergy of materials design and development, their intelligent adaption on construction components and the advanced manufacturing Smart Envelope Systems which can comply with EU standards and norms as well as climatic conditions across EU. The application of a wide range of nanoenabled materials (insulating to thermal storing properties) will make available a customizable element design for each building architectural style and enhancing it with powerful renewable energy generation features, optimizing their efficiency by maintaining the optimum operational parameters, modernizing the envelope and increasing its economic and aesthetic value. Digital solutions for the analysis and planning of construction and renovation solution as well as adapted 3D printing technology to cope with the necessities of each application will be also provided.

Innovation 4: BIM application for generating construction and renovation schemes - The Exploit4InnoMat OITB will develop a data centralisation and sharing framework supported by advanced BIM analytics which will analyse and evaluate the optimal cost-effective solutions. BIM analysis, fast track modelling and simulation tools will provide a semi-automated tool to be used at construction and retrofitting design stage to optimize the retrofit solution based on building physics, operation providing a dedicated Decision Support System for evaluating the technologies to be applied from continuously updated data platform of Exploit4InnoMat technologies and solutions available on the market.

Innovation 5: Benefits for SMEs and Industry in the field of solutions for the building envelope

Exploit4InnoMat Ecosystem main innovation lays on the **creation of a safe and neutral environment for start-ups, SMEs and industry to access facilities and manufacturing services for small series production of building envelope materials and components and thus, project with accuracy the cost and performance of such solutions in the industrial scale**. The OITB in this sense will provide ultimately also an estimate of what could be the projection of the

manufacturing cost at full scale. Certainly, the potential clients of the OITB needs to know if their materials/products concepts work well – both in technical (e.g. properties and functions) as well as in non-technical (e.g. compliance with standards and safety regulations) aspects and what is the cost of the process in order to proceed with solid business plans.

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Under this sense, **Exploit4InnoMat** is defined as a network of experts that will support the maturation and transfer of excellent technology to industry, in order to apply innovative technological opportunities. This will boost their position within the European materials Innovation Radar spurring competing products, and shortening time-to-market, with the objective of advancing competitiveness at global scale. The ecosystem intends to become an important and strategic platform for cross-cutting dialogue, cross-discipline cooperation, **validating/promoting the formation of new supply chains** preparing thus, a solid market penetration strategy towards niche markets.

It should me mentioned that currently **there are no similar structures across Europe**. Of course, large enterprises have their own manufacturing test beds inhouse, however these are not shared. Thus, there is no possibility for SMEs to access to production facilities able to provide this kind of processes at the scale needed for small series production. This, is blocking the introduction of advanced materials into innovation and sustainable products.

Innovation 6: Operation of the Exploit4InnoMat OITB

Exploit4InnoMat is a one-stop- shop offering all the services needed to make commercial applications, based on open technological infrastructures, crucial to the technology maturation. It has been decided that the Ecosystem will be represented by a **Single Entry Point**. The SEP will be established in month 12 of the project and will have the mandate to represent the OITB, closing agreements and signing contracts with the potential customers, **being in accordance with national and EC regulations**. Thus, the administrative burden for the customers will be minimum as they will come in contact only with the SEP structure. The main services that will boost the innovation capacity of the ecosystem are **(A) technological services:** providing access to expertise, technology, and infrastructures, including manufacturing and product development. This includes for instance: collaborative RD&I, technical support on scale-up, commercial infrastructure, testing and validation, etc., **(B) ecosystem services:** developing and driving open innovation ecosystems with strong cooperation between research and industry and good connections between users and suppliers of innovations across the value-chain. This includes for instance: dissemination of information, innovation strategy development, brokerage, inter-collaboration, etc. and **(C) business-oriented services**: supporting businesses to commercially apply these innovations and transform their business, including access to financing and investors. This includes for instance: incubator/accelerator support, access to finance, business/market intelligence, etc. Innovation management in the open innovation environment will enhance information sharing and diffusion of expertise for all actors involved to increase the chances of success, lower risks, and reduce costs. To achieve these objectives and ensure technology uptake, both technological and non-technological services are crucially compulsory.

1.1.4 Positioning and Technology Readiness Level

Exploit4InnoMat starts from individual technologies standing at TRL 4 that will be improved, developed and integrated into the *Exploit4InnoMat* project and be stepped up to TRL 7 scaling up at an operational environment.

Table 1.1: *Positioning of Exploit4InnoMat project*

Technology		Current TRL	Activities for Increasing TRL	Final TRL
Renovation Materials/Elements	Active elements	5	SOLI is a PV manufacturer seeking innovative technological solutions for building applications able to host PVs. SINT has a vast experience in developing (TRL 5) façade and roof elements but also high-end PVs and PVTs (combined electrical and thermal production) systems. Exploit4InnoMat will enable the smart combination of these technologies in upscaling (TRL 7) production of nano-enabled components, incorporated with solar thermal photovoltaics. Prototypes of PVT prefabricated elements combining insulation and energy production capabilities will be produced.	7
	Structural elements	5	CETMA is interested on new cement structural products with enhanced functionalities and reduced environmental footprint. UNST has an extensive know how on the production of fibers reinforced cement-based elements and in the application of advanced manufacturing processes enabling the use nanoenabled additives (TRL 5). Through this project UNST will use a combination of a 3D printing and an advanced robotic spraying system to produce structural elements of complex geometries and enhanced functionalities (TRL 7). CDW aggregates will be also used to further improve the environmental footprint of the new structural elements.	7
	Cladding elements	4	NTUA has developed a new process for designing and producing 3D printed gladding element with complex geometries and advanced functionalities (TRL 4). This process will be further optimized and upscaled (TRL 7) to produce new market oriented lightweight cladding elements. CETMA has developed and demonstrated the successful production of lightweight inorganic polymers (TRL 4). This technology will be suitably adapted and upscaled to produce inorganic boards and panels for building facades (TRL 7).	7
	Multifunctional ceramic products	4	SCHL is a big bricks producer with much experience in insulating bricks. CSIC has extensive expertise in developing ceramic products with tailored designed internal geometries and technology to successfully introduce insulating and multifunctional	7

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			additives in the core of the ceramic component. These technologies will be appropriately adapted, optimised and upscaled to enable the industrial production of a new generation of bricks offering improved insulation and thermal storage properties.	
	Nanoenabled dynamic glazing elements	4	NTNU has developed and tested technologies adding electrochromic functionalities to glass elements. Moreover, it has experience in the development of innovative insulating aerogel glazing elements (TRL 4). The target is to upscale the technologies to produce prototype glazing elements with dimensions suitable for their application in buildings.	7
	Nanoenabled mortars	4	CETMA is an important market player on mortars and cement-based ready mix and is interested in high innovative insulation and thermal storage products. NTUA has developed lightweight inorganic fillers with insulation and/or thermal storage properties in the frame of three EC funded research projects. There is a pilot line for the production of lightweight inorganic fillers that will be upgraded (TRL 7) to enable the use of new ultrafine and nano additives materials that will greatly improve the mortar and ready mix insulation and thermal storage properties.	7
	Nanoenabled coatings	4	KANS is seeking efficient new coatings with advanced reflective and antifouling properties which will improve the environmental and energy building performance. RISE has significant expertise and infrastructure for the development of nanoenabled coatings (TRL 4) and will use its know-how to further improve of their functional properties. The pilot line will be adapted to produce prototypes of the new multifunctional coatings (TRL 7) and increase its capacity and improve its monitoring system.	7
	CDW aggregates	5	All end users participating in the project, are interested on new construction products with improved environmental footprint. Several Partners have developed and tested an optical sorting production line able to recognize and separate materials based on their morphology and synthesis. The pilot line will be upgraded and optimized for sorting alterative CDW streams and separate finer materials.	7
Pilot Lines	Capabilities, capacity and safe operation	4-5	Although the pilot lines available in the project have the main capabilities required for producing and evaluating the new construction products, it is important to be upgraded in terms of production capacity, capability to process nanomaterials and safety. PLs will be appropriately equipped with infrastructure for efficient and uniform dispersion of additives and supporting equipment for fulfilling health and safety requirements. PL2 and PL7 will proceed also to software upgrade for enabling more complex processing, with increased production rate and minimum materials waste. STAM, IRIS and POLM have a great expertise on process development and optimization which will upgrade PLs following the project requirements	7
Living Labs for Building	Sensors, monitoring and communicating infrastructure	5	Exploit4InnoMat is hosting three innovative living labs (SINTEF, NTUA, FACH) equipped with intelligent monitoring systems for accurate evaluation of new technologies efficiency. Calibrated models are already developed able to be revised and evaluated by adapting the new renovation solution. Living labs will be further improve their capabilities by adding more sensoring devices, logging systems and gateways for delivering the data needed to the service provides	7
Long term evaluation &	Micro scale to building modelling, simulation and calibration	4	AMS has a vast know how on materials modelling, UNAV and FACH have developed accurately calibrated models for complex energy systems at both building and grid level. UNIPD has a great experience on BIM technologies while NTUA has a core role on LCA analysis in more than nine research projects. These partners will deliver calibrated models able to evaluate materials efficiency, to deliver a holistic solution for building envelope and to evaluate the efficiency and the environmental impact of the new technologies. Efforts will be dedicated to the long term prediction of the solution developed facing also degradation impact.	7

1.2 Methodology

1.2.1 Overall research approach - concept

Management and Operation of the Exploit4InnoMat OITB

The ***Exploit4InnoMat*** OITB consists of **22 core partners.** The OITB will be governed by the **OITB Council**, consisting of one person from each of the core partners, having equal voting rights. ***Exploit4InnoMat*** will also establish three Operational structures

(a) the **SEP**: The partners have already decided that the most efficient way for the proper operation of the OITB is to **establish a new Joint Venture Entity to act as the SEP** in which each partner will be able to have shares. The founding of the Organization will take place during the project implementation, by month 12. The structure and operational procedures will be set-up and followed from the initiation of the project. In case that one or more partners decide not to be a shareholder of the Joint Venture their involvement in the OITB will be ensured through their participation in the Council and through specific contracts (see Section 2). The responsibilities of the SEP will be to receive, evaluate and analyze the feasibility of the customers' test cases for accessing the ecosystem and proceed with offering NDAs, contracts and access fees based on the established pricing policy of the ecosystem. Also, to proceed in the promotion of the OITB and create synergies with other OITB and EC Initiatives.

(b) the **Working Groups:** The SEP will be supported by the **Working Groups** that will consist of members of the Groups of Services. The main target of this structure will be to develop the projects' workflows in collaboration with the customers and the involved member of the OITB.

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(c) the **Data management support**. This will include the whole IT infrastructure needed to support the activities of the OITB, including the web platform, the data sharing points and systems etc.
The potential customers will be able to reach the ***Exploit4InnoMat*** OITB though its **SEP** that will be realized through a **web platform** in which they will be able to apply for access providing the necessary data. Access through direct contact of the member of the Ecosystem will be not allowed.
Finally, it is important to refer on the **decision-making procedures** of the OITB that will be defined during the first period of the project and will include the following main aspects:

1. Evaluation of test cases and translation of the customers' needs to be based on characterization workflows
2. Pricing of ecosystem services. The members of ***Exploit4InnoMat*** are committed to provide their services at fair prices and thus, competitive offers will be given to the customers. It is expected that due to the synergies that will be developed in the ecosystem leading to a **one stop shop**, the total revenues that will be created will be higher comparedto the sum of the revenues that each partner can create on its own (**economy of scales**).
3. Internal conflict mechanism resolutions will be developed for possible conflicts inside the ecosystem.
4. Intellectual property issues and confidentiality aspects will be arranged between the ***Exploit4InnoMat*** and the clients

The ***Exploit4InnoMat*** project will have a duration of 48 months and is divided into the following **three phases**:

Phase 1: **Set up the structural and operational aspects** (WP2). During this phase the Exploit4InnoMat Ecosystem will be set up. The services of the ecosystem will be organized while internal and external interactions will be developed. The **legal and operational procedures of the Ecosystem** will be defined in WP2. In parallel, the Ecosystem partners will work to develop the appropriate procedures for the operations of the Ecosystem, the decision-making mechanisms, the internal structures (SEP, communication channels) and the external relationships with the clients. **Important milestone of this stage will be the foundation of a new legal entity that will be the SEP of the OITB**. During this phase the **tools of the Ecosystem** will be also set up. The core of the Ecosystem are its 9 pilot lines. Thus, all necessary upgrades and modifications to the pilot lines will be implemented.
Phase 2: Validation of the ecosystem (WP5). During the second phase the developed ecosystem will be validated. The validation will take place in **two parts**. During the **first part** selected test cases will be analysed, targeting to cover and evaluate most of the services that ***Exploit4InnoMat*** can offer.

Table 1.3: *Short presentation of the Exploit4InnoMat test cases, including starting TRL*

Partner	Test case	*Short test case description* & Unique Selling Points	PL(s) involved	Lighthouse/ Demo
SOLI	PVT systems	*Single solution of energy production systems (PVT) integrated in lightweight cement based façade elements.* - Up to 87% efficiency by capturing thermal energy increases conversion levels over PV alone - Lower energy payback time & lower operating costs	PL1, PL7	SINTEF, NTUA,
KANS	Smart Glazing	*Glazing surfaces with embedded electrochromic mechanism and enhanced durability* - U-value of the glazing element 1.1W/(m²K) - Solar factor 0.36-0.12 - more than 10^5 colouring/bleaching cycles	PL4	NTNU
CSIC	Bricks	*High insulating bricks based on inorganic additives (silicon based loose fills materials) including CDW secondary raw materials* - Single rigidity and insulating solution - Gross Dry Density less than 800 (Kg/m^3) - Thermal Conductivity less than 0.2 W/(mK) - Initial Rate of Water Absorption less than 2 (Kg/m^2 -min) - Reaction to Fire Category: A1	PL6, PL9	NTUA
NTUA	Cladding systems	*Prefabricated insulating geopolymer panels and boards* - Excellent environmental footprint (more than 60% reduction on single score in comparison with conventinal materials) - Thermal Conductivity less than 0.3 W/(mK) - Density less than 1200 kg/m^3	PL7	YCC, SINTEF,
		2. 3D printed multifunctional cladding systems presenting high thermal storage (PCMs), lightweight and anti-fouling protection - Complex geometries able to follow various architectural designs - Thermal conductivity λ less than 0.11 W/(m K) - Easy installation due to lightweight components (450-600kg/m^3) and embeddedanchoring mechanisms	PL2, PL3, PL5,	
KANS	Paints	*Anti-bacterial, anti-virus & IR-reflectance paints* - Emmisivity between 0.87 to 0.92 at full color range - Solar reflectance more than 0.5 at bright yellow range - Able to be compatible with a wide range of substrates	PL5	RISE, KANS

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		- Long lasting properties		
CETMA	Mortars	*Low CO_2 footprint mortars of high insulation capacity and durability* λ value 0.050 W/mK or lower, density 300 Kg/m³, compressive strength > 1.5 MPa, low water uptake (0.001-0.008 Kg/m.s$^{1/2}$), vapour permeability 5±2 ng/Pa·s·m. As finishing solutions, Solar Reflectance will be >0.75, thermal emittance >0.85, and the photocatalytic properties will lead to 5% NOx reduction in continuous flow measurement.	PL3, PL9	NTUA
UNIST	Prefab elements	*1. Load bearing lightweight prefabricated elements for new constructions* - Steel fiber reinforcement - Decreased environmental footprint (CDW replacement > 15%) - Euroclass A1 fire behaviour for materials, > EI60 for panel - Surface weight < 360kg/m², U-value < 0.26 W/m²K	PL7, PL9	SINT
		2. Prefabricated elements for refurbishment/external cladding - Thermal conductivity concrete layer λ<2.30 W/mK - Sandwich panel surface weight <160kg/m2, thickness <13 cm - Sandwich panel fire resistance > EI60		

It should be mentioned that during the 2nd phase of the project, besides the access to pilot lines all the other services (e.g. characterization, testing, LCA/LCC, application of modelling and consultancy in standardization, regulation compliance, H&S, training and innovation management, installations to Lighthouses) will be provided to the 7 selected test cases. The end users will provide the necessary feedback to the Ecosystem (e.g. assessment of communication with the ecosystem and the elements, quality of services, adherence to time-schedule and workflows, flexibility, risks identification, final assessment of the experience) targeting in the improvement of its services and the internal communication between the Groups of the ***Exploit4InnoMat***.

The second part of validation will start at month 14 during which ***Exploit4InnoMat*** will proceed in an open call to attract additional test cases/end users for accessing free of charge the OITB through the established SEP (web platform). It is intended **that 6-8 more test cases will be selected** based on the assessment of SEP. The selection will be based on the evaluation of technical feasibility, market impact and TRL and their implementation in the OITB will start at Month 36 until month 48.

Phase 3: Ecosystem extroversion and sustainability (WP11,12). The ***Exploit4InnoMat*** OITB will proceed in a series of actions promoting extroversion and targeting to its sustainability. Dissemination activities will take place (e.g. workshops, participation in fairs, exhibitions & conferences, development of leaflets, posters, videos, establishment of ***Exploit4InnoMat*** Biannual International Conference), clustering with other similar projects and initiatives, including European standardization bodies and established clusters (e.g. EPPN, ECTC, EMMC, EMCC, EC4nano, nanosafety cluster) and liaison with stakeholders and possible investors will be planned. Targeting in the sustainability of the ecosystem post-project the initial business plan will be continuously updated.

1.2.2 Methodology

In order to respond to current challenges of the global competitive environment, European SMEs, which are the pre-dominant type of companies in the construction industry value chain, need simultaneously ***access to knowledge, technology, capital and markets***. EXPLOIT4INNOMAT aims to provide exactly this access to companies by leveraging upon the multiplier potential of cross-border/cross-sectoral collaboration and the ***systemic approach which homogenizes services towards the overall aim to place new products/services in the market.*** The three pillars of this approach are **(i)** the innovative selection and funding scheme; **(ii)** the three large scale demonstrators and **(iii)** the holistic portfolio of supporting services. This could be the ***start of a new Eco Construction economy*** - one that reflects more competition and more innovation, provides opportunity for a broader group of investors, and is more dynamic and responsive than the industrial model that has dominated for decades. Europe needs to understand this trend and adapt to the new circumstances, by ***fostering the diverse and emerging industries enabled ecosystems, which are the space on top of this new Construction economy.***

From a conventional value chain (Newton's Cradle) ***To a symbiotic ecosystem (Coral Reef)***

Figure 1.2: EXPLOIT4INNOMAT Strategic Impact

In this rapidly changing environment conventional value chains that look like a Newton's cradle reach their limits: the balls hit each other in a linear way, and if one fails, all will fail. Energy is needed from the outside to get and keep the balls going.

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EXPLOIT4INNOMAT takes into account the combination of technology trends, consumer needs and global competition to provide the infrastructure (in terms of knowledge, technology, market orientation and risk capital) that facilitates the transition towards a new symbiotic novel eco materials ecosystem that looks more like a coral reef. Our approach recognizes the fact that a reef is not always a cozy place, as there are hunters and prey, hence multifaceted support is needed, particularly now; in the dawn of the new Construction economy. EXPLOIT4INNOMAT infrastructure will have a long-lasting impact in supporting SMEs of the ecosystem to share similar values, adapt to change and survive global competition.

Four partners within the EXPLOIT4INNOMAT consortium are clusters with proven track records in identifying the needs of their ecosystem and delivering tailor-made support services. These partners can guarantee the interest of a critical mass of their member SMEs to be part of the EXPLOIT4INNOMAT ecosystem. At the same time a diverse and strong presence of five SMEs – technology providers within the consortium will guarantee that the technological infrastructure for the large scale demonstrators will be delivered by actors who understand the needs of the community. The innovative selection method and the corresponding algorithms will be provided by an RTD organization with long experience in organizing and running Open Calls and competitions for SMEs.

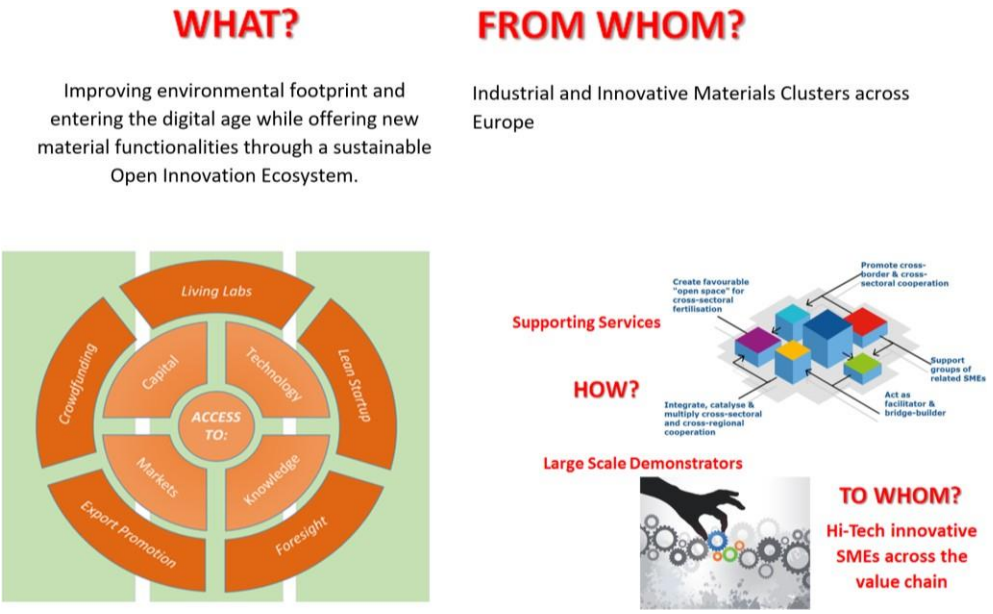

Figure 1.3: Addressing the needs of the ecosystem

Three key pillars of the Exploit4InnoMat concept

The purpose of this chapter is to present in detail three key pillars of EXPLOIT4INNOMAT concept. In addition, three key pillars have been presented in the figure below.

Figure 1.4: Three key pillars of the Exploit4InnoMat Concept

For an efficient and sustainable operation, the core partners of the Exploit4InnoMat OITB have decided the following:

1. Administrative Plan

The members of the Exploit4InnoMat OITB have already decided that the most efficient way for the proper representation and operation of the OITB is through a **Single Entry Point (SEP).** The SEP will undertake the whole administrative,

financial and legal issues of the OITB (e.g. marketing activities, contracts with customers, secretarial and administrative tasks, facilitation to additional funding etc). As already mentioned, the SEP will be **Joint Venture company** in which it will be possible for the core members of the OITB to have shares.

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For a more efficient implementation of the project, the SEP will be included in the consortium at ~M24 through and amendment with a total budget of 250.000€. This budget will be attributed initially to the coordinator (UNIS) and after the amendment it will be transferred to the SEP in order to undertake the corresponding tasks.

The decision making for the SEP related to the market strategy of the OITB will be based on the **OITB Council** consisting of a total of 22 people (one person from each one of the OITB core members). The **Council** will act as the **supreme advisor of the SEP** and will be responsible for the preparation of all strategy and policy matters of the OITB. The Council will be chaired by the president that will have a 3 years' service having the right to re-elect for another term. The **Executive Manager (EM)** of the SEP will manage the company's assets, optimize financial operation, provide leadership to all staff and ensure the legal and tax compliance. The EM will be accountable to the **Board of Directors (shareholders)** and will be advised by the OITB Council. **Although some of the core members may not be shareholders on the Joint Venture, their presence in the OITB will be ensured through their participation in the Council as well as through the establishment of the legal formalities that will be analyzed in the following paragraph**. Besides the **SEP staff**, the OITB will establish **Working Groups (WG)**. The WG will be linked with the six groups of the services provided by the OITB (Figure 1.1). The WG will be staffed by experts of the members of the OITB based on the target and mission of each working group. The responsibilities of the WG will be to provide an initial assessment of the feasibility of the customers' test cases for accessing the ecosystem and proceed and to develop the necessary workflows for each test case. With the progress of time the WG will include high expertise employees hired by the SEP.

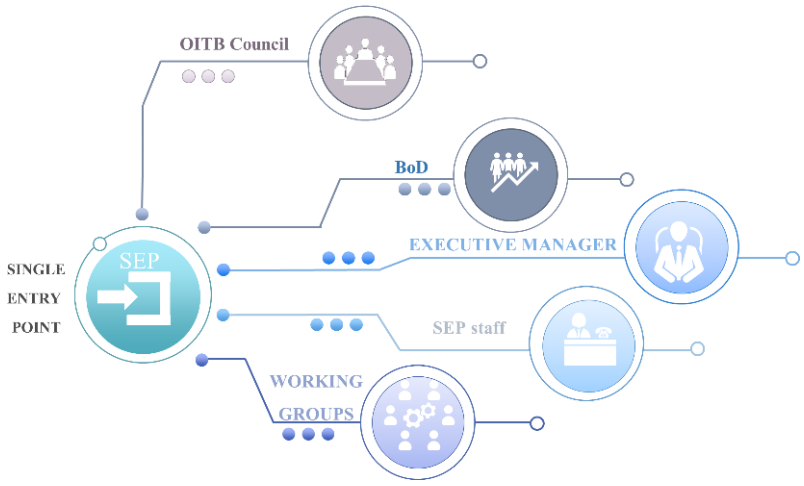

Figure 1.5: *SEP operation*

The above propose scheme will allow to the leadership of the OITB as well as to the technical experts and innovation managers to provide high quality services on the clients. As the OITB businesses will grow, it will projected that representative office will be established in order to facilitate local markets to reach the OITB.

2. Legal Scheme

As said, the SEP will have the mandate to sign contracts with clients on behalf of the OITB. The legal scheme for this will be based on two basic agreements:

Framework Agreement between the SEP and the OITB core members.

The SEP will be assigned to facilitate the negotiation of the services to the end users and accelerate the adoption of the Exploit4InnoMat technologies to the market. Thus, it is essential the **SEP to establish a Framework Agreement (FA) with the core members of the OITB**. The FA **will form the contractual framework for the provision of the Exploit4InnoMat services via the SEP**. In the FA, it will be described in detail the services that each core member will provide (e.g. access to Pilot Lines, modelling, characterization, consultancy etc) as well as other general terms (e.g. legal aspects, confidentiality terms, organization of work, pricing policy etc). Each Party of the FA may – with the mutual agreement of the respective other party – amend or add information to the pilot/services covered by the Framework Agreement and also add new Pilot/services sheets. New pilot/services will be welcome, provided that the terms and conditions set forth in each new Pilot/service sheet do not contradict those of the Agreement. The FA will include a **Statement of Work (SoW)** that will be activated each time a new customer will require the services of the OITB. In the SoW that will be signed between the SEP and separately by each involved member of the OITB will provide the details of the required service for the specific end user (analytical tasks, costs, persons involved, resources required).

Agreement between SEP and customers of Exploit4InnoMat OITB.

The access to the Exploit4InnoMat Pilot/services by the end users will be based on the **Services Agreement (SA)**, which the SEP will conclude with each individual customer. The SA will include aspects related to the tasks required by the end user (e.g. which pilot lines will be accessed, need in characterization and other services, the cost for these services, the duration of the project and of course confidentiality issues and IPRs). The SA will be based on the following basic principles:

- ✓ The SEP provides the access to Pilots and services to properly registered users via an online platform within the scope of a technology services solution.
- ✓ The services are provided to the customers against payment of a specific amount agreed with the actual service provider members of the Exploit4InnoMat OITB, for the duration of its use ("pay-per-use" business model).
- ✓ The access to Pilots and services available on the Ecosystem is usually third-party service to which the SEP merely provides access. The use of Pilot/services may be subject to deviating or additional regulations (e.g. due to national regulations), which the SEP will point out to the User prior to use.

3. Operational procedures for accessing the OITB

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Clients will apply for access to the OITB through a dedicated web platform operated by the SEP. Awareness of potential customers concerning the services of the OITB will be done through various channel (e.g. table 2.1) , while the necessary material in order to understand the potentiality of the pilot lines and other services will be free of charge for the users/customers. The procedure for accessing the ecosystem will be the following:

- ✓ The customer/end user make an application in a non-confidential form via the web-platform of the SEP. The application concerns the completion of a questionnaire consists of closed and open type questions.
- ✓ The application is assessed by the SEP and the customer is contacted. An NDA is signed between the SEP and the customer and more conventional details of the concept are provided as well as of the required services.
- ✓ The SEP in collaboration with the WG are setting up an **offer** including the technical workflow, a detailed cost estimation of the services described in the workflow and the handling of new IPRs.
- ✓ Upon the approval of the offer by the client, Service Agreement between the SEP and the customer is signed, while a Statement of Work between the SEP and the involved members of the OITB is activated.

Management and Operation of the Exploit4InnoMat OITB

The ***Exploit4InnoMat*** OITB consists of **25 core partners.** The OITB will be governed by the **OITB Council**, consisting of one person from each of the core partners, having equal voting rights. ***Exploit4InnoMat*** will also establish three Operational structures

(a) the **SEP**: The partners have already decided that the most efficient way for the proper operation of the OITB is to **establish a new Joint Venture Entity to act as the SEP** in which each partner will be able to have shares. The founding of the Organization will take place during the project implementation, by month 12. The structure and operational procedures will be set-up and followed from the initiation of the project. In case that one or more partners decide not to be a shareholder of the Joint Venture their involvement in the OITB will be ensured through their participation in the Council and through specific contracts (see Section 2). The responsibilities of the SEP will be to receive, evaluate and analyze the feasibility of the customers' test cases for accessing the ecosystem and proceed with offering NDAs, contracts and access fees based on the established pricing policy of the ecosystem. Also, to proceed in the promotion of the OITB and create synergies with other OITB and EC Initiatives.

(b) the **Working Groups:** The SEP will be supported by the **Working Groups** that will consist of members of the Groups of Services. The main target of this structure will be to develop the projects' workflows in collaboration with the customers and the involved member of the OITB.

(c) the **Data management support**. This will include the whole IT infrastructure needed to support the activities of the OITB, including the web platform, the data sharing points and systems etc.

The potential customers will be able to reach the ***Exploit4InnoMat*** OITB though its **SEP** that will be realized through a **web platform** in which they will be able to apply for access providing the necessary data. Access through direct contact of the member of the Ecosystem will be not allowed.

Finally, it is important to refer on the **decision-making procedures** of the OITB that will be defined during the first period of the project and will include the following main aspects:

5. Evaluation of test cases and translation of the customers' needs to be based on characterization workflows
6. Pricing of ecosystem services. The members of ***Exploit4InnoMat*** are committed to provide their services at fair prices and thus, competitive offers will be given to the customers. It is expected that due to the synergies that will be developed in the ecosystem leading to a **one stop shop**, the total revenues that will be created will be higher comparedto the sum of the revenues that each partner can create on its own (**economy of scales**).
7. Internal conflict mechanism resolutions will be developed for possible conflicts inside the ecosystem.
8. Intellectual property issues and confidentiality aspects will be arranged between the ***Exploit4InnoMat*** and the clients

1.2.3 The Exploit4InnoMat Living Labs

In the following paragraphs the services of the Exploit4InnoMat OITB are presented in details. At the beginning of each group of services (in italic) the industrial need for the services is briefly explained.

A. Group of Pilot Lines *(PLs owners supported by STAM, IRIS, ABIS, POLM)*

Many material and components concept never reach the market due to lack of appropriate infrastructures for upscaling. These types of infrastructure are necessary to be open to the industry and versatile in order to able to handle different formulations, use of nanomaterials and produce materials in volumes that will allow validation in the real environment.

The first Group is offering the access to Pilot Lines for upscaling materials and products concepts. The Group contains 9 Pilot Lines, covering a wide variety of materials related to the building envelope. These existing Pilot Lines have been developed in previous funded projects (EC or inhouse) and will be now further upgraded in terms of processing capacity, monitoring systems and process control and will be able to successfully integrate nano-enabled technologies. The following Table 1. presents the 9 pilot lines.

Table 1.4: *Exploit4InnoMat Pilot Lines*

	Pilot Lines 1	**PVT façade elements**
	Location	Norway
	Partner	SINT
	Products	Active façade elements, thermal and electric energy
	Description	

SINTEF has a vast experience in upscaling production of nano-enabled components, incorporated with solar thermal photovoltaics focusing on miscellaneous energy-efficient solutions and energy production capabilities. **PL1** will enable the adaption of multiple technologies and nano enabled materials for developing PVT prefabricated elements. These elements will utilise a wide range of nano enabled back envelopes, providing properties like light weighting, insulating, enhanced thermal capacity and fire proofing, optimising PVTs operation based following each application requirements. Materials and technologies which will be used in such systems are nano insulation materials, aerogels incorporated in lightweight panels and phase change materials. For the overall characterization of the materials, advanced infrastructure and experimental procedures will be applied. Additionally, further characterization can be performed at large-scale to full-scale building level) through large-scale hot box apparatus evaluating the thermal transmittance (U-value) and thermal conductivity. Climate chambers can run accelerated climate ageing simulation for evaluating heat and ultraviolet ageing, moisture and wind-driven rain exposure. It's worth to mention that SINTEF was label awarded during last year with "*Solar Impulse Efficient Solution Label*" which act as a new and reliable label that can be used on products, processes and services, representing a guarantee of quality for those wishing to implement clean solutions.

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Main Innovations during Exploit4InnoMat project: SINTEF will proceed with the upgrade of the laboratory infrastructure (e.g. fume hoods, etc.) as well as the equipment for more efficient and automatic materials synthesis and component fabrication and assembling. In the field of the monitoring systems, further upgrade will be performed in respect with synthesis and characterization equipment from nano- to large- scale for allowing better control of various parameters, mainly electrical functions, temperature and moisture. The overall actions will result in the improvements of the development and upscaling of high-quality end-products as well as the application of environmentally friendly processes.

Pilot Line 2	**3D printed semi-inorganic elements**
Location	Greece
Partner	NTUA
Products	Active inorganic elements

Description

PL 2 is a 3D-extrusion hybrid manufacturing system able to handle semi inorganic filaments mixed with nano enabled binders. The unit hosts two heads, from which the first one builds the geometry required through semi inorganic filaments (like ABS or PLA mix with inorganic minerals and other additives), whereas the second head fills the closed cellular structures with loose fill or binder-additives pastes. The simultaneous 3D printing and extrusion process significantly increases the production rate and simultaneously provides the opportunity to adapt multiple technologies in a single element. This technique can evaluate the performance of new binders and nano additives in 3D printing process, but also considering sustainability issues, materials in which traditional cement-based ingredients will be replaced with environment friendly ones. Additionally, the pilot line is also able to develop complicated shapes, which is one of the biggest advantages for most architects.

Because the freedom in geometric complexity applies also to the inner part of the elements manufactured with additive processes, it is possible to customize inner geometries to integrate multi-functionalities. For example, operational mechanisms and embedded components can be fabricated directly, enabling the control of heat exchange between the indoor and the outdoor environment, by integrating geometries for thermal insulation and heat storage, while guaranteeing structural strength.

Main Innovations during Exploit4InnoMat project:
PL2 will be upgraded in order to increase its productivity and the dimensions of the elements produced. On this direction it's horizontal platform will be upgraded to 1x1m while the vertical axis will be extended to 80 cm. A second printing head will be also installed for simultaneously fabricating of printing layers fed from two independent premixing lines. Peripheral equipment will be also installed for dust control and monitoring. The upgrade will enable the production of LEGO type prefabricated active elements with embedded active mechanisms for adapting technologies for energy harvesting and storage with a production rate of one element per day (1 m x 1 m x 0.8m).

Pilot Line 3	**Nano enabled mortars and premix products**
Location	Greece
Partner	NTUA
Products	Mortars with various insulating fillers and PCMs

Description

PL3 will be used for development and manufacturing of novel, multifunctional cement-based mortars and other cementitious materials. NTUA is operating a "smart" concrete pilot line comprising of three silos (250 kg each) with automated weighting apparatus, feeders and a mixing unit of 500 L volume capacity. The mixing unit in particular is equipped with sensors for monitoring the moisture content as an indicator for the homogeneity of the mixture, while additional monitoring sensors have the possibility to be installed, if needed, for the better monitoring and quality control of the produced materials. Furthermore, PL3 is equipped with an automated weighting apparatus, energy consumption smart meters, water consumption monitoring systems while all parameters are

harvested in a one single datalogging system. Based on the capacities of the "smart" concrete pilot line, it is possible to develop and produce materials tailored to each client's requirements through the optimisation of material and process parameters while ensuring each time robust process and quality control. This can be achieved by integrating i) sensors in the mixing unit that are characteristic of each product additional functionality, such as viscosity or temperature and ii) suitable sensors to the storage and transportation units for measuring critical raw material quality properties such as moisture. PL3 is also equipped and experienced in the use of nano-materials in liquid dispersions, thus can utilise a variety of additives ranging from lightweight aggregates, fibers and PCMs to any nanomaterial that can provide additional functionality to the product. Concerning lightweight additives, a sub-pilot line of PL3 is able to produce ultralightweight aggregates originating from volcanic glasses including perlite, pitchstone and obsidian, or other inorganic materials. The sub pilot line comprises of size reduction and size control units (crushing and grinding devices), size distribution control units (sieving devices) as well as two prototype expansion furnaces thatcan be used for the production of lightweight aggregates from different raw materials: the vertical electric furnace (VEF) and the IR furnace. Maximum productivity for the VEF is 500 L/h and for the IR Furnace 300 L/h for certain qualities. It must be noted that PL3 have also the capacity to provide high quality characterization services both for wet-mixture and for hardened product properties, thus the final PL3 product will be accompanied with all the necessary characterisations, according to the products requirements.

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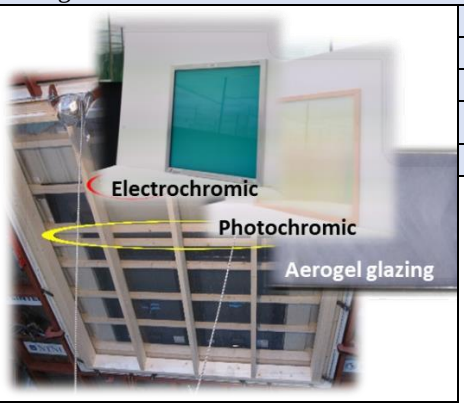
Main Innovations during Exploit4InnoMat project: The pilot line will be suitably modified in order to accommodate the new functionalities. More specifically:

Upgrade: Even though PL3 has already established a silo dedicated to lightweight aggregates, in case of raw materials with other specific property and storage requirements (such as PCMs or fibers), new storage facilities (silos, tanks etc) will have to be established. Furthermore, the transportation processes and the feeding equipment will either have to be upgraded or new, more suitable equipment will have to be installed.

Quality control: In order to ensure quality control in the storage section of PL3, moisture sensors and controllers will be established to all raw material silos and containers. Additionally, in the case that the performance of the advanced raw materials is dependent on an additional property (such as temperature), the new silos will have the opportunity to integrate more sensors destined for quality monitoring. In the case of liquid solutions of nanomaterial dispersions, it is possible that ultrasonification apparatus will have to be installed in order to ensure solution homogeneity and to enhance quality control. Regarding the mixing process itself, as already mentioned, sensors and other monitoring systems (e.g. , viscosity meters etc), can and will be integrated to the mixing unit, supplementing the already existing monitoring capacities, in order to ensure quality and processes control even for the most challenging and demanding products.

Health and safety: Advance ventilation dust control and sound insulation will be installed in order to ensure the correct management of fine and ultrafine materials and to implement the H&S standards for the pilot personnel.



Pilot Line 4	**Smart glazing and windows manufacturing**
Location	Norway
Partner	NTNU
Products	Windows system

Description

Pilot line for preparation of materials and components for development of smart windows manufacturing incorporating advanced technologies as electrochromism. Through this PL, different material/system compositions will be tested and evaluated in order to achieve reversible and persistent changes of their optical properties in a controllable way. In this approach, high end systems will ensure the dynamic control of the solar radiation and. best performance of solar radiation glazing factors by emphasizing on solar radiation for daylight, heating and electricity generation, and avoidance of undesirable solar heating. The development of windows system will be based in a high-quality testing of properties and identification of materials' chemical composition. Advanced analytical instruments, such as FTIR, UV-VIS-NIR, GC-MS, LC-MS, LC-MS-MS-QQQ, including FT-ICR, FFF/ICP-MS-MS will be applied for define the chemical composition of the initial materials in nano- and micro-scale (Mass Spectrometry Laboratory).

Main Innovations during Exploit4InnoMat project: The PL 4 will be mainly upgraded in order to provide an innovative production and assembly approach from nanoscale to large-scale structures with corresponding equipment as well as the adaptation of new functionalities. All laboratory infrastructure (e.g. fume hood, relevant synthesis and characterization equipment, etc.) will be upgraded for increased automatic materials synthesis and component fabrication and assembling as well as more efficient control and evaluation of parameters (electrical, temperature, moisture, etc.). The upgrade of all monitoring systems will allow the application of environmentally friendly processes. This approach will be performed by taking into account the usage of variable materials, which will be synthesized and produced following a combination of variable chemical composition, crystal structure, size, dimensionality at nanometer-scale and consequently different properties in order to result in high-performance, adaptive and controllable glazing systems.

Pilot Line 5	**Functional nano-enabled coatings**
Location	Sweden
Partner	RISE
Products	Paints and coatings

Description

RISE has an excellent infrastructure to develop and investigate multifunctional coatings. RISE has fully equipped labs for processing nano additives including 50L Stainless steel reactor, 100L enamel reactor, ultra high shear fluid processors (M700

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microfluidizer) for pilot production of surface-engineered nano-additives. RISE is equipped with a complementary expertise in coating application facilities including spraying, curing (thermal and IR) and testing equipments such as weathering equipment's. Compatibility of nano-additives with the coating formulation will enhance the performance of materials, increase the material efficiency and thus decrease costs. The nano-additives pilot line at RISE is designed to improve the dispersibility and compatibility of nano-additives in the coatings as well as enhance the designed function. These surface-engineered nano additives produced in PL5 can serve other pilot lines including PL3 (mortar and pre-mix), PL6 (Advanced Ceramics) and PL7 (3D printing and robotic spay) in the current project. The multifunctional coatings will be produced by formulating surface-engineered functional nano-additives into the coating to obtain a nano-enabled multifunctional coating with enhanced hydrophobicity (polymer chemistry, nano-Al_2O_3, nano-SiO_2 and carbon black), UV resistance (nano-TiO_2, nano-ZnO, nano-CeO_2, nano-FeOOH), IR reflectance (nano-Al_2O_3, nano-talc, nano-TiO_2) and corrosion resistance (nano-cellulose, nano-clays).The best performing coatings will be tested and validated in the light houses for their efficiency. RISE QA/QC labs are equipped with characterization tools such as ESEM, STEM, TEM, BET, AFM, XRD, UV-VIS, zeta probe, plate reader, QCM, foam analyzer, contact angle, rheometer, mastersizer, nanosizer, optical microscopes, turbiscan, XPS, Raman, NMR, SAX/WAX GC, LC/MS, TOF-SIMS etc. The capacity of the PL is 10 kg per day of nano-additives, 100L per day of dispersion of nano-additives and 30-50L per day paint formulation

Main Innovations during Exploit4InnoMat project: The PL 5 will be mainly upgraded in terms of capacity of the overall production. Thus, equipment such as larger homogenizer reactor, and larger bead mill (up to 150L) will be installed, including the necessary control systems.

Pilot Line 6	**Advance ceramics**
Location	Spain
Partner	CSIC
Products	Tiles, bricks and other ceramics

Description

The pilot line of advance ceramic materials (CSIC) will be able to produce new formulation for ceramic materials for the building envelope, mainly bricks and tiles. Its capacity will include advance mixing procedures, clay conditioning, formation processes, drying and sintering. CSIC has great experience in the part of mixing and clay conditioning including production and preparation of mixtures clay, adjustment of respective water contents, addition of porosifying agents, admixtures, recycle materials (e.g. from PL 9) and secondary fuels and conditioning and homogenization of plant mixtures. Crushers, mills and mixers as well as equilibrium moisture and plasticity measuring instruments are available for the production of the clay mixtures. In this pilot line infrastructure for formation of the final products will be also available, providing various geometries based on dies or extrusion processes. CSIC will also provide the possibility of application of coatings in the final products (e.g. from PL5) in order to adjust further functionalities on the ceramic materials. The developed material may characterized by means of XRD, SEM, spectroscopy techniques & porosimetry. CSIC will also study the effect of nanomaterials on the microstructure of ceramic and cementitious materials using adsorption-desorption isotherms, laser technology, nanoindentation and measurement of reflectance and / or transmittance, among others.

Main Innovations during Exploit4InnoMat project:
Advanced equipment and state of the art software for the control of the dispersions mixtures and emulsions and the analysis of nanoporous samples will be purchased. These solutions will allow the accurate identification of the new phases formed in high performance insulating brick due to the incorporation of innovative solutions. Moreover, the pilot line will upgrade the spray coating line in order to host a wide range of nanomaterials, to increase the production rate and minimise material losses. Additionally, the pilot line will be equipped with safety infrastructure for nanomaterials handling and an automated dust control and monitoring system. The upgrade plan will make capable the production of advanced ceramics in commercial dimensions under a sufficient production capacity.

Pilot Line 7	**3D printing, robotic spraying and bio concrete**
Location	Germany
Partner	USTUTT
Products	Cement and cement substitute structures

Description

The production platform can briefly be split into two main parts, being a hybrid manipulator for material application as well as the concrete mixing and conveying devices. The former consists of a gantry type system with a Stewart-Gough platform attached. This allows for arbitrary positioning of the spraying nozzles within the working space. The implemented concrete technology provides the possibility to convey the mixed ingredients of the concrete in a dry state pneumatically through hoses. At two entry points inside the conveyor line, water can be added to complete the dry mixture. The two mixtures are blended either inside the hoses in a mixing unit and applied through a single nozzle or by overlaying the spraying cones of two separate nozzles. With these techniques, a continuous material gradient can be created. Similarly, a wet-mix conveying line can be used to build homogeneous concrete components or components with a structural gradient through insertion of hollow bodies. The internal structure of the building component is designed as micro- or macro-graded structure using specially developed software tools linked to the control system of the production platform. Based on a digital blueprint of concrete component the production process is planned for layer by layer manufacturing employing a process-simulation model. The PL has the capacity of 2x2x0.3 m component per day.
The second part of the PL is dedicated to the use of natural microbiological processes to consolidate aggregate into so-called bio concrete represents a promising and potentially CO2-neutral alternative to cement-bound concrete. In bio concrete, instead of cement, the aggregate is bound by calcium carbonate crystals precipitated by the activity of bacteria. The process is known as microbiologically

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induced calcite precipitation (MICP). Bio concrete is already being applied in some areas of construction, such as sealing cracks in concrete, fixing sandy soils, and making bricks and tiles. To date, however, no processes exist by which bio concrete can be used to manufacture structural components. In our own preliminary work, test specimens of bio concrete with a compressive strength of up to 50 MPa were produced (Nething et al. (2020)[5]). The best results in terms of cementation depth, homogeneity and strength were obtained with a production method in which the sand-bacteria mixture is flushed with the cementation solution under pressure. The cementation process took 3 to 5 days. The aim of the pilot line is to modify the methodology to be able to produce in one step a mixture which can cure without further feeding in formwork. Necessary aggregates will be investigated that can act as ion carriers in the bio concrete. They should be able to store large amounts of ions and release them slowly into the cementing solution so that the ion concentration of the solution does not exceed a certain limit at any time. These include both calcium-containing clays or clays charged with calcium ions with a high cation exchange capacity (CEC) and superabsorbent polymers (SAP), which can absorb and release dissolved urea.

Main Innovations during Exploit4InnoMat project:
The pilot line will be upgraded in order to host multiple feeding lines for concrete ingredients including bio concrete mixtures with automated online weighting equipment. The upgrade will provide the capability for producing at a single stage element with multi-layered functionalities. For example, the envelope of the element can act as a structural layer through the addition of reinforcement ingredients, when the core of the element can act as insulating barrier. Finally, the upgrade will enable the application of a cement-based coating providing antifouling, reflectance, hydrophobicity and self-cleaning functionalities, depending on end user requirements. The routines for evaluating the interior design of the construction element will be expanded to deal with differing hollow body types and sizes considering multiple load cases. The option of integrating different reinforcement materials and methods will also be of research interest.

Pilot Line 8	**Cement-free materials and components**
Location	Italy
Partner	CETMA
Products	Cement-free (alkali activated) materials and components (pre-casted solutions) with recycled or natural aggregates

Description

The pilot line can produce construction pre-casted components, divided in two subsections. The first section is equipped with mechanical mixing devices (drum type and a pan type), whereas the second section hosts moulding equipment able to be re-designed and re-dimensioned depending on the target component. The drum mixer is used to prepare small volumes of concrete for preliminary evaluations (quality assessment, production of samples for testing), while the pan mixer is used to prepare larger volumes of concretes with improved homogeneity, compactness and suitable workability for the following casting operations. There is also a dedicated lab equipped to perform the main characterization tests of the produced components, both on fresh state (e.g. consistency measurement) and hardened state (e.g. mechanical performance), thus assessing the suitability of the materials produced for the target application. In this line, the application of recycled carbon non-woven mat from aerospace sector will be also examined in order to provide components with high technical performances. The capacity of the current PL is production of 2 medium size pre-casted components per day (slab 60x60x10 cm) and 5 medium/low pre-casted components (i.e. beam 15x15x60 cm).

Main Innovations during Exploit4InnoMat project: CETMA will proceed in upgrade of the infrastructure (PL) as well as on the monitoring systems for the products. Thus, additional mixers of bigger volume and able to operate on alkaline environments, will increase the production capacity of the line by 100%. It is also foreseen to increase the capacity and to make fully automated the raw materials feeding lines with accurate weighting devices. Reusable moulds made of plastic or steel, easy to be assembled and disassembled for the next re-use will be designed and manufactured. In the field of monitoring systems and QA, a multichannel datalogger will be installed to log some of the of the process data (T and RH) of the mixtures inside the moulds. In order to ensure the optimum conditions for un-moulding procedure, an infrared camera-based system will be integrated for fast and efficient evaluation of the curing readiness level.

1.2.4 Related research and innovation activities - Baseline Assets

The following table provides details of national and international research and innovation activities whose results will feed into the Exploit4InnoMat project, and synergies will be established. Baseline Assets will be adapted as Exploit4InnoMat components and tested in the LLs so their maturity will increase.

Most of the partners are already involved in relevant National or European projects. The following table describes R&D initiative related to *Exploit4InnoMat* where for running one synergy with these initiatives will be developed.

Table 1.5: *Exploit4InnoMat synergies with the R&D&I activities*

Project	Correlation with the Exploit4InnoMat
EXPERL	**NTUA** was involved in the EXPERL project dedicated to the efficient exploitation of EU perlite resources for the development of innovative and high added value micro-perlite based materials for Chemical, Construction and Manufacturing.
LightCoce	**NTUA, STAM** and **AMS** are members of the LightCoce OITB for upscaling nano-enables lightweight materials of concrete and ceramic base.
MF-RETROFIT	**AMS** participated in this project devoted to the development of a lightweight, durable, cost effective and high performance panel for façade retrofitting.
H-HOUSE	**RISE** (Katarina Malaga), coordinated the H-HOUSE project aiming at the development of multifunctional and flexible components for the building envelope and internal walls, for new buildings and renovated one.

[5] Nething et al. (2020) Materials and Design 195 (2020) 109032, https://doi.org/10.1016/j.matdes.2020.109032

SUS-CON	**NTUA** and CETMA (coordinator) worked on the integration of waste materials in the production cycle of concrete related to new technology routes for the development of innovative light-weight, eco-compatible construction materials.
RE	**CETMA (coordinator), RISE, and STAM** worked on the integration of CDW in a prefabricated energy-efficient building concept that can be easily assembled and disassembled for future reuse.
PVadapt	**SINTE & NTUA**are involved in the PVadapt project for developing of adaptable and multifunctional BIPV system with substantially lower cost than conventional solutions.
NanoLEAP	**Stam** was responsible for the optimization through the implementation of a dedicated control system of 10 PLs to produce nanomaterials.
Eco-Solar	**SINTEF** was participated in Eco-Solar project for integrating a vaue chain to manufacture and implement solar panels in the most environmental way by maximizing resource efficiency, taking into account reuse of materials during production and repurposing solar panel components at end of life stage.
Robust, EffWin, ZEB	**SINTEF and NTNU** have cooperated for developing robust envelop construction solutions, improved window energy efficient building and nZEB technologies
ECLICESOL	**CSIC** has developed technologies for Eco-efficient light ceramics: green energy and alternative raw materials
EXC 2120 IntCDC, CRC 1244	**STUTT had a core role on Collaborative** Research Centre 1244, for building a roadmap on adaptive skins and structures for the built environment of tomorrow. Clustering activities are also undertaken, like the Cluster of Excellence Integrative Computational Design and Construction for Architecture (IntCDC

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1.2.5 Inter-disciplinary and trans-disciplinary approach

Exploit4InnoMat approach is fully interdisciplinary as it combines a wide range of IT technologies, engineering knowledge and know-how in order to develop tailored solutions for the development, upscaling, characterisation and testing of climate neutral materials. In more details, the proposed multidisciplinary methodology: (i) utilises the potential of IT technologies for data recording, remote sensing and quality control (ii) valorises the combined technological knowledge and know-how for the development processing and treatment of climate neutral materials, (iii) promotes interaction between research and industries and (iv) attempts to gather climate neutral technology providers in one site and v) promotes pan-European collaboration. In this regard, **Exploit4InnoMat** approach can be used by any project that needs hybrid solutions that combine business, technology, and research interaction. Leveraging on IT technologies, open innovation models, the SEP that is foreseen to be created as a major outcome of the project will act as an open-source super structure able to adapt, expand though the interaction and collaboration with other OITBs. By ensuring the participation of key stakeholders and groups of interest within the Circular Economy and Climate Neutral concepts, **Exploit4InnoMat** SEP is expected to have a long-lasting impact. The approach developed in **Exploit4InnoMat** for collaboration between the different pilot lines and services can be used as a paradigm of methodological approach for interaction of different technological and industrial areas.

1.2.6 Gender dimension

The **Exploit4InnoMat** consortium supports the objectives of the European Commission to promote gender equality in research and innovation as outlined in Article 16 of Horizon 2020[6]. Women' under-representation in science due to the lack of attention to gender aspects constitutes a major concern in the EU as stated within the Commission's Action Plan on Science and Society, adopted in 2001[7]. The checklists and methodological tools of the Gendered Innovations project and the gender-sensitive research cycle (EC Toolkit Gender in EU-funded research) will be followed to identify gender sensitive aspects in the project. Gender dimension will be monitored in WP1 (project management). We will analyse the relevance of sex/gender when it comes to the different expectations males and females may have towards **Exploit4InnoMat** innovation. We will consider the gender dimension and cater for the needs, motivations and differences between females and males where applicable in particular in Requirements` analysis in WP2; Throughout the development **building materials and Components** in WP2 and WP3 and infrastructure preparation in WP4; During the trials and the evaluation of results in WP6. Where possible, we will sex-disaggregate the data, as per the recommendations of the EIGE[8], to find out possible different expectations regarding the user interfaces, features, functionality, etc., exist. In our **Business Modelling & Exploitation, Communication, Dissemination, Stakeholder Mobilisation** (WP7), we will consider gender in training and education materials, in target groups of the community building and outreach activities as well as in the business models. Consortium members are themselves aiming to achieve gender balance, e.g. Tieto targets 40% female employees by 2026 and 50% by 2030.

1.2.7 Open science approach

The materials that will be developed by **Exploit4InnoMat** will be available on the projects' website. As boundary condition, users have to commit themselves to use the brand, when adapting the available material at national level and use the business models, labelling and qualification schemes. All other knowledge, materials and tools will be available for all professionals for free (with a small maintenance fee after the project duration). All the data that will be delivered will be curated and preserved during and at the end of the project on the **Exploit4InnoMat** platform. Here the maintenance and exploitation of the project results will take place, i.e., the developed and gathered technologies from relevant running and coming H2020 projects, also for other/new technologies, concepts and products, during and after project completion. Furthermore,

[6] H2020 Programme "Guidance on Gender Equality in Horizon 2020", 2016
[7] SEC(2001)771, "Women and science, the gender dimension as a leverage for reforming science
[8] European Institute for Gender Equality | European Institute for Gender Equality (europa.eu)

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Exploit4InnoMat will not generate open research data. The consortium expects a variety of professional publications to result from the project. The beneficiaries will be extremely keen to pursue "open access" publications for scientific papers and for all the other types of scientific publications (some of which may, in some cases, not be peer reviewed) including monographs, books, conference proceedings and grey literature (informally published written material not controlled by scientific publishers, e.g. reports). In order to comply with the "open access" requirement, the beneficiaries will ensure that their publications can be read online, downloaded and printed. This will be implemented depositing publications into a repository (online archive) and providing open access to it. Services promoted by the **Open Research Europe platform** will be also exploited. **Exploit4InnoMat** is willing to engage in the Open Research Data pilot (Article 29.3). This deliverable (see Task 1.4) will include relevant information about the development and update of a Data Management Plan (DMP), including procedures on how to deposit project data in a shared data repository, and insurance for third parties to freely access, mine, exploit, reproduce and disseminate project data.

- **Open Access Publications**: Open access to all the peer-reviewed scientific publications of the project will be provided. Selected publications will be made available with the highest standard (Gold Open Access). The rest publications will be also made available in the project's website and Fair Data Marketplace, but also in OpenAIRE's Zenodo.org European open access repository, thus supporting an economically sustainable transition from a subscription model towards an open and well-functioning publication model.
- **Open Peer Review**: **Exploit4InnoMat** will allow self-selected reviewers to provide comments on the project's scientific outputs (i.e., publications), beyond reviewers selected by the Open Access journals (or other forums), where the results will be made available. Likewise, partners will participate as open peer reviewers to open peer reviews of results from related projects such as projects funded under the same call.
- **Open Research Europe Publishing Platform (OREPP)**: Consortium will publish a minimum of three (>=3) articles in the OREPP, including one project overview article at the beginning of the project and a concluding article providing a summary of the project's main research offerings and achievements at the end of the project.
- **Open Data and FAIR Data**: All the project's datasets will be anonymized and made available as open data in various FAIR repositories (e.g., EOSC, OpenAIRE, re3data.org, EUDAT, DataHub). Similarly, the project will share the constructed AI models and data analysis flows in open repositories like OpenML, AI4EU.
- **Open Innovation**: EXPLOIT4INNOMAT will promote open innovation in national and international collaborations and networks. It will actively utilize the potential of collaboration with partners' academic and business networks as well as with other Horizon EU clusters and collaborate with stakeholders in the public and private sector across diverse domains (e.g., ICT technologies, industry, environment), aiming to adapt and/or extend the EXPLOIT4INNOMAT developed systems and techniques, whilst sharing the EXPLOIT4INNOMAT ideas, knowledge, and best practices towards their adoption among these collaborators.

2 Impact

2.1 Project's Pathways Towards Impact

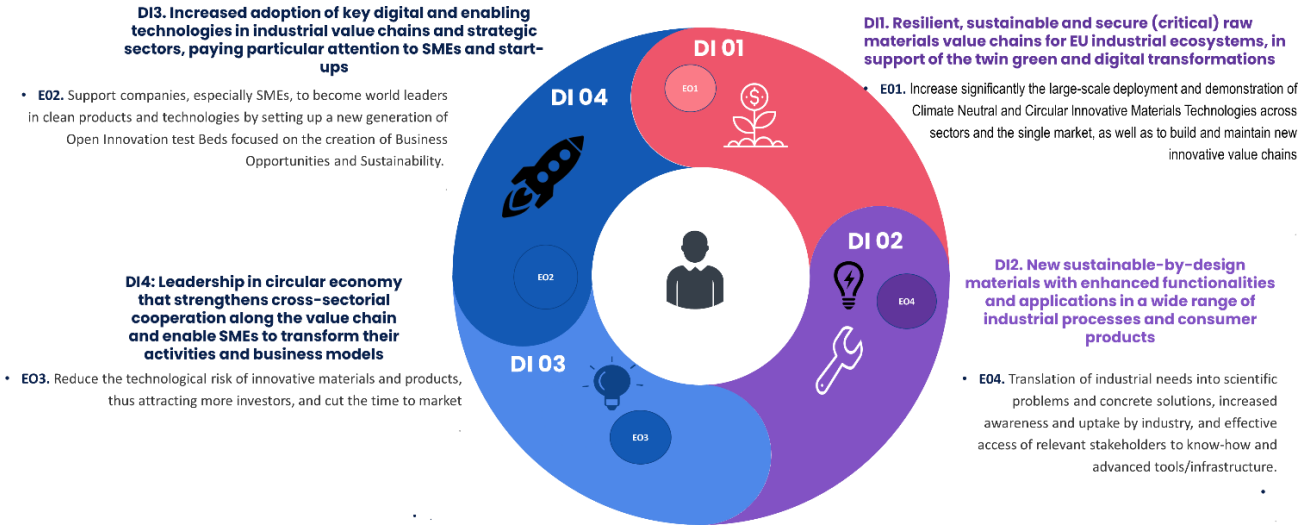

Figure 2.1. Expected mid-& long-term Exploit4InnoMat Impacts

Call impact: Exploit4InnoMat results and pilot activities will contribute to the 4 expected outcomes (EO) asked for by the call, thus contributing to destinations of Cluster 4 'Digital, Industry and Space'

EXPLOIT4INNOMAT answers to this call of CL 4, which falls under destination '***Destination 2 – Increased Autonomy in Key Strategic Value Chains for Resilient Industry***. Proposals for topics within this destination have to cover several expected impacts, which we refer to as 'Destination Impact' (DI). Figure 2.1 shows the call expected outcome (EO) links to 3 DIs from CL4 as well as to one DI from CL 6.

EO 1: Increase significantly the large-scale deployment and demonstration of Climate Neutral and Circular Innovative Materials Technologies across sectors and the single market, as well as to build and maintain new innovative value chain;

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Impact Analysis:
For the Large Scale deployoment of **Climate Neutral and Circular Innovative Materials Technologies it is essential** the European SME and industry to be provided with world class, reliable OITB services leveraging products including climate neutral and circular innovative materials in the EU key value chain. The clients of the OITB need to enjoy open access to a number of physical facilities and infrastructures as well as a series of services in order to fast and accurate evaluate and validate materials and components concepts. On the other hand, software applications (e.g. BIMs, Energy+, materials modelling and simulation) will be available facilitating the design and evaluation of the materials. Having access to these types of infrastructures and implementing upscaling projects, it is mandatory to ensure the necessary funds in order to cover the costs.

EXPLOIT4INNOMAT Contribution: The Exploit4InnoMat presents 9 pilot lines in 7 countries around Europe and is targeting in the development, characterization and testing of advanced materials for building envelopes. The OITB proposes a framework composed of: business networks, knowledge integration, and innovation management [9] and will be easily accessible, through a single entry point (SEP), from SMEs and industry that want to invest in conducting a full development of their business idea. The Exploit4InnoMat will aim to externally focused, collaborative innovation and will be open to interested companies to access it at fair conditions and costs, providing them vast opportunities to tap into a broad network of experience and expertise to develop new products and services as well as for enhancing their operations. The Exploit4InnoMat OITB during the implementation of the project will form a Joint Venture Organization that will have as shareholders members of the core group of the ecosystem. The single entry point will be associated with this Organization. The exploitation strategy and dissemination actions foreseen will ensure that potential customers will be informed about the test beds existence, services, and access conditions.

EO 2: Reduce the technological risk of innovative materials and products, thus attracting more investors, and cut the time to market;.

Impact Analysis: In order to overcome the structural challenges that result from the number and diversity of SMEs in the European Advanced Material sector companies need to cooperate and look for possible external support. The chain reversal process that puts the consumer at the center of production implies that companies have to find new innovative technological solutions and find new business models. Therefore, the latest important changes in the construction sector demand and in supply chain organization in a more competitive environment have made innovation a fundamental corporate activity extremely relevant to the profitability of the industry. Innovative SMEs that want to increase their knowledge may need to associate with other actors of the supply chain as well as with external possible partners such as: universities and research centers or other industries such as biotechnology, preservation, technology and nanotechnology.

a. Exploit4InnoMat Contribution: Access to finance:

The members of Exploit4InnoMat have already recognized that access to the services of the OITB might have a cost that is above the capacity of potential customers despite the fact that the services will be provided at fair prices. This might be more evident in the case of SMEs and start-ups. Thus, the Exploit4InnoMat ecosystem, the SEP will undertake specific actions, targeting in facilitating the access to additional finance, including private and public funding, of such cases. Such cases will be posted in a dedicated page of the OITB's web platform, providing non-sensitive information in order to become visible in investing groups so that can come in direct contact with them. The Exploit4InnoMat will work on the following directions:

1. **Private funds** (Venture Capital Funds, Business Angels etc) that have interest to invest in the field of construction materials. The Exploit4InnoMat ecosystem will provide a safe and neutral environment for private investments. The 4 Lighthouses will have a significant role towards this direction as the clients will have the opportunity to showcase already validated technologies, minimizing thus, the risk of investments.
2. **Crowdfunding platforms** concerning innovative idea derived from SMEs and startups (e.g. Crowdcube, Seedrs, Kiss kiss bank bank, Companisto) as well as worldwide platforms (e.g. Indiegogo, RocketHub, Fundable, Ulele). During the project, these platforms will be contacted in order to establish collaboration with the Ecosystem. The target is to undertake the showcasing of the concepts need additional funding following the "all-or-nothing" funding model.
3. **Bank firms** established in Europe, targeting to facilitate the acquisition of low interest loans by the partners.
4. **National or European funding mechanisms**. Of particular importance could be considered funding schemes such us the SME instrument, FTI and regional projects dedicated in innovation.
5. Finally, Exploit4InnoMat OITB will act as a bridge between upstream and downstream industry. Large enterprise and end users can provide direct funding to SMEs and start up for accessing the ecosystem

In all above cases, the potential customers can take advantage of the services provided by the ecosystem concerning development of dedicated business plan, market analysis, showcasing videos etc.

Exploit4InnoMat will cover this need offering an open access innovation environment, in which SME will be able to proceed in technological innovation in a fast and low cost way. This decreased cost of innovation, compared to other existing alternatives (e.g. in-house research, customer-seller collaboration with individual RTOs) lays on the synergies that are developed inside the ecosystem, knowledge and good practice experience, as well as higher chances of commercialization and adoption of new products by the market due to the presence of important players from relevant

[9] Boccardelli, P. et al. (2018). Learning and Innovation in Hybrid Organizations: Strategic and Organizational Insights. 10.1007/978-3-319-62467-9.

fields (e.g. large construction companies). These advantages makes ***Exploit4InnoMat*** OITB extremely attractive for SMEs that are ready to innovate. The initial business plan presented below, shows the expected impact on European industry.

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EO 3: Support companies, especially SMEs, to become world leaders in clean products and technologies by setting up a new generation of Open Innovation test Beds;.

Impact Analysis:
SMEs play a fundamental role in investment and innovation in Europe. Despite macroeconomic challenges such as weak domestic demand, difficult access to finance and lingering economic uncertainty, they continue to be an essential part of the productive sector. Besides globalisation, SMEs face a number of other challenges such as increasing digitalisation, rising energy prices and competition from non-European, low labour cost countries, which require sensible investment and innovation strategies.

EXPLOIT4INNOMAT Contribution: The target of the Exploit4InnoMat proposal is to set up and operate a fully sustainable OITB related to the upscaling of materials and products for the building envelope. The Ecosystem will be represented by the SEP. The SEP will be a new legal entity (a Joint Venture Company) with limited liability as it will be involved in contracts and financial transaction and will have the mandate by the members of the OITB to sign contracts on behalf of them, while at the same time it will be the central point of an **Open Innovation Community** that will be created around the OITB. In this community, stakeholders from all aspects of the sector (SMEs & enterprises covering the supply chain, technology providers, policy makers, clusters, association etc) will be invited to participate, sharing ideas, concepts and providing expertise and IPRs, under specific conditions and agreements, targeting in boosting the innovation on the building sector.
A common commercial model for the OITB will be adopted, aiming to strengthen the inclusion of further technology providers and the exponential growth of the end-users. The SEP point concept will assist in consolidating the technology, connecting people and facilitating collaboration. The OITB aims to be open and to promote the collaboration among the stakeholders, in order to connect people and businesses with one another, beyond the borders of single European countries focusing on European SMEs without being limited to be used by European companies only. In order to have a sustainable OITB the focus will be given in two important issues.

EO 4: Translation of industrial needs into scientific problems and concrete solutions, increased awareness and uptake by industry, and effective access of relevant stakeholders to know-how and advanced tools/infrastructure;

Impact Analysis: The challenge that SMEs and start-ups face in terms of shorten their product development cycles has been addressed by the Lean Startup method, proposed by Ries. This method Ries proposes a combination of business-hypothesis-driven experimentation, iterative product releases, and what he calls "validated learning". Ries' overall claim is that if SMEs start-ups invest their time into iteratively building products or services to meet the needs of early customers, they can reduce the market risks and sidestep the need for large amounts of initial project funding and expensive product launches and failures. The main idea is that partner SMEs will develop large Scale Pilot Cases on the top of which beneficiary SMEs will create new products/services. Thus large scale demonstrators are an integral part of EXPLOIT4INNOMAT strongly interacting with other key activities of the project such as training, networking, crowdfunding campaigns and technology watch. **In that aspect the technology platforms are expected to act as the connecting tissue for large scale demonstrators, to attract beneficiary SMEs to the eco-system and provide them with the required technological infrastructure, enabling the development of market-ready products and services, without the need to re-invent the wheel.**

EXPLOIT4INNOMAT Contribution: Our approach is clearly influenced by Lean Startup method, since EXPLOIT4INNOMAT proposes a similar, iterative approach (from teams to products and from ideas to market validation), that aims exactly to accelerate the pace of launching new products services in the market. EXPLOIT4INNOMAT will have clear and measurable results as expressed by specific KPIs in the Project Objectives . Moreover, in medium term, EXPLOIT4INNOMAT will provide business support structures, including the matchmaking facility and reward/equity crowd-funding platform that are self-sustainable and ***ownership of these structures can easily be handed over to the ecosystem itself*** to ensure wider impact in the medium-term and sustainability of EXPLOIT4INNOMAT results.

2.1.2 External conditions influencing the expected impact: requirements & barriers

It is evident that although the last two decades **a high number of research projects has been dedicated in the development of new materials for the building envelope** (e.g. according to Scopus, more than 1000 publications concern this field in the last 5 years)**,** only a very limited number of such cases has been finally commercialized and reached the market. There are many reasons that these concepts have not been adopted by the market and this phenomenon is usually described as '*valley of death*'. Table 2.2 shows the plan of the ***Exploit4InnoMat*** ecosystem for **overcoming the upscaling and market adoption barriers** though its services that will be provided to SMEs and Large Industry.

Table 2.1: *Mitigation plans of **Exploit4InnoMat** for overcoming barriers for upscaling of materials concept*

Barrier 1: Lack of available infrastructures
Pilot Lines are of utmost importance for upscaling of innovative materials solutions. Utilizing such PLs, results derived from the lab can be transferred to a pre-industrial level and thus, develop robust production technologies. Unfortunately, such pilot lines are not always available or open to companies since their acquisition cost is high, needs extra expertise

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in monitoring and control systems, and SMEs are not eager to proceed in such investments unless they can decrease the risk. On the other hand, large industries, are not eager to stop their production lines to carry out tests for upscaling of new materials and products. ***Exploit4InnoMat*** *OITB* will provide a group of flexible pilot lines for upscaling as well as infrastructure for testing, monitoring and process optimization of production of the new materials covering the whole of the building envelope

Barrier 2: Need of expertise in characterization and evaluation of results
Fast and accurate characterization and testing of materials or components and evaluation of the corresponding results is essential for the development and upscaling of new products. Nevertheless, this expertise exists mainly in universities and research institutes, in which access might be proven of high cost if it is not a part of a scheduled innovative plan. ***Exploit4InnoMat*** experts will provide technological and development expertise; advance, accurate and fast characterization techniques for the innovative materials, including 3 lighthouses in which the components will be able to be tested in real conditions for adequate period of time, combined with advanced modelling techniques linking the crucial parameters of production steps with the final properties, lifetime and environmental impact of the new materials.
Barrier 3: Industry needs standardized and validated solutions
A lot of highly promising technical solutions often remain in a lab scale validation as they are not in compliance with standards and regulatory frameworks. Especially, when it comes to building materials where the non-uniform codes across Europe regulatory and standards are becoming a major barrier as the regulatory framework has not been yet assimilated by the industry. Thus, it is possible that end users will ignore products that are not harmonized with the latest standards. ***Exploit4InnoMat*** ecosystem will provide all the necessary services and activities related to standardization, compliance with existing regulatory framework and safety/nanosafety/standardization aspects of the solutions.
Barrier 4: High perceived risk for investors
In order to bring a promising idea to the market it is essential that additional funding sources are needed. Most often, SMEs cannot afford to invest such amounts on their own, while investors are reluctant to invest without securing the previous mentioned aspects (e.g. compliance with standards of the new materials, lifetime, performance). Inside the ***Exploit4InnoMat*** ecosystem proper conditions for bridging investors with SMEs will be provided. Thus, activities, such as translation of knowledge of discoveries developed usually in the academic community, in a context that is relevant to the industry investors will take place.
Barrier 5: Environmental and Energy burden
Building's material are in general energy-intensive materials in which energy costs and associated emissions are major factors. Energy costs improvement is considered one of the main drivers for efficiency and potential improvements. Likewise, emission savings could also benefit from the improvements in the energy efficiency i.e. through heat recovery. In the Exploit4InnoMat OITB a comprehensive Lifecycle approach will be provided. Therefore, the contribution of the products developed in terms of resource and energy efficiency can be assessed and improve within the OITB and especially in the dedicated Lighthouses and demonstration sites. The main barriers for the decarbonization of the building envelope's material industry have been identified as: i. Long investment cycles (given materials and components life of up to 50 years) [10] and high capital costs of new technologies, ii. Low margins affecting ability to invest and limiting the intention for innovations iii. Increasing concern about the security of energy supply where an interruption can cause major damage taking months to repair, iv. Threat of rising energy prices, v. Requirement for very high rates of return or short payback time on all projects including energy efficiency, vi. Shortage of proven, financially viable and demonstrated energy-efficiency technologies, vii. Lack of government support (e.g. financial) for sector research, development and demonstration, viii. Regulatory uncertainty on energy, climate, environmental and innovation issues

2.1.3 Scale and significance of the project's contribution to the expected outcomes and impacts

The European construction industry is expected to record a CAGR of 4.4% to reach 2,567.7 billion € by 2023. The residential construction industry in value terms increased at a CAGR of 3.8% during 2014-2018. The commercial building construction market in value terms is expected to record a CAGR of 3.6% until 2023, while infrastructure construction was estimated to be 558.3 billion € in 2018, posting a CAGR of 1.7% during review period. Specifically, the Building Envelope market is competitive since the industry has a range of elite players as well as new entrants. This has also resulted in tremendous fragmentation. The number of new players joining the Building Envelope market is at an all-time high which is making the market highly competitive. They leverage the use of convergence, buy, partnership, product launch, creativity and other approaches as strategic moves. On the other hand, the recent outbreak of COVID-19 pandemic is having a massive impact on construction projects, causing delays and disruption, in some countries pause of works, as well as new legislations and standards. [11] Historically, after economic crisis, the R&D funding is increased, as innovation is considered a secure way of development. [12] In the following a draft market search study is provided related to the Exploit4InnoMat activities. Besides the market segmentation on the field of materials for the building envelope, it is important to have insights on the R&D intensity of the sector, as the Exploit4InnoMat OITB will target this specific sector. Compared to the economic significance of the construction industry, R&D investments are quite small. Construction industry is on the lowest category

[10] A. Grant "Impact of building service life models on life cycle assessment", Building Research and Information 41 (2), 2013
[11] N. Bouchardie, I. Madalena, "COVID-19: the current impact on construction and engineering projects" Whitecase, April 2020
[12] J. Rehm "Ten years after the economic crash, R&D funding is better than ever" Nature, September 2018

with less than 1% of net sales to be used for R&D. Many construction business owners and project managers can be quite creative when it comes to solving construction-related problems. However, getting involved in an R&D project or process is not an easy decision. Many practitioners think that research is too distant from the everyday to yield any business benefits. There are also structural reasons for the low interest in R&D. The industry is comprised of large supply networks, which means that one company's development efforts do not make much difference if other players do not comply. Client behavior is not encouraging R&D investments either. Most contractors claim that when it comes to choosing vendors, price, not value, is the defining factor. However, that may be based on the fact that the price is actually the only differentiating factor between vendors. [13]

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Exploit4InnoMat OITB aspire to overcome this trend and be positioned in the R&D reality of the construction sector due to its value proposition. The OITB will cover aspects of materials upscaling and validating across the supply chain of the construction industry and thus, will attract the interest of innovative companies.

Financial projections:

The target market of the Exploit4InnoMat OITB are companies in the field of construction and more specifically in companies related to materials for the building envelope. The ecosystem will cover part of the R&D needs of such companies focusing on the upscaling and validation of new materials concepts. Exploit4InnoMat Ecosystem concept is in full accordance with the R&D&I reality in the field of materials for the building envelope. As already depicted in the SWOT-PEST analysis, various factors (environmental, financial, regulatory) pushes towards the use of more efficient and environmental friendly materials. This shift, has created a high interest from both SMEs and larger enterprise to adopt such concepts in their portfolio. Thus, Exploit4InnoMat ecosystem will attract the interest of such companies and other stakeholders. The consortium has set the **following KPIs for a period of 5 years after the project completion** concerning the express of interest from possible customers of Exploit4InnoMat OITB. These numbers will be achieved by the targeted marketing activities that will be undertaken by the members of the OITB as well as based on the current clientele of the core members of the ecosystem

Table 2.3: Exploitation KPIs (5 years after the project end) – contract agreements

Description	# achieved
Contract agreements with SMEs in the relevant fields for access in the OITB	~250
Contract agreements with L.E. in the relevant fields for access in the OITB	50
Contract agreements with SMEs and L.E for ***Exploit4InnoMat*** technologies transfer	30
Associated members (by year 5) of the ***Exploit4InnoMat*** open innovation community	500
Events carried out by the Ecosystem (conferences, workshops, training etc)	15

Based on the above KPIs, the Table 2.5 presents the financial projections for the ***Exploit4InnoMat*** ecosystem for a period of 5 years after the finalization of the project. The main clients of the OITB will be materials and components producers, constructors and other stakeholders (e.g. research institutes etc). The following assumptions/estimations have been taking into account for the set up of the table. The presented costs are based on the experience of partners in pricing industrial projects in their entities.

- The average cost of a test case concerning SMEs is ~150.000 €. This cost concerns the access to the PLs and provision of the additional services and corresponds to a duration of ~6 months.
- The average cost for a test case concerning L.Es is ~250.000€ and corresponds to a duration of 6-9 months.
- Patent licensing or technology transfer average revenue is calculated to 200.000 € per case
- Membership fees for the community per year is set at 2.500€
- An average revenue per organized event is set at 60.000€ including sponshorships. It should be mentioned that although the events organized by the Ecosystem will not have a significant contribution in the overall revenues, they are considered as of high importance due to their significant role in the dissemination and networking activities.

Table 2.4: Exploit4InnoMat Ecosystem financial projections

	2025		*2026*		*2027*		*2028*		*2029*	
Revenue stream	#	**Revenue**	#	**Revenue**	#	**Revenue**	#	**Revenue**	#	**Revenue**
Contracts SMEs	15	2.250.000 €	24	3.600.000 €	30	4.500.000 €	38	5.700.000 €	45	6.750.000 €
Contracts Les	6	1.500.000 €	12	3.000.000 €	17	4.250.000 €	25	6.250.000 €	30	7.500.000 €
Patent licensing	2	400.000 €	5	1.000.000 €	9	1.800.000 €	13	2.600.000 €	16	3.200.000 €
Community fees	150	375.000 €	220	550.000 €	300	750.000 €	400	1.000.000 €	500	1.250.000 €
Events fees	4	240.000,00 €	4	240.000 €	5	300.000 €	5	300.000 €	5	300.000 €
Total Revenues		4.765.000 €		8.390.000 €		11.600.000 €		15.850.000 €		19.000.000 €
Cumulative revenues		4.765.000 €		13.155.000 €		24.755.000 €		40.605.000 €		59.605.000 €

The ***Exploit4InnoMat*** ecosystem targets to meet the requirement set by the EC for creating revenues being 4 times the funding provided to the project by the EC. The average scenario of sales as presented in the previous table, project a total cumulative revenue of 60 million €. Taking into account that a new job is created for every 200.000€ of additional revenue it is expected that ~**300 new jobs will be created within the Ecosystem, including also the NPO**. However, the impact on jobs creation will be much higher by including also the **new jobs positions concerning the clients of the Ecosystem**. With a moderate estimation of creation of **3 new jobs for each of the clients entering the Ecosystem** (due to their business growth attributed in novel and competitive products they will develop) a total of additional ~**730 new jobs** will be created.

[13] A. Heiskanen, "Is it time to invest in Construction R&D?", AEC-Business, February 2014

2.2 Measures to maximise the impact – Dissemination, exploitation and communication

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To maximize the exploitation and the impact of the project results, the consortium is committed to following a clear and comprehensive innovation management and exploitation strategy supported by a focused dissemination and communication plan. While communication activities will focus on raising awareness regarding project work and results, and will be directed to a broad audience, dissemination will target specific stakeholders to enable the use and uptake of project results and be closely linked to the exploitation and sustainability strategy. This facilitates not only the commercialization of the project results, but also the potential knowledge and technology transfer to interested stakeholders.

2.2.1 Exploit4InnoMat Stakeholders

Dissemination activities will begin in the early stages of the project (M1). Already, the consortium has worked on designing a detailed dissemination plan, that will be further elaborated on and advanced in order to successfully reach the largest groups possible with the aim of inform EU stakeholders about the BUILDING SHIELD project's objectives and results. Dissemination activities will aim at paving the way towards the future market penetration of the BUILDING SHIELD materials and technologies and to ensure the successful exploitation of the project outputs (see Draft Business plan). Moreover, the consortium members will promote the project's approach and encourage interested stakeholders to adopt the solutions developed.

For an effective public outreach, both standard dissemination activities dedicated to public information, and more targeted engagement activities- where dialogue (communication) and alliances are created-, dynamically co-exist. The methodology includes (a) the definition of the subject to be disseminated, (b) the identification of the target audiences that will be interested in, or could impact from project results and findings, (c) the definition of the most effectives ways (methods, channels and tools) to reach the audience, (d) the most appropriate time to disseminate, and (e) the dissemination policy and rules.

Table 2.5. Dissemination outline and strategy

Item	Description
Target Audiences	Specialized audience (scientific and technical); general public; owners of buildings, public services, engineers; constructors, architects, technology industry; policy makers, regulatory & standardization bodies; stakeholders from the value chain
Key messages	- Technical Community: Innovation Cycle for a specialized manner (public authorities, building and ICT experts, Utilities providers, and other stakeholders) - Non-Technical Community: the language has to be able to inform about aims, objectives and activities of the project to non-professionals (citizens, householders, ICT/Building technology SMEs, other).
Dissemination channels	Scientific publications in peer review journals; publications in technical journals (e.g. building materials, energy efficiency, constructors); engagement activities; Conferences; social networks; school classes, press releases; poster presentations in conferences; marketing material; videos; e-newsletters.

2.2.2 Exploitation of project's results

Exploit4InnoMat Virtual Community/Ecosystem "Materials" Helix: An Impact Accelerator

The already existing vibrant, international, virtual community/ecosystem, named the "Materials helix", will be built during the Exploit4InnoMat project. The Materials Helix will be an engine for impact, exploitation, clustering and dissemination activities.

The Materials Helix: The Materials Helix provides an Open Innovation platform for experts seeking international collaborators to carry out research and innovation across the following topic areas:

- Nanomaterials, including additives, encapsulation, photonics, surfaces, & coatings
- Nanofabrication (including workers' exposure safety)
- Graphene
- Biomaterials including synthetic biology, industrial biotech, & bioinformatics
- Materials modelling & characterisation
- Smart materials including catalysis, environmental treatment, energy transport and storage
- Material from Cradle to Cradle (including Bio-based, Re-use, & Recycling)

The Crowdhelix Network: Exploit4InnoMat consortium will benefit from the CHX network that gathers **500+ member organisations** from **51 countries** (all EU member states). It has the potential to reach **8,000+ research and innovation collaborators** from its existing membership base. This network, supported by a customised technology platform, language processing and machine learning tools, has **40** virtual thematic areas/clusters (called "Helixes"), that cover all the research and innovation fields.

Building the Materials helix: Through the helix, the consortium will create a network of at least **150** organisations (M36), including: **Exploit4InnoMat partners**, members from existing **relevant Helixes** (i.e., *Nature Building*, *Raw Minerals*, *Manufacturing*, *Digital*, *Smart Cities*, *Circular Plastics*, *Mission Climate*, *Energy* helixes, etc.), **researchers** (i.e. *experts in circular economy, innovative materials, sustainable manufacturing*), **industry actors** (i.e. *Building companies, SMEs, world leaders in clean products manufacturing, technology providers*), **ethic and policy makers** (i.e. national and regional agencies and authorities), **civil society** (end users), participants of **other relevant projects** to promote and develop clustering activities (i.e. *sister projects and other relevant EU projects and initiatives "details in section 1.2"*), **investors**

(i.e. InvestEU, Single Market programme) and **relevant actors/associations** across **Europe/International** (i.e. European Institute of Innovation and Technology (EIT) Manufacturing, EIT Digital, EIT Raw Materials, Smart Specialisation Platform for Industrial Modernisation (S3P-Industry, S3 Partnership on Sustainable Buildings, Cluster Eco Construction, European Builders Confederation (EBC) etc.). We will integrate policy-makers and public administrations to provide **feedback to policy measures**. CHX will organise 2 helix events to promote collaboration between the Materials helix members.

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Project sustainability engine: To ensure **sustainability,** CHX will commit to maintaining the Helix for at least **2 years after the project** has concluded. When the project ends, the Helix will continue to be hosted on Crowdhelix's platform, and everyone signed during the project up will be able to retain their **accounts for free** (as external organisations), as well as have the option for their organisations to join as full members on the network. The CHX Helix Manager will identify a member, from the Helix, that would be interested in strategically leading this Helix after the end of the project. The new helix will be supported by the CHX team. The activities performed in the Helix are shown in figure X.

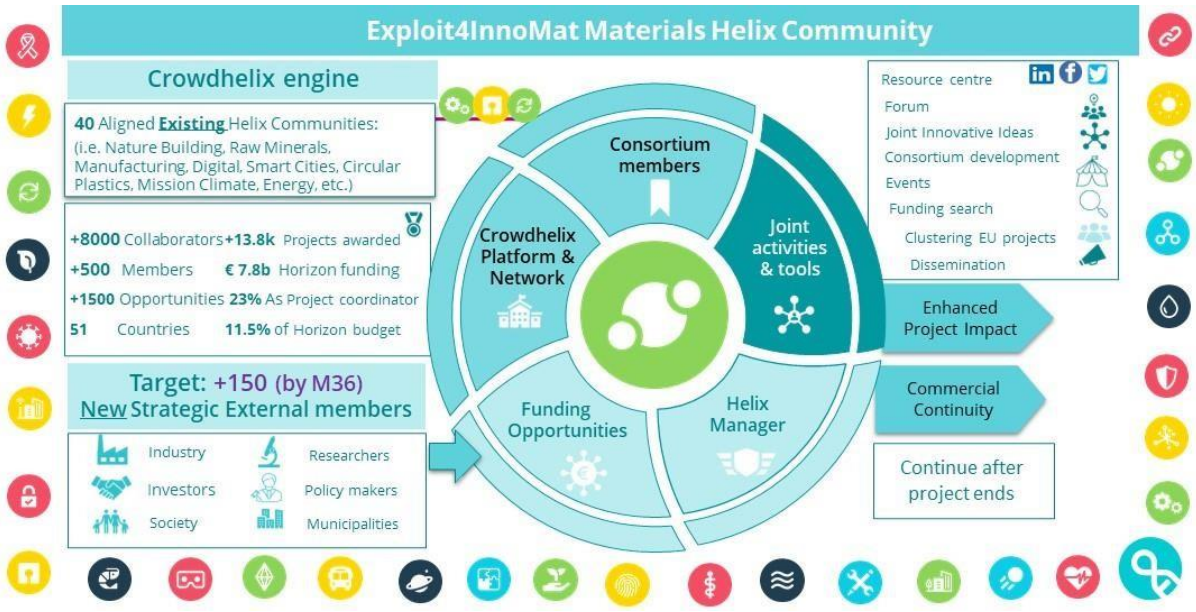

Figure 2.2. Exploit4InnoMat Materials Helix Ecosystem: Virtual Community

Exploit4InnoMat Clustering plan

Exploit4InnoMat cross-projects collaboration plan is divided in 3 main sections: Collaboration board (S1), Joint activities (S2), and Future collaboration plan (S3).

In Section 1. Collaboration board, the projects with similar aims and synergies will be identified (**1.1**), including: a) the projects funded under the same call (sister projects), b) other European projects and c) national and regional projects. Exploit4InnoMat will collaborate with relevant European projects (*details in section 1.2*) and **relevant actors/associations** across **Europe/International** (i.e., EIT Manufacturing, EIT Digital, EIT Raw Materials, S3P-Industry, S3 Partnership on Sustainable Buildings, Cluster Eco Construction, EBC etc.). All the projects and key stakeholders will be contacted (**1.2**) and the Collaboration Board (CB) will be created (**1.3**). The CB will define the Key Performance Indicators (KPIs) (**1.4**) and the main ways of a fluent and frequent communication (**1.5**). A common identity (including logo and templates for communication and reporting) will be developed (**1.6**).

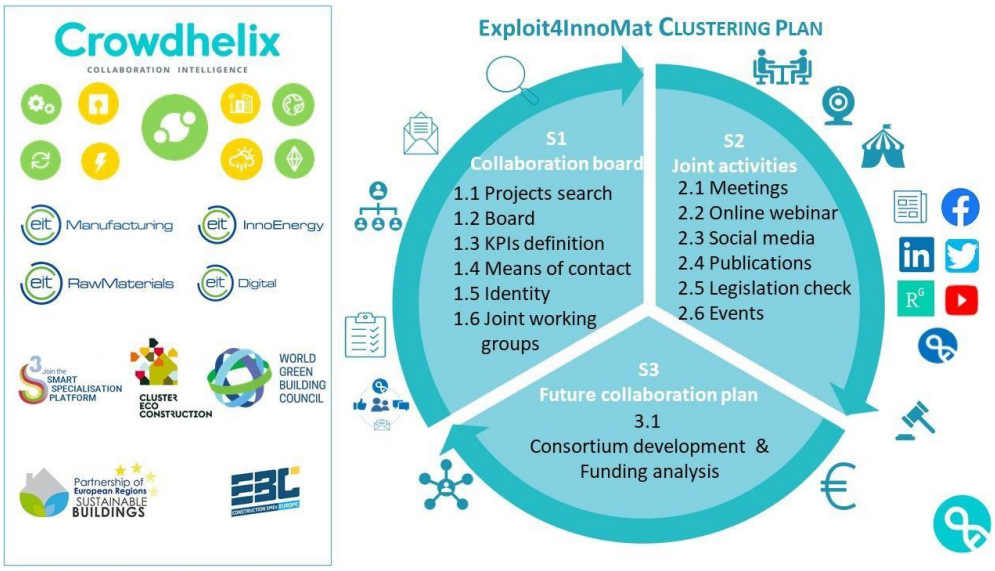

Fig. 2.3 1: Exploit4InnoMat Clustering plan

In Section 2. Joint activities, the Exploit4InnoMat CB members will *promote and encourage* collaboration activities with the different projects identified in Section 1 (mainly with the sister projects). These activities include: regular meetings (sister projects will be invited to Exploit4InnoMat Kick Off meeting) (collaborative ideas and work are expected to emerge thanks to these meetings/workshops) (**2.1**), joint participation and attendance to webinars and local info days (organised by projects) (**2.2**), active participation in projects social media sites (Facebook, Twitter, LinkedIn, and similar tools) (**2.3**), development of joint scientific publications (**2.4**), supporting the development of future legislation around Climate Neutral

and Circular Innovative Materials Technologies, specially for Sustainable Building (**2.5**) and joint participation in scientific conferences and business tradeshows (**2.6**). Finally, in Section 3 Future collaboration plan, a virtual workshop will be organised to promote the development of interdisciplinary research and innovation ideas between the projects, to develop new consortiums, and to identify target calls aligned with the consortium strengths. The **Materials Helix** will be an engine of clustering activities.

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2.2.2.1 The Exploit4InnoMat Business Case

Exploit4InnoMat value proposition is the **operation of a new, OITB (Ecosystem) targeting in the upscaling of solutions (materials and components) for the building envelope that will serve the near Zero energy targets set by the EC**. The following figure presents the CANVAS business model of the Exploit4InnoMat OITB.

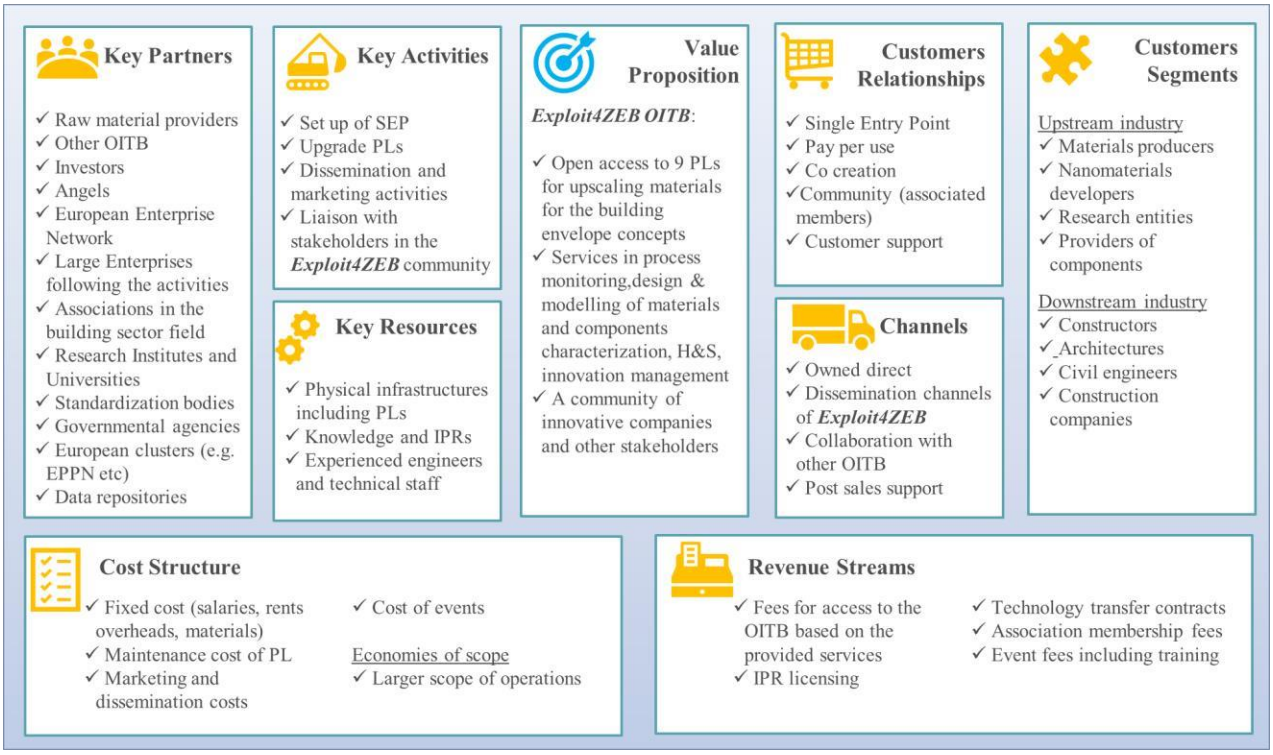

Figure 2.4: *Exploit4InnoMat CANVAS business model*

The Exploit4InnoMat is based on the collaboration of high expertise groups across Europe in the fields of construction materials, and manufacturing, monitoring, testing, characterization, modelling, environmental assessment, standardization, nanosafety, innovation management and training. The ***Exploit4InnoMat*** **OITB will provide innovative services to SMEs, midcaps, large industries and academia** interested in developing new applications and products for the construction industry, focusing on building envelope, by means of improving their functional properties (e.g insulation, durability, energy harvesting or energy storage), reducing their environmental footprint (e.g. by utilizing CDW as secondary raw materials or developing higher efficiency production processes).

2.2.2.2 SMEs engagement focus

Exploit4InnoMat has a strong SME participation in the project and a commitment to promote use of open-source components by innovative SMEs and to encourage third party SME open-source products in the Open EXPLOIT4INNOMAT Framework.

2.2.3 Dissemination and communication of results

Communication activities will be oriented to raise the awareness and outreach of the project activities to the general public, while dissemination will focus on enabling the use and uptake of the project results, by addressing all potential stakeholders detailed in Section 2.2.1.

Exploit4InnoMat is aligned to the Rome Declaration on Responsible Research and Innovation in Europe[14] and is fully committed to the EC DG Research and Innovation's main objective of fostering public engagement and sustained two-way dialogue between science and the civil society. Moreover, general public awareness and acceptance of theiractive role in the renewable energy markets is crucial for the success of the energy transition.

The communication plan will be consistently implemented throughout the whole lifetime of the project and will focus on two main areas and objectives:

- **Internal communication management:** The main objective is to keep a fluid interaction between project partners and to establish transparent and accessible information sharing mechanisms. Communication between different partner subgroups to work on common goals or tasks will always be performed in an orderly and detailed way, making use of some of the many technological tools for teleworking, and following a clear methodology to

[14] *See participant portal FAQ on how to address communication activities in Horizon 2020*

establish at all times how work must be done together. Communication between partners will be supported mainly by telephone and videoconferences, email and face-to-face meetings. Besides, a restrictedaccess area of the project's website will allow project documents sharing and progress status monitoring. The project coordinator will be responsible for launching and keeping this private area updated.

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- **External communication activities:** Communicating relevant knowledge about the project's latest activities and achievements to the relevant identified **Exploit4InnoMat** target audiences is certainly a way to keep all partners actively involved in the project and the **Exploit4InnoMat** consortium harmoniously acknowledges that communication is not something that should be seen as a side task; on the contrary, it is one of the topmost aspects, priorities and prerequisites of running a successful research project. In this context, the draft communication plan for promoting this project and its results, includes an efficient and effective mix of both interpersonal and mass communication tools. Moreover, it shall be emphasized, that for each targeted audience,a distinct strategy using targeted messages, means and language has been planned. Namely, specific audiences and communication methods have been defined for promoting **Exploit4InnoMat** and its main results:
    - Direct proactive communications and physical demonstration: *Attendance at seminars and conferences,one-to-one communication, e-mailing stakeholders, periodic newsletters, etc.;*
    - Clustering activities: *Contacting parallel related projects, cross-field events;*
    - Mass and general communication: *Project website, press releases, posters and leaflets;*
    - Scientific excellence: *Publications in highly ranked journals, presentations in conferences;*
    - Academic: *Conveyance of the new knowledge into University curricula, publications of MSc and PhD theses.*

Dissemination will focus on describing and assuring results available to others to use which is essential for realising EXPLOIT4INNOMAT impact objectives. In order to efficiently and effectively communicate at an early stage, a dissemination and communication plan will be elaborated at the beginning of the project. This plan will be updated throughout the project and include the full details of the following important communication aspects:

1. Description of the target group of each communication activity.
2. Formulating the key milestones that will give rise to dissemination of information.
3. Determining the content of the communication / channels in relation to the target groups.
4. Description of the details of the materials to be used for distribution (e.g., seminar, journal article, blog, newsletter), including the communication rules as established by HE.
5. Planning of project events.
6. Participation in external events.
7. Six monthly evaluation and review of the plan.
8. Further information (project information, meetings, newsletters…) of the EXPLOIT4INNOMAT project will be posted at project and partner's websites, social media (LinkedIn, Twitter, YouTube).

1.1.1.1 Communication and dissemination activities and KPIs

Objective	C&D channels	Actions	KPIs
Create a business-oriented image of the project and raise awareness of the benefits of the EXPLOIT4INNOMAT project within the targeted communities and the general public	Corporate identity	Development of the overall project design, templates, presentations, etc.	Project design completed and templates available to partners M1
	Project website	Website creation and continuous update	EXPLOIT4INNOMAT website online – 500 unique visits/month expected by the end of the project
	Social media	LinkedIn and Twitter presence	By the end of the project:.300 contacts in LinkedIn 200 followers in Twitter
	Public dissemination materials	Creation of brochures, posters, white papers, footage/videos, etc. Scientific publications	At least: 1 project brochure, 1 poster/rollup per partner, 1 white paper 2 project videos, 4 scientific publications
	Press material	Creation of press-releases templates, writing press releases and articles	At least: 5 press releases, 5 articles in magazines
	Events	Stakeholder workshops with CI operators and institutional stakeholders	
		EXPLOIT4INNOMAT Conference, society in general (local, regional and national institutions, entrepreneurs)	Self-organised: 1 conference in Europe attended by 150+ participants M18 including video recordings for keynote speakers
To disseminate and promote project activities among targeted stakeholders	Online marketing	Website search engine optimization, for community building, use web analytics	Continuous work, at least 2 social media posts per week
	Newsletters	Regular publishing a dedicated newsletter and contributions to external newsletters	Newsletters: once every quarter 100 registered receivers by the project end
	Press campaigns	Press releases in diverse media; print, radio, press, TV…	At least 5 press releases

	Conferences	Participation in selected conferences	At least 10 presentations in conferences
Liaison activities	Workshops and roundtables	Co-organization of workshops and roundtables	At least 4 events

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2.2.4 *Intellectual Property management*

EXPLOIT4INNOMAT responds to a recent imperative from the EC in boosting patent protection of strategic and commercially sensitive innovation from EC funded R&I projects [15] [16] [17] [18] (Figure 2.4). The EC's imperative is evidenced in recent EC communications, as well as recently enacted EC tenders that promote the objective of boosting the commercialisation of EU research results[19] [20] [21]. The Commission underline protection of IP as crucial for the EU's ability to stimulate innovation and to compete in the global economy and advocate that formal IP protection is one of the principal means through which companies, creators and inventors generate returns on their investment in knowledge and creation. The Commission's impetus is further endorsed by EC studies that estimate that IP-intensive activities account for around 42% of EU GDP (circa EUR 5.7 trillion annually), generating 38 % of all jobs, and contributing to circa 90% of EU exports[22].

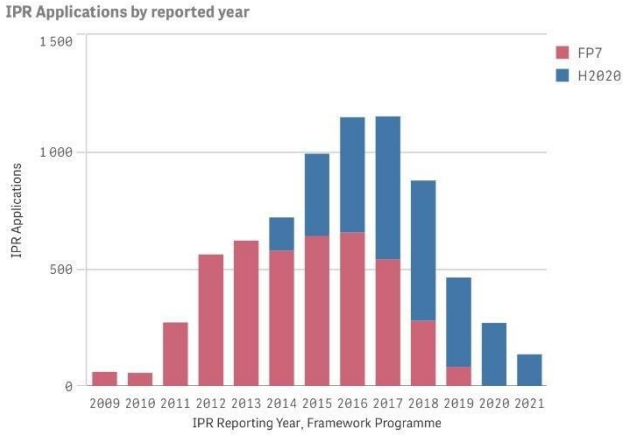

Fig. 2.4 - Declining Patent Trend in EC Funded R&I

Exploit4InnoMat has strategically important and commercially sensitive innovation with strong licencing and commercialisation potential. The project aligns with the EC's strategic IP vision for Europe and commits to filing 2 Patents with the ultimate goal of protecting and retaining the commercially strategic/sensitive IP to the benefit of the consortium members, in turn giving the project partners a competitive advantage on a world-wide stage. This will be led by ICP, whose IPR leader (Dr. Pat O'Sullivan) is cited in Wikipedia's list of worldwide lead inventors with over 370 successfully granted patents in his name and thus intimately familiar with the path to success[23]. **Exploit4InnoMat** also *recognises the need to find a balance towards open sourcing aspects of the project's R&D, to allow other EU researchers to build on and from the R&D created in the spirit of continued progress of EC funded initiatives, as well as in the spirit of open dissemination and sharing across EU member states.* Hence, the project commits to significant parts of the project being open-sourced, a full implementation of the EC's FAIR data principles[24], as well as adhering to EC guidance on open access conferences and journals[25], thus enabling continuity and sustainability of the projects results, advancements and strategic direction in support of continued research excellence in Europe.

Knowledge management

The management of knowledge and IPR will be governed by the Consortium Agreement (CA) which will implement and complete Grant Agreement regulations defining the distribution of EU funds and Intellectual Property Rights protection. The CA will be signed by all partners before the start of the project and will implement the following preliminary agreements: Background knowledge to the purposes of the project will be specified in the CA and will be made available to consortium members on a royalty-free basis where necessary to perform project activities. Foreground knowledge is owned by the partner generating such information or result. Each partner will make their foreground knowledge available, on a royalty-free basis, to other partners to the extent that such information is necessary to produce their own foreground results within Exploit4InnoMat. If it is not possible to precisely determine the ownership of new foreground knowledge generated, i.e., several partners participated in the specific development, ownership will be shared among the contributing contractors in equal parts by the pro ratio effort invested by each partner. Background and foreground knowledge will be made available: (*i*) to the other project partners for dissemination, research, demonstration, interoperability of software and academic purposes in respect to the intellectual property rights of the partner generating this knowledge on a royalty-free basis; (*ii*) to the other project partners for exploitation purposes under fair and reasonable conditions, and with respect to the normal commercial conditions applied by the granting partner. The CA will also make explicit reference to important administrative points such as decision procedures within the project, risk management strategies, legal aspects regarding software to be used/produced in the project, trademarks, patents and rights of each partner in the exploitation of results.

[15] https://ec.europa.eu/growth/industry/intellectual-property_en
[16] https://ec.europa.eu/docsroom/documents/26581
[17] http://ec.europa.eu/trade/policy/accessing-markets/intellectual-property/
[18] https://ec.europa.eu/docsroom/documents/26583
[19] EASME/H2020/2018/008 - https://etendering.ted.europa.eu/cft/cft-display.html?cftId=3518
[20] 2018/RTD/B3/OP/PP-7141-2018 - https://etendering.ted.europa.eu/cft/cft-display.html?cftId=3655
[21] JRC/SVQ/2018/B.6/0024/OC - https://ted.europa.eu/udl?uri=TED:NOTICE:354602-2018:TEXT:EN:HTML
[22] EUIPO, Intellectual property rights intensive industries and economic performance in the EU, 2016
[23] https://en.wikipedia.org/wiki/List_of_prolific_inventors
[24] https://ec.europa.eu/info/sites/info/files/turning_fair_into_reality_0.pdf

[25] http://ec.europa.eu/research/openscience/index.cfm?pg=openaccess

2.2.5 Summary of key elements of the impact section

SPECIFIC NEEDS	EXPECTED RESULTS	D & E & C MEASURES
a) Engineers & Technical staff (Trained People at Bio Based materials, constructors; b) Developers Programmers; c) Financial Resources; d) Existing IPR (Patent ownership rights); f) Physical Infrastructure including PLs; g) Technical and financial capability; h) Partner municipalities in Bulgaria Greece, Italy, Latvia, Romania and Israel. **EXPLOIT4INNOMAT** partners with strong links in the target sectors, Effective dissemination and marketing plans and strong presence in the EU R&D community, EU client portfolio	**a) Materials and Components: 1)** Minerals Aerogels; 2) Hybrid PCM's; 3) Mortats, Eco Concrete & Geopolymers; 4) Ecological Multifunctional Coatings; 5) PVT Systems; 6) Bio Based Panels; 7) Thermoreflective, insulating, ionizing external plasters and finishing; 8) Gypsum based indoor insulating plaster and skims; 9) Prefabricated modular panels consist of a timber frame structure; 10) Smart Glazing **b) EXPLOIT4INNOMAT Platform.** **c) Fast 3D Scanning methods: 1)** Services in process monitoring, design & modelling of materials and components characterization; 2) A community of innovative companies and other stakeholders. 3) Open Access to 9 PL's for upscaling materials for the building	a) Website and social media. b) Articles in academic journals and key magazines, Periodic newsletter, Direct contact with key stakeholders using workshops and Training, Presence at key international conferences. c) Presentation of project results at local and internationals conferences. d) Dissemination towards the industry: Recommendations white book for industries, scientific publications, yearly Helix event, Final conference.

TARGET GROUPS	OUTCOMES	IMPACTS
a) Raw Material Providers. b) Sustainable Material and components Companies. c) Constructors. d) Architects& Civile Engineers. e) Large Enterprises following the activities. f) Investors. g) European Enterprise Network. h) Association in the building sector field. i) Standardization Bodies. j) European Clusters. k) Governmental Agencies. l) Customer Segments. m) Upstream Industry. n) Materials Producers. o) Nanomaterial's developers. p) Research Entities. r) Providers of components. s) Downstream Industry. t) Constructors; s) Architectures; u) Civil Engineers. v) Construction Companies	High efficiency in Construction processes with high potential productivity improvements, Potential annuallabor, raw material, energy and environmental savings,estimated payback, Efficient on-site technologies and applications, Modular and flexible installations and maintenance, Increase in high-end technology, Durability and long service lifetime, Low installation and operation cost, Online Capacity Building and Training to labor force.**Adoption in small- and medium-sized manufacturing:** >5 European SMEs start adopting the toolkit and recommendations demonstrated by project end. **Increased awareness and interest** for the principles demonstrated during the project. **Develop policies** incentivizing national, regional and EU for wood industry end to end chain as well as in other industries. **Increased awareness of and demand for sustainable products and services**.	**Scientific:** New breakthrough discoveries on: 1. Methodology for the material service life assessment 2. Novel precise indoor hybrid positioning system based on enhanced fusion methods to obtain cm-level accuracy; 3. Self-organized ONN with the generative neuron model that can optimize different operations for training a network with less computational complexity 4. Strong technological and upscaling capacity for providing solutions and fast prototyping 5. New smart materials value chains, including development and commercialization 6. Cross sectoral market approach **Economic:** a) Consultancy Services to Construction Industry; b) Direct Sales of building Materials c) Registration fees for the use of platform by the Bio Based Material Providers; d) Pay per use of the **EXPLOIT4INNOMAT** Platform; e) Technology Transfer Contracts; f) Association Membership fees; g) Event fees including training **Environmental:** a) Decreased wood waste by 50%; b) 100% Product lifecycle traceability **Economic/Technological:** EUR 150 MLN in the project lifetime for each piloting enterprise (10% increase from current revenue) and EUR 3 BLN generated by project end across enterprises adopting the toolkit. New markets for wood modelling tools, DTs, and reusable temporary houses enabled first in pilot countries, then beyond. **Societal:** Lower climate impact of manufacturing (including wood material sourcing and wood waste management). Emergence of a new customizable reusable wooden houses market.

3 Quality and efficiency of the implementation

3.1 Work plan — Work packages, deliverables

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The partners have already discussed and agreed about the management structure and the necessary procedures for the successful implementation of the Explot4InnoMat project. In line with the technological/marked/innovation objectives that Explot4InnoMat has to achieve in this 48 months programme the work to be accomplished has been structured in 14 Work packages (Figure 3.1), each of which has been sub-divided into a number of tasks, which are in details listed in Figure 3.2 (Gant Chart). Each of the work packages will be led and managed by a project partner who is the expert in the relevant field. The whole development is targeting on demonstrating the technology in operational environment and supported by targeted innovation activities. The objectives are planned to be in complete alliance with the issues raised by the topic CL4-2022 and will be implemented in a reasonable, scalable and cost-effective manner, justified by integrated production methods. Appropriate synergies and collaboration are scheduled between the WPs. Feedback will be provided from the test cases WPs in order to continuously optimize the Ecosystem operations and tools. Due and monitoring of the progress of the project, as in detailed described in paragraph 3.2. Figure 3.1 depicts to the high total number of the WPs appropriate measurements have been taken in order to ensure the proper management the interconnections between the work packages and tasks of the Explot4InnoMat project.

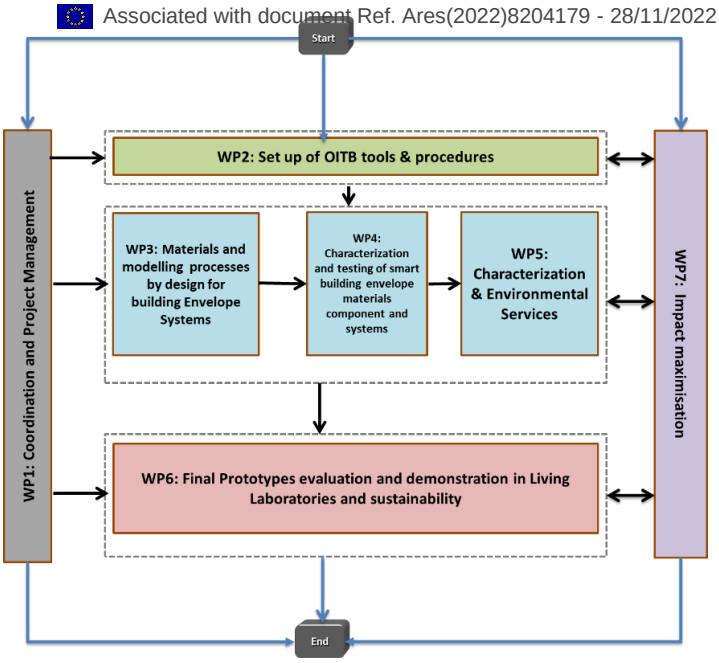


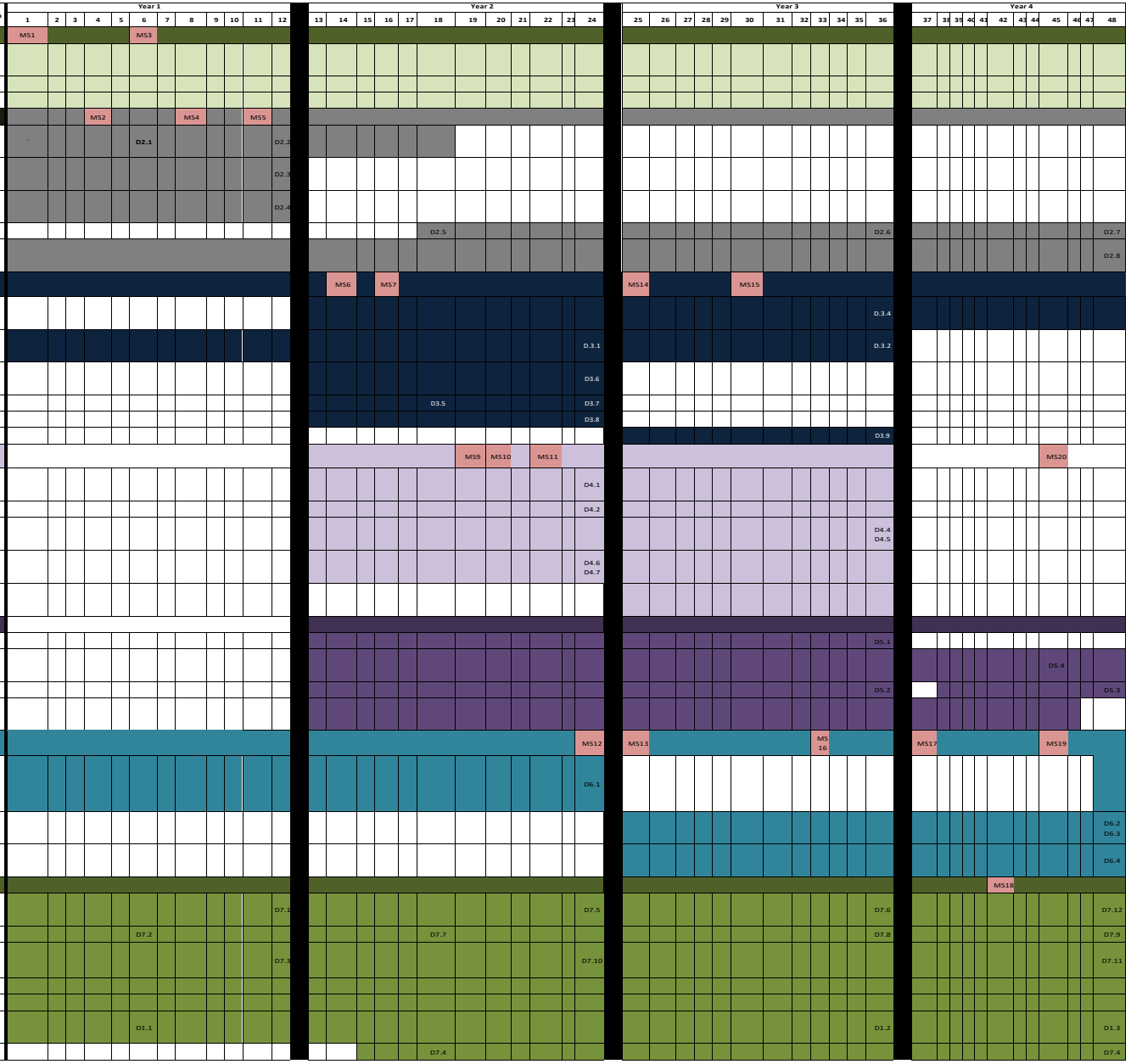
Fig. 3.1 2: ***Explot4InnoMat Pert Diagram***

Table 3.1a: *List of work packages*

WP No	*WP title*	*Lead partner No*	*Lead partner*	*Person months*	*Start month*	*End month*
1	Coordination & Management	1	UNIS		1	48
2	Set up of OITB tools & procedures	11	STAM		1	48
3	Materials and modelling processes by design for building Envelope Systems	5	RISE		13	48
4	Characterization and testing of smart building envelope materials component and systems	4	NTNU		13	36
5	Characterization & Environmental Services	20	UNIVPM		13	48
6	Final Prototypes evaluation and demonstration in Living Laboratories and sustainability	3	NTUA		1	48
7	Impact maximisation	1	UNIS		1	48
			Total			

Figure 3.2: *Gantt chart of Explot4InnoMat project*

ID	WPs and Tasks	Leader	Contributors	Start Month	End Month
WP1	Coordination & Management	UNI	ICCS, all	1	48
T1.1	Administrative and financial project management	UNI	all partners	1	48
T1.2	Technical project management	ICCS	UNI, all partners	1	48
T1.3	Quality Assurance and risk management	UNI	all partners	1	48
WP2	Set up of OITB tools & procedures	STAM		1	48
T2.1	Set up of the OITB administrative, operational tools and procedures	F6S	all core ecosystem partners	1	18
T2.2	Upgrade and modification of pilot lines	STAM	IRIS, Pilot Line owners	1	12
T2.3	Preparatory phase for the test cases	NTUA	SOLI, UNIS,FORTH, PL owners	1	12
T2.4	Fine tuning of the OITB tools & procedures	F6S	all partners	18	48
T2.5	'Standardization & Regulatory services	UNIST	all core OITB partners	1	48
WP3	Materials and modelling processes by design for building Envelope Systems	RISE		1	48
T3.1	Multiphysics and multiscale materials modelling	AMS	USTUTT, end users, PL owners	13	48
T3.2	Process modelling services and Building Information Modelling Services	STAM	PL owners	1	36
T3.3	Multifunctional coatings for building envelope	RISE	KANS	13	24
T3.4	Lightweight insulating mortars	NTUA	CETMA, BIBM, CSIC	13	24
T3.5	High performance insulating bricks	CSIC	CETMA, AMS	13	24
T3.6	Open call test cases for Materials PLs	CSIC	RISE, NTUA, AMS	25	36
WP4	Characterization and testing of smart building envelope materials component and systems	NTNU		13	36
T4.1	PVTs embedded in lightweight structural elements	SINT	SOLI, UNST	13	24
T4.2	Smart windows	NTNU	SOLI	13	24
T4.3	Prefabricated insulating lightweight walls	UNST	FORTH, YCCM, NTNU	13	24
T4.4	Multifunctional cladding systems	NTUA	UNST, FORTH, ULIG, NTNU	13	24
T4.5	Open call test cases for test cases for Components and smart manufacturing PLs	NTNU	SINT, UNST	25	36
WP5	Characterization & Environmental Services	UNIVPM		13	48
T5.1	Structural & morphological characterization	UNIPVM	PL owners	13	36
T5.2	Functional properties characterization	FALU	PL owners, end users	13	48
T5.3	LCA/LCC analysis	CETMA	CETRI, all partners	13	48
T5.4	Recyclability assessment and circular economy aspects	NTUA	Pilot Line owners	13	45
WP6	Final Prototypes evaluation and demonstration in Living Laboratories and sustainability	NTUA		1	48
T6.1	Set up of lighthouses	SINT	NTUA, PL owners, FORTH, SOLI, TECH, BEIA, YCCM	1	24
T6.2	Demonstration activities of Exploit4InnoMat products	UNIS	all partners	25	48
T6.3	Development of the Exploit4InnoMat products database	NTUA	all partners	25	48
WP7	Impact maximisation	UNIS		1	48
T7.1	Dissemination, communication and capacity building	UNIS	all partners	1	48
T7.2	Exploitation and sustainability	DIAD	all partners	1	48
T7.3	Collaborative Ecosystem: Building the "Materials" Helix	CHX	all partners	1	48
T7.4	Liaison with policy makers and stakeholders	DIAD	all partners	1	48
T7.5	Training activities	WHITE	all partners	1	48
T7.6	IP Management and IP Protection - Patent Filing	ICP	all partners	1	48
T7.7	Launch of the open call	ICP	all partners	14	48



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3.2 Consortium as a whole

The Exploit4InnoMat Consortium includes well known organizations across 9 European Member States (more specifically: Greece, Spain, Germany, Ireland, Italy, Belgium, Cyprus, Romania and Lithuania).
However, there is a direct and clearly identified need to involve close nations such other European countries in order to maximise/fully increase the overall European competitiveness; as such, Norway (2 Partners), United Kingdom (1 Partner), Turkey (1 Partner) -associated countries to the EU- as well as Switzerland (1 Partner) -non-associated third Country as per the EU List of participating countries- have joined forces and are an integral part of the Consortium, based on their niche expertise, technological/industrial know-how and close connection to their corresponding national manufacturing ecosystems. These Organizations will allow to communicate and disseminate the Exploit4InnoMat Consortium message to industrial organizations in these important industrial countries in Europe.**A. 11 SMEs**

SOLI is an SME producing BIPV with a strong research capacity. Over 60% of their BIPV panels are exported in Sweden, Norway and Germany proving their strong position in Europe. Together with SINTEF, they will commercialize a new series of PVT systems, suitable for integration in building renovation projects. **ICP** has great experience with commercially oriented skillsets that uniquely supports innovation teams on their journey from research to Commersalization, through developing a robust IPR Strategy that will lead to Patent Submission. **STAM** is an engineering company, which performs all stages of the product design cycle, from conception through validation with CAD and FEA tools, to component and subsystem design and specification, down to the definition of tools, and production cycles. It will undertake the set-up of Exploit4InnoMat Ecosystem tools. **TECH** is highly specialised in robotics integration, production process monitoring, manufacturing and programming industrial control devices. **BEIA** specialized in business strategy of companies with the aim of building a solid innovation strategy driven by the market demand. It will lead the Innovation Management services of Exploit4InnoMat. **WHITE** provides smart solutions, and insight to technology and innovation landscape, whereas bridges engineering excellence, generating cost-effective solutions. **DIAD** is an ICT company with strong expertise in materials modelling and simulation and specifically for materials for building applications. of Exploit4InnoMat project. **CHX** will disseminate project results through the Green Materials Helix platform and network, promoting innovative & sustainable data-driven solutions in novel materials manufacturing & production, contributing to CO2 neutral production of new products beyond the lifetime of the project. AMS will provide solutions to a growing market of engineering services. Advanced Material Simulation is a high-tech enterprise, which has specialized media and a team of professionals whose constant goal is to provide a personalized service tailored to customer needs.

IRIS is an an advanced engineering Group specializing in the manufacture and integration of Process Analytical Technology (PAT)-based real-time quality monitoring solutions for the process industries and IoT systems for smart manufacturing. **YCCM** is a Cypriot company and a market leader in commercial available coatings an anti-COVID , antibacterial coating from YS Cypriot Catalysts will be provided which contains Cu nanoparticles. This coating will be also investigated in order to be combined with the existing ones that were mentioned above.

B. 12 RTDs

RISE is the Swedish Research Institute and innovation partner that offers unique expertise and about 100 testbeds and demonstration facilities, instrumental in future-proofing technologies, products and services. Under Exploit4InnoMat. **NTNU** is bringing the knowledge of smart windows and smart glazing. They will deliver high performance and aesthetics smart windows that will be able to be integrated in pre-fabricated components. **NTUA** is the largest Technical University in Greece, the specific group has a great research experience in designing building materials and characterization of their performance, whereas it has fully equipped laboratories for carrying out the most challenging tests for building materials and components, approaching the concrete technology both from chemical and mechanical side. NTUA will be the Technical Coordinator. **UNST** has a strong expertise in the field of construction materials utilizing advance technologies. They will work on the structural cement-based components. **CETMA** is specializing in developing innovative solutions related to materials, components and systems for industrial and services applications, whereas in this project it will develop innovative pre-casted cement-free construction components. **SINT** is the largest research institute in Norway. The participating group has a strong background in the field of BIPVs and PVT systems. Together with SOLT they will work for the development of a new generation of such systems. **UNPD**, **UNAV** and **UPDM** have both experience in the modelling of buildings and delivering models that will assist in the purposed of Exploit4InnoMat by applying Computer-Aided Design, modelling and simulation processes that encompass the Building Information Modelling Tools. **CSIC** is Spain's largest public research institution mainly specializing in high-end ceramics products. They will collaborate with SCHL for developing innovative products for the purposes of the project. FORTH-IESL will be used for development and manufacturing of novel, multifunctional 3D printed nanocomposites. FORTH has great experience in the synthesis and mixing of polymeric nanocomposites for various applications, varying from photocatalytic devices heating insulators for buildings and constructions.

D. 2 Association

BIBM is Federation of the European Precast Concrete Industry and the major precast concrete industry hub which can deal with European-wide issues. BIBM provides a platform for a network of professionals, experts and

industrialists involved in the precast concrete industry to come together and address jointly some key issues. It will lead the clustering and training activities of Exploit4InnoMat.

F. 3 Large Enterprises

UNIS, is a company with great expertise in the coordination of EU projects. Their research activities in the project will be a cornerstone for the Exploit4InnoMat project. UNIS is specialized in applied innovative solutions for cloud-based technology areas, providing services for creative design, digital marketing, web & mobile application development and business software implementations. UNIS, will be the coordinator of the project and will lead the Exploitation, Dissemination and Communication activities. **KANS** is one of the biggest paint companies in Turkey, which provides products to more than 500 customers in Turkey, Europe, Middle East and Africa, whereas it is ISO certified for its production. In collaboration with RISE, it will develop multifunctional paints/coatings. **Dr Eleftherios Kiamilis (Project coordinator)** will be the intermediary between the partners and the EC and shall perform all tasks assigned to it as described in the Grant Agreement and the Consortium Agreement. Furthermore, the coordinator will take over the day-to-day management of the project consisting of both strategic (technical management, risks, external opportunities,) and operational (animation, communication, reporting) management tasks (see WP1 for details). The internal organization of the consortium is represented in to achieve the expected objectives and outcomes of the call and work programme (See Sections 1.1.1 and 2.1.1). **Management Support Team (MST)** members are proposed by the Coordinator as seen in Table 3.2 and validated by the Executive Board who lead the daily activity. They will assist and facilitate the work of the Executive Board and the Coordinator for executing the decisions of the General Assembly and carry out high level day-to-day management of the project.

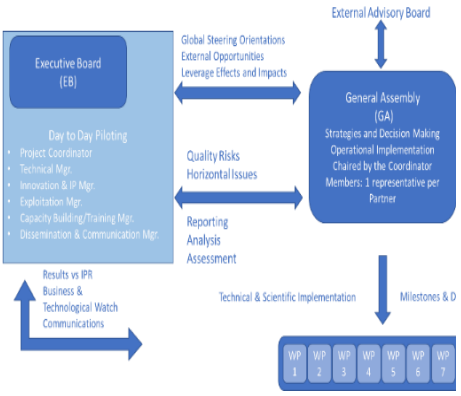

Fig. 3: Organizational Structure of Project

Table 3.2 The Executive Board consists of the Coordinator, WP leaders and the managers in the table below.

EB manager members	Names
Technical Manager	Dr Antonis Peppas (NTUA)
IP Manager	Dr Pat O' Sullivan (ICP)
Exploitation Manager	Dr Georgios Athanasiou (DIAD)
Skills development/Training Manager	Dr Sonia Saracino (CETMA)
Diss. & Comm. Manager	Dr Estefania Ledesma (CHX)
Social Media Manager	Dr Estefania Ledesma (CHX)

Research & innovation activities reporting will be implemented on a 6 months basis and presented during General Assembly meetings who will assess the progress achieved by the project. Operational progress will be monitored by the Project Coordinator and WP leaders at the level of WP every 3 months. Physical meetings gathering the General Assembly will be organized on a 6-monthly basis in order to review and assess the project achievements. Project mid-term and final reviews will be implemented as stipulated in the Grant Agreement. The General Assembly will organize this mid-term meeting with all partners, and the EC if appropriate.

Decision making, the GA will be the ultimate decision-making body for the project. The principles of the decision-making mechanisms to avoid status-quo or blocking situations, as describedin the CA will be applied:

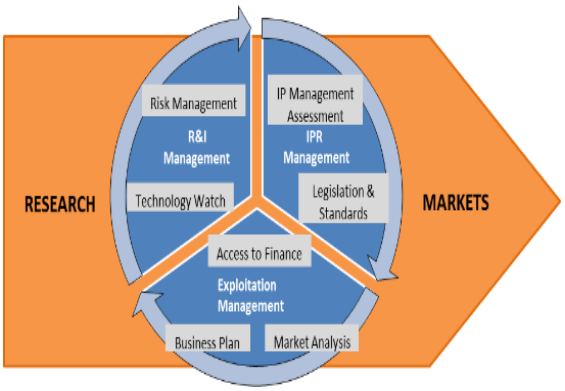

Fig. 4: Project innovation management

Knowledge management, Project documentation will be stored and exchanged with a private and secure platform that will be administrated by the Project Coordinator. All project members will have a registered access for project internal document exchanges. Draft documents will be edited and stored following the project proposed quality chart. Such platform will allow an efficient knowledge management ensuring data and information exchanges between project members without unnecessary risk of external disclosure.

Innovation management involves different resources to successfully bring technologies generated from research activities to the markets. Such process requires a strong interlink between Research & Innovation activities, Intellectual Property Rights management and Exploitation management that are covered in the project.

4 Ethics and Security

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The beneficiaries will ensure that all ethics issues related to activities in the grant are addressed in compliance with ethical principles, the applicable international and national law, and the provisions set out in the Grant Agreement. This includes the ethics issues identified in the Ethics report and any additional ethics issues that may emerge in the course of the grant. In case any substantial new ethics issues arise, beneficiaries shall inform the granting authority.

Ethical dimension of the objectives, methodology and likely impact

The Exploit4InnoMat Consortium is fully aware of the ethical implications of the proposed research and respects the ethical rules and standards of HORIZON EUROPE, and those reflected in the Charter of Fundamental Rights of the European Union. Ethical, social and data protection considerations are crucial to this project and will be given all due attention to them. Exploit4InnoMat will involve human participants and might deal with personal data during the activities of researching the human knowledge to the domain and the decisions taken, towards creating more trusted, explainable and robust DMR. However, it will be ethically driven and privacyaware by design: the collecteddata belong to the business, who will remain the owner of them and will be able to freely and consciously choose what to share and with whom and what to remove. During the implementation phase, the consortium will guaranteethe proper handling of ethical issues and the adherence to national, European and international law, regulations and directives and more specifically: - Regulation (EU) 2016/679 (General Data Protection Regulation) - The UniversalDeclaration of Human Rights and the Convention 108 for the Protection of Individuals with Regard to Automatic Processing of Personal Data - The European Convention for the Protection of Human Rights and Fundamental Freedoms. - Directive on Privacy and Electronic Communications (2002/58/EC) and the upcoming Privacy Regulation, aiming at repealing it for reinforcing trust and security in the Digital Single Market, as well as at ensuringalignment with GDPR new rules; - Article 19 Ethical principles of Regulation No. 1291/2013 /EC of the European Parliament and of the Council which states the fundamental principles of the H2020 Ethics in research; - Directive 96/9/EC of the European Parliament and the Council of 11 March 1996 on the legal protection of databases; - The Charter of Fundamental Rights of the European Union (2000); - Directive EU/2016/1148 NIS Directive, and the upcoming Regulation on Privacy and Electronic Communications In order to conform to national and international ethical guidelines, every participant involved in the project will receive an introduction and continued support from the ethical board. The partners involved in the pilots will receive this information before involvement in any Exploit4InnoMat activities dealing with data related to piloting or data they will use for research. This is the responsibility of each partner organisation. For what concerns the use of AI, the main ethical issues so far identifieddo not directly focus on lawful AI but instead translated some ethical principles for ethical and robust AI into sevenrequirements that Trustworthy AI should meet: 1. human agency and oversight: protection of fundamental rights, theinteraction between humans and AI systems; 2. technical robustness and safety: resilience, accuracy, reliability of AIsystems; 3. privacy and data governance: data protection, data management, privacy rights; 4. transparency: traceability, explainability, communication; 5. diversity, non-discrimination and fairness: accessibility, lawfulness;
6. environmental and societal well-being: sustainability, social and societal impact; 7. accountability: audibility, reporting, responsibility.

Compliance with ethical principles and relevant legislations

The research activities during Exploit4InnoMat involve the participation of humans to the extent that the project will interact with individuals in the living labs and gather, process and share data on their working environment. The purpose of this research fully respects human dignity and ethical values are fully to the side of the individual who will be involved on a voluntary basis: the project will contribute to improving the general quality of work of business experts and will serve them with value-added information and services. The free and aware choice of the data subjects is assured by the provision to them of the risk assessment and privacy exposure report regarding their shared data. In this way, both in the validation phase and in the post-project uptake of Exploit4InnoMat technologies, it will be concretely ensured that the individual's consent is freely given, specific, informed and unambiguous, as requested by GDPR art. 7 and Recitals 32. Likewise, situations, where individuals feel pressured to share, are avoided, while safeguarding human dignity at the maximum extent. The data processing foreseen by Exploit4InnoMat does not involve the collection of any sensitive personal data, tracking and observation of participants. In such a context, the results of the research have zero risk and burden to the participants and will benefit the individual or the group represented by the participant. Actually, the individuals are recognized as the starting point and key players of Exploit4InnoMat value chain and their significant role in the business decision making is coupled with the attribution of data's (and its derivatives) value to them. They will gain benefits as employees (e.g. through insights provided by better and more explainable AI models). Furthermore, they will even profit from Exploit4InnoMat as members of a group of an educated workforce, which will take advantage of the improved data-driven efficient, effective and value-added services, thanks to the availability of additional tons of personal data, complementing the ones they already manage in order to adhere to the ethical principles, the Exploit4InnoMat Ethical Policy will ensure that the activities are compliant with the EU / national legal and ethical requirements. - Ethical Issues will be addressed at a first level within the consortium, for potential ethics issues arising from the activities to be performed during the project - Organization of open discussion and consultation with stakeholders on the privacy and ethics issues arising from project research. These participatory events will also address ethical and legal concerns and expectations; - All the

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subjects involved in the demonstration activities will be strictly volunteers. The voluntary participants will be recruited from the pilot organizations following recruitment procedures and inclusion/exclusion criteria previously set, including also details on the sampling and recruitment process for each of the demonstrators. The Exploit4InnoMat (scientific and technological) objectives and project advancement, as well as use cases refinement and pilots planning and operation, will drive the identification of voluntary subjects. The demonstrators will serve as a basis for identifying the number of volunteers (i.e., the sample size), as well as relevant statistical parameters of the sample. All necessary steps will be taken to eliminate bias within the selection process in order to avoid discrimination based on physical and cognitive aptitude and political, social, religious and cultural, gender orientation. Under no circumstances will vulnerable subjects be selected as Exploit4InnoMat voluntary participants; this includes persons under the age of 18 and any other person unable to give informed consent. - Informed consent procedures for the participation of humans will be defined and implemented. Exploit4InnoMat will maintain transparency, i.e. it will never intentionally deceive, mislead or withhold information from participants as to the purpose and nature of the investigation. All participants in the demonstrators will be informed of all aspects of the activities that might reasonably be expected to influence willingness to participate. The foundation of ethical research is the principle of informed consent. The voluntary participants, who agree to take part in a specific exercise, after the described information session and delivery of an information sheet, will be asked to fill and sign a consent form. The informed consent form provided to, and signed by, the volunteers involved in the use cases will explain all details of the use case or validation activity according to specific National and EU legislation.

ANNEX A - Participation of UK & Switzerland partners

One (1) partner from the United Kingdom will participate in Exploit4InnoMat. Their budgets will not be covered by EU funding, but, in case of a successful proposal, a request of funding will be elaborated by them to the United Kingdom Research and Innovation (UKRI). Here below is a recap of contributions required by them.

No 17 partner [TECH INSPIRE LTD]

- Personnel costs: 383.500,00 €
- Travel: 16.000,00 €
- Services: €
- Indirect costs: 99.875,00 €
- Total costs: € 349.563,00
- Maximum funding request (100% of reimbursement rate as UK partner): 349.563.0 €

One (1) partner from Switzerland will participate in Exploit4InnoMat leading Task 5.2: Functional properties characterization /WP5. Their budgets will not be covered by EU funding, but, in case of a successful proposal, a request of funding will be elaborated by them to the State Secretariat for Education, Research and Innovation (SERI)). Here below is a recap of contributions required by them.

No 27 partner LUCERNE UNIVERSITY OF APPLIED SCIENCES AND ARTS

- Personnel costs: 372.112,00 €
- Travel: 15.000,00 €
- Services: 14.000,00 €
- Indirect costs: 100.000,00 €
- Total costs: 500.000,00 €
- Maximum funding request (100% of reimbursement rate as Swiss partner): 500.000,000 €